THE EUROPEAN UNION SPACE ACT PROPOSAL

AN INSTANT BOOK



Edited by
Monica Auteri
Francesco Morandi
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Consumatori e Mercato 19



Università degli Studi Roma Tre
Dipartimento di Giurisprudenza

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Università degli Studi Roma Tre
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Roma Tre Research Center
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Monica Auteri, Francesco Morandi, Sirio Zolea

Introduction

The publication of the European Commission's proposal for a Regulation on the safety, resilience and (environmental) sustainability of space activities in the Union — COM(2025) 335 final, 25 June 2025, immediately designated by commentators as the EU Space Act — marks a turning point in the governance of outer space that extends well beyond the internal market rationale invoked in its explanatory memorandum. For the first time, a supranational legal order is attempting to regulate essential elements across the entire lifecycle of commercial space operations — from launch authorisation to end-of-life deorbiting, from cybersecurity obligations to light pollution mitigation — to a single, directly applicable regulatory framework, binding on operators established within the Union and, through the logic of market access, on those established outside it.

The ambition is not modest. Europe has watched the space economy undergo a structural transformation that renders the regulatory vocabulary of the 1967 Outer Space Treaty increasingly inadequate. That vocabulary was drafted when satellites were few, government-owned and measured in units; today a single private operator maintains a constellation of several thousand orbital assets, and mega-constellations are reshaping the physical and electromagnetic environment of low Earth orbit. The EU Space Act is the institutional response to this transformation: a regulatory instrument conceived in the tradition of the GDPR and the AI Act, designed to impose European standards on the space sector on a global scale through the leverage of market access.

In relation to the diverse array of domestic laws, the European legislative proposal claims 'internal' sovereignty over a specific area of competence vis-à-vis the Member States. This is arguably accompanied by an ambition to assert 'external' sovereignty over the rest of the world, in the sense of an attempt to reaffirm the 'Brussels effect' (the export of regulatory standards unilaterally set by the EU). That effect has now run into difficulties in the digital sector for reasons that could resurface with equal force in the space sector: the lack of industrial leadership capable of serving as a substantial foundation for ambitions of global regulatory

hegemony. The stated legal basis for the proposed Regulation is Article 114 TFEU, which provides for the adoption of measures to ensure the establishment and functioning of the internal market. That basis is accompanied by the choice of a regulation — an instrument binding down to the finest detail — rather than a directive, justified on the ground that ensuring a uniform level of implementation throughout the Union will prevent divergences from hampering the provision of space activities and space-based data within the internal market. The declared objectives are legal certainty and transparency, the uniform protection of rights and obligations for all EU and third-country providers of space services in the internal market, and consistent implementation across all Member States. This use of Article 114 arguably circumvents the clear and specific provision of Article 189 of the same Treaty, which is intended to prevent any harmonization of Member States' legislative and regulatory space provisions.

Moreover, the proposal raises the question whether such (hyper-) regulatory approach can be beneficial to the European space industry — that is, the very internal market that an EU regulation is ostensibly intended to support. The costs of adapting to the new standards appear enormous, with the burden falling largely on the sector's companies — whether public, semi-public, or private; the economic benefits of adopting common standards, by contrast, seem uncertain and far off in the future. Meanwhile, if we consider the close integration of Europe's most successful companies into international space supply chains — particularly those led by the United States — the unilateral adoption of standards that Europe does not seem to have sufficient industrial clout to impose on other players could result in significant losses of contracts and, in the long run, in the replacement of European companies within these supply chains by competitors from other countries. European countries that are party to the Artemis Accords — linked to the U.S.-led Artemis program, which aims to establish a permanent human presence on the Moon and Mars in the coming years — might suffer even greater harm. Nor does the EU, despite its growing direct space ambitions, seem able or willing to absorb the same level and variety of contracts within a reasonable timeframe.

Therefore, it is particularly important that legal scholars, in collaboration with experts in space-related fields from other disciplines — starting with economics — should promptly offer their views on the legislative proposal, already at the current stage, while the text is still being reworked in the appropriate institutional forums, highlighting

the strengths and weaknesses of the draft to inform the institutional and political debate and the public opinion. The contributions collected in this volume examine the proposal and its impact and implications considering several complementary disciplinary vantage points — comparative and international law, public finance, corporate governance, space policy, and geopolitics — and, taken together, offer a critical anatomy of an instrument that is simultaneously ambitious and structurally constrained.

The opening contribution, by Victoria Anfuso and Giovanni Maria Riccio, situates the EU Space Act within the broader process of *actification* of European law — the systematic preference for directly applicable regulations, short branding titles, and Article 114 TFEU as the legal basis — and interrogates the coherence and the limits of transposing to the orbital domain the regulatory grammar forged for the digital sphere. Their analysis identifies a central paradox: while the Space Act faithfully replicates the *form* of European digital regulation, it encounters an irreducible *aporia* at the point where regulation meets the physical and geopolitical reality of outer space. The GDPR can reach a data controller in California through the market-access mechanism because data processing is localisable; the imaging of European territory from low orbit is not. The sovereignty that the European lawmaker exercises over the internal market dissolves precisely where the atmosphere ends. Anfuso and Riccio further document two substantive lacunae that compound this structural limit: the near-total silence of the proposal on privacy and personal data protection in the context of Earth observation — a silence that becomes increasingly untenable as commercial constellations achieve sub-metre resolution with multiple daily revisits — and the absence of any governance framework for autonomous and AI-driven space systems operating in human-out-of-the-loop configurations.

The second contribution, by Beatrice Ermini, shifts the perspective from the supranational to the firm level, and from the legal to the financial and managerial. Working from Italian Law No. 89/2025 — entered into force on the same day as the Commission's proposal, with remarkable temporal coincidence — Ermini develops a methodological framework for translating space regulatory obligations into structured ESG risk indicators capable of integration within corporate Enterprise Risk Management systems. The framework is applied, as a conceptual mini-case, to Avio S.p.A., whose 2024 sustainability disclosures constitute the first reporting cycle falling within the CSRD perimeter and the last to precede the entry into force of the national space law. The contribution's

central analytical claim is that regulatory obligations in highly governed sectors such as aerospace are not merely external compliance constraints: they are endogenous generators of risk exposure across the Environmental, Social and Governance dimensions. An authorisation regime administered by the Presidency of the Council of Ministers introduces timing uncertainty that directly affects revenue recognition on multi-year ESA contracts; a dual-track liability structure generates insurance and provisioning variables that feed directly into the financial materiality assessment required under ESRS 2; debris mitigation obligations shape capital allocation in R&D for upper-stage passivation and deorbiting. The framework thus illustrates, through the lens of an operator that sits at the intersection of national space law, European sustainability reporting and the prospective EU regulatory framework, how the convergence of space regulation and sustainability governance is producing a new category of legally grounded ESG indicator — one that is neither a pure compliance metric nor a conventional financial variable, but a structured bridge between the two.

The third contribution, by Frans von der Dunk, addresses what is perhaps the most conspicuous omission in the Commission's proposal: the treatment of international third-party liability. Von der Dunk documents, through a systematic comparison of the thirteen national space laws currently in force among EU Member States, the substantial variation in both the scope of national authorisation regimes — who qualifies as a 'national' operator subject to licensing requirements — and the structure of recourse obligations in the event of an international liability claim under the 1972 Liability Convention. The analysis reveals a landscape in which most Member States limit operator recourse to approximately €60 million per incident, but in which Sweden retains comprehensive reimbursement, Luxembourg applies unlimited liability, and several jurisdictions defer limits to further regulation. This fragmentation is precisely the kind of internal market distortion that the Space Act invokes Article 114 TFEU to remedy — yet the proposal is entirely silent on third-party liability. Von der Dunk's explanation of such silence is analytically compelling: harmonising recourse obligations would require taking a visible and politically sensitive position on the ceiling that operators can be required to bear, and would simultaneously constitute an unmistakable instance of the harmonisation of national laws that Article 189(2) TFEU explicitly prohibits in the field of space policy. The draft EU Space Act thus chooses to regulate what is politically achievable while leaving untouched the dimension most

directly relevant to the private sector's competitive calculus.

The fourth contribution, by Rafael Moro Aguilar, provides both historical context and a real-time chronicle of the legislative process. Moro Aguilar traces the intellectual genealogy of the Space Act back to the EU's International Code of Conduct for Outer Space Activities, developed between 2010 and 2014 and ultimately shelved for want of international support, observing that the three pillars of the current proposal — safety, resilience, sustainability — reproduce with striking continuity the objectives that the Code of Conduct had articulated a decade earlier. The failure of that earlier initiative and the current trajectory of the Space Act illuminate a persistent tension in European space governance: the Union's aspiration to set global standards through the authority of its regulatory acquis, and the resistance of non-European actors — above all the United States — to what they characterise as regulatory overreach and protectionism. Moro Aguilar tests this tension against the specific case of the protection of dark and quiet skies, a provision included in the original proposal at the recommendation of researchers and scientific societies worldwide, and subsequently softened in successive compromise texts under sustained pressure from the US government and industry. The trajectory of this single article — from a hard seventh-magnitude brightness threshold to a plan-based, technology-neutral approach aligned with existing FCC requirements — encapsulates the broader dynamic of the entire legislative process: European regulatory ambition calibrated downward in the encounter with geopolitical and commercial realities.

The fifth contribution, by Alfonso Giordano, reads EU space policy as a problem of economic-political geography: how the Union seeks to reconcile a single-market template for space with a strategic autonomy trajectory in an orbital environment that is increasingly crowded and commercially dominated by non-EU actors. Applying a quadruple analytical filter (feasibility, desirability, legitimacy, sustainability), it identifies three recurring fault lines in EU space governance: an unresolved competence-and-legitimacy question, a persistent core-periphery landscape across the European space economy, and a structural tension between strategic autonomy and the long-term sustainability of an increasingly occupied orbit. In Giordano's article, legal instruments - the Treaty on the Functioning of the European Union, the draft EU Space Act, the Regulation establishing IRIS², the Joint Communication on a European space strategy for security and defence - are treated throughout as materials for analysing governance trajectories, territorial and industrial

geographies, and an emerging 'space society' in orbit, rather than as the primary object of doctrinal legal analysis. The general thesis of the article is that EU space governance is not a coherent strategy converging on a settled endpoint, but an unresolved compromise between single-market and strategic-autonomy rationales, pursued through overlapping instruments and an institutional architecture inherited from an earlier, less contested period of European space cooperation.

The sixth contribution, by Dario Elia Tosi, begins with a brief review of the current provisions governing space activities at the international level, and then reconstructs the evolution and the present configuration of the governance of the space sector at the EU level. Based on that, the study sketches out the most important features of the governance model provided for in the draft Regulation submitted by the Commission for the Parliament and the Council last year and its possible consequences for the domestic models adopted by certain Member States. In fact, countries like France and (recently) Italy that have already established a comprehensive system of space governance might be compelled to reconsider it, in case of approval of the European Act.

The seventh and final contribution of the volume, by Stefania Paladini, extends the analysis from the internal architecture of the Space Act to its external geopolitical consequences, providing the broadest systemic frame of the volume. Paladini situates the proposal within the longstanding fragmentation of international space law — five core UN treaties drafted in the 1960s and 1970s, manifestly inadequate for a sector now driven by private operators and over eighty spacefaring nations — and traces the incremental evolution of EU space governance from the establishment of ESRO and ELDO in the early 1960s through the Lisbon Treaty's structurally ambiguous Article 189 TFEU to the present regulatory ambition. Her analysis focuses with particular depth on the sustainability pillar, which she identifies as the most geopolitically consequential dimension of the Act and the one least reducible to technical standard-setting. By mandating full life-cycle assessments (LCAs) for space missions, connecting those assessments to the Ecodesign for Sustainable Products Regulation and its Digital Product Passport framework, and interlocking the entire architecture with the Critical Raw Materials Act's domestic content targets, the sustainability pillar functions, Paladini argues, as a comprehensive industrial sovereignty instrument — one that shapes who can supply the European space market and on what terms, independently of tariff structures. The CRMA's requirement that no

single third country supply more than 65 per cent of EU consumption of any strategic raw material, combined with the Life Cycle Assessments' obligation to disclose and ultimately reduce the use of virgin materials, creates a regulatory gateway that is particularly consequential for Chinese rare earth supply chains and for the titanium recycling loops that currently traverse transatlantic buy-back agreements. Paladini documents with equal rigour the differentiated reactions of Washington, Beijing and New Delhi: the United States, whose operators account for approximately three-quarters of all active satellites, perceives the giga-constellation threshold and the asymmetric authorisation procedure — under which non-EU operators must obtain authorisation directly from EUSPA, the same agency involved in the operation of the competing IRIS^2 constellation — as structural non-tariff barriers dressed in environmental language; China, confronted with the mandatory transmission of data to an EU-administered Environmental Footprint Database, anticipates technology transfer risks and is responding by accelerating domestic standards development through ITU channels; India, whose rapidly liberalising space sector is increasingly oriented toward the European downstream market, faces a structural 'framework gap' — the absence of any domestic LCA equivalent — that translates into bespoke compliance costs of several million euros for a medium-sized operator. Whether the Brussels Effect in space will function as a convergence mechanism toward higher global sustainability standards or as a sophisticated instrument of protectionism, depends critically, Paladini concludes, on amendments still to be introduced through the ordinary legislative procedure.

Read together, these seven contributions converge on a diagnosis that is at once sobering and instructive. The EU Space Act is conceived as a legally sophisticated and institutionally coherent instrument — although it remains to be seen to what extent its initial structure and content will be reflected in the final text, given that the draft already appears to be undergoing significant revisions as the process unfolds. Its safety and cybersecurity architecture seeks to supersede the fragmented national landscape. Its sustainability provisions, even in attenuated form, establish a normative baseline intended to exert regulatory pull beyond the Union's borders. And yet the proposal discloses, in each of the dimensions examined here, the same structural limit: the power to regulate access to the European market is not equivalent to the power to regulate access to outer space in general. The Brussels Effect — real in some sectors, ineffective in others — sees its effective radius contract precisely where it

is most needed — in the face of operators who can choose not to access the European market while continuing to observe, connect and monetise it from above; in the face of debris-generating actors whose satellites predate the regulation's temporal scope; and in the face of geopolitical rivals whose response to European standard-setting is not alignment but the accelerated construction of alternative frameworks.

This volume does not aspire to offer definitive answers to the questions that the Space Act Proposal raises. It aspires, rather, to provide the analytical tools — legal, economic, managerial, and of international politics — with which legislators, practitioners, scholars, and policymakers can try to look for and find solutions to those questions with greater precision. The space economy is still an *adolescent* sector: old enough to generate consequences of global scale, not yet mature enough to have attracted governance commensurate with those consequences. The EU Space Act is one of the first serious attempts to bridge that gap. Whether it succeeds will depend less on the quality of its drafting than on the political will to address the structural limits that the chapters in this volume so clearly document: the sovereignty paradox that no regulation can overcome by itself, the liability gap that political prudence has chosen to leave open, the tensions with the national and international regulatory layers, and the geopolitical fractures that a unilateral Brussels Effect risks deepening rather than healing.

Victoria Anfuso, Giovanni Maria Riccio

EU Space Act, Data Protection and Artificial Intelligence: Continuity and Discontinuity with the EU Digital Regulations

Summary: 1. Introduction – 2. The EU Space Act as a manifestation of the "actification" of Union law – 3. Substantive lacunae: what the Space Act leaves unregulated – 4. The sovereignty paradox: regulating what cannot be controlled – 5. Mega-constellations and the dilution of regulatory control – 6. The Starlink case: a paradigmatic stress-test for the European legal arsenal – 7. High-resolution Earth observation and the spatial dimension of privacy – 8. Conclusion: between regulatory ambition and geopolitical realism.

1. *Introduction*

On 25 June 2025 the European Commission unveiled COM(2025) 335, the proposal for a Regulation on the safety, resilience and environmental sustainability of space activities in the Union — popularly designated as the EU Space Act[1]. The initiative aspires to consolidate a hitherto fragmented landscape characterised, according to the Commission's own assessment, by thirteen divergent national regimes;[2] it pursues the simultaneous objectives of legal certainty, competitiveness and the harmonisation of safety, resilience and sustainability standards. With its envisaged entry into force on 1 January 2030, the proposal is positioned within an unmistakable lineage: the decade-long stratification of European digital law that has produced, in sequence, the General Data Protection Regulation (2016), the NIS2 Directive on cybersecurity (2022), the Digital Services Act (2022), the Data Act (2023) and the Artificial

[1] European Commission, *Proposal for a Regulation of the European Parliament and of the Council on the safety, resilience and sustainability of space activities in the Union*, COM(2025) 335 final, 25 June 2025. For an institutional framing see C. Evroux & A. Papunen, *European Space Act*, EPRS Briefing, European Parliament, September 2025.

[2] European Commission, *EU Space Act: enhancing market access and space safety*, Press release IP/25/1583, Brussels, 25 June 2025.

Intelligence Act (2024).[3]

The Commission's underlying ambition is no longer concealed. The Space Act extends to the orbital domain the regulatory grammar that has become the Union's distinctive contribution to global rule-making: maximal harmonisation by regulation, extraterritorial reach grounded on market access, and the aspiration to set worldwide standards. Once adopted, the instrument is expected to inaugurate a single market for space activities, to impose sector-specific cybersecurity obligations modelled on the *Digital Operational Resilience Act* (DORA) and on NIS2, and to bind not only Union operators but also third-country actors providing space services within the internal market.

The present contribution examines the dialectic of continuity and discontinuity between the Space Act and the broader corpus of European digital regulation, foregrounding the structural limits that emerge once the regulatory technique forged for the digital sphere is transposed to outer space. The thesis highlighted in these pages is straightforward: while the Space Act faithfully replicates the *form* of European digital law, it simultaneously discloses an irreducible *aporia* once confronted with the geopolitical and physical reality of space activities. The Space Act, in short, embodies the paradox of a European regulatory sovereignty that ceases precisely where outer space begins. If — to borrow the formula that orients the analysis — for space *ambition* the sky is not the limit, for European *regulation* it is, and so it shall remain.

2. *The EU Space Act as a manifestation of the "actification" of Union law*

The Space Act inscribes itself within the broader process of "actification" that has shaped Union normative production since the GDPR.[4] The phenomenon is recognisable from a set of recurring formal traits: the systematic preference for regulations over directives, ensuring uniform and direct applicability; the recourse to short titles facilitating public uptake of the instrument; and the grounding of the legal basis on Article 114 TFEU

[3] For an introductory framework on the legislative landscape of space law see R. Jakhu & P.S. Dempsey (eds.), *Routledge Handbook of Space Law*, Routledge, 2017, especially the chapter on *Introduction, Factual Background and Sources of National Space Laws*.

[4] V. Papakonstantinou & P. De Hert, *The Regulation of Digital Technologies in the EU: Act-ification, GDPR Mimesis and EU Law Brutality at Play*, Routledge, 2025; A. Miglio, *The EU Space Act proposal: A GDPR for outer space?*, in *European Law Blog*.

on the internal market, which side-steps the political constraints inherent in Article 189 TFEU — the Treaty provision specifically dedicated to space policy, which explicitly excludes any harmonisation of national legislation.[5]

This formal coherence is not ornamental.[6] It betrays a deliberate strategy aimed at unifying the regulatory architecture of the digital and the space economies, treated as two declensions of a single challenge: the governance of activities intrinsically transnational, technologically sophisticated, dominated by powerful private operators, and presenting systemic risks for security, fundamental rights and the environment.[7]

The Space Act likewise borrows the extraterritorial mechanism that defines the GDPR. Just as Regulation (EU) 2016/679 captures any processing of personal data of individuals located in the Union, irrespective of the controller's place of establishment, so does the Space Act assert jurisdiction over any operator providing space services or space-based data within the Union, whether or not established on Union territory.[8] The technique is that of the so-called *marketplace principle*:[9] entry into the European market triggers the obligation to comply with

[5] M. Kellerbauer, *Article 114 TFEU*, in M. Kellerbauer et al. (eds.), *The EU Treaties and the Charter of Fundamental Rights: A Commentary*, OUP, 2019, 1235. Some authors have questioned whether reliance on internal-market provisions to regulate space — particularly where the relevant rules intersect with national-security considerations — may stretch Union competence beyond its constitutional bounds: see B. Jacobs, *Understanding the EU's competence to harmonise space law*, in *Air and Space Law*, 49(4), 2024.

[6] S. Benvenuti & S. Zolea, *Legal Comparison as a Tool to Sketch New Rules and Institutions for a Space Society*, in *International Journal for the Semiotics of Law*, available at https://doi.org/10.1007/s11196-026-10485-y.

[7] Article 114 TFEU empowers the Union to adopt measures for the approximation of national laws affecting the establishment and functioning of the internal market, typically invoked where disparities among Member-State frameworks generate obstacles to trade or distort competition. Article 189 TFEU, by contrast, establishes Union competence in the field of space policy but explicitly precludes the harmonisation of national laws: under this provision, the Union may support, coordinate or complement Member-State action — including in research and technological development — but it lacks the authority to impose binding uniform rules on domestic regulatory regimes.

[8] Court of Justice of the European Union, Case C-230/14, *Weltimmo* (2015), broadly construing the concept of "establishment" under Union data-protection law: an organisation may be deemed established in a Member State where it carries out any real and effective activity, however minimal, through stable arrangements.

[9] See, for instance, *Report on the proposal for a regulation of the European Parliament and of the Council on the protection of individuals with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation)*, 21 November 2013, COM(2012) 11.

European rules. The reproduction is virtually identical, complete with the creation of a mandatory *Union Register of Space Objects* (URSO) under Article 24 of the proposal, which conditions access to the internal market.

The transposition, however, is not unqualified. The much-debated extension of the *Brussels Effect*[10] to the space sector encounters limits that the comparison with the AI Act usefully clarifies. Three factors should be highlighted. First, the AI Act presupposes administrative capacity, sectoral expertise and enforcement infrastructures that are unevenly distributed across jurisdictions, particularly in the Global South. Second, divergent political and economic priorities — innovation, digital inclusion, technological catch-up — reduce the incentives for non-EU jurisdictions to align with European templates. Third, the economic leverage that propelled the global diffusion of the GDPR is significantly less pronounced in the AI domain, where market access does not yet hinge on compliance with a single regulatory model.[11]

The transposition to space activities encounters even more obstacles. The space sector is structurally fragmented: a small number of highly specialised actors, pronounced state involvement, and acute geopolitical sensitivities weaken the market-mediated dynamic that underpinned the success of the GDPR. Unlike the processing of personal data — ubiquitous and difficult to geographically segment — space activities remain territorially anchored to launch infrastructures, are subject to national licensing regimes, and are deeply enmeshed with sovereignty concerns and strategic interests. A further limit is the pre-existence of parallel international regimes — pre-eminently those developed under the auspices of the *United Nations Office for Outer Space Affairs* (UNOOSA) and the body of UN space treaties — which, however imperfect in their enforcement mechanisms,[12] generate a form of normative pluralism that dilutes the Union's capacity to set unilateral global standards. In the field of data protection, by contrast, no equally consolidated international regime preceded the GDPR, allowing the Union to occupy a regulatory vacuum.

[10] C. Ryngaert & M. Taylor, *The GDPR as Global Data Protection Regulation?,* in *AJIL Unbound*, 114 (2020). For a critical perspective qualifying the Brussels Effect as a form of protectionism, see M. Bauer & D. Pandya, *EU autonomy, the Brussels Effect, and the rise of global economic protectionism*, ECIPE Occasional Paper No 1/2024.

[11] O.O. Cherednychenko, *The Public and Private Enforcement of EU Digital Space Private Law*, in *European Review of Private Law*, 33 (2025), 1073.

[12] UNOOSA, *Guidelines for the Long-Term Sustainability of Outer Space Activities*, adopted by the Committee on the Peaceful Uses of Outer Space (COPUOS) in 2019 and consolidated in *United Nations International Space Law Booklet*, Vienna, 2025.

In the cybersecurity dimension, the Space Act explicitly positions itself as *lex specialis* with respect to the NIS2 Directive. The proposal complements NIS2 by addressing the *space segment* proper — the satellites themselves — and by securing the resilience of space infrastructures against cyber threats throughout the mission lifecycle.[13] Articles 75 to 92 establish a sectoral resilience regime that prevails over the general NIS2 obligations for those space operators classified as essential or important entities.[14] These provisions refine and articulate the cybersecurity, resilience and incident-management duties of operators of satellites, ground stations and downstream services in light of the peculiar risk profile of space systems: orbital congestion, signal interference and the systemic dependence of countless terrestrial services on space-based assets.

NIS2, by contrast, applies horizontally across multiple sectors — including, to a limited extent, space-related operators — and is generally confined to medium and large entities (those exceeding fifty employees and an annual turnover of ten million euros).[15] The cascade architecture — sector-specific regulation superimposed on a general horizontal framework — is directly modelled on DORA. The Space Act even imports the most stringent of DORA's instruments, mandatory *Threat-Led Penetration Testing* (TLPT) prior to launch and triennially thereafter.

3. *Substantive lacunae: what the Space Act leaves unregulated*

The impressive formal coherence of the proposal nonetheless conceals substantive omissions that distinguish the Space Act from its digital antecedents. The most conspicuous concerns the wholesale absence of provisions on privacy and personal data protection.

While the GDPR devotes ninety-nine articles to a comprehensive data-

[13] F. Casaril & L. Galletta, *Space cybersecurity governance: assessing policies and frameworks in view of the future European space legislation*, in *Journal of Cybersecurity*, 11 (2025).

[14] A critical concern, raised by academic commentators and industry stakeholders, relates to the duplication of obligations between the future Space Act and NIS2: legal certainty and uniform rules must be prioritised in order to avoid burdening operators with overlapping requirements from different Union frameworks.

[15] I. Baumann & K. Prall, *The EU Space Act's Global Reach: Legal Considerations on the Envisaged Application to Non-EU Operators*, in *Global Policy*, 28 October 2025; L. Orešković & S. Grgić, *The new EU space regulation: one small step or one giant leap for the EU?*, in *Croatian Yearbook of European Law and Policy*, 17 (2021), 77.

protection framework, the AI Act imposes specific obligations on systems processing biometric data, the Digital Services Act regulates targeted advertising and recommender systems, and the Data Act establishes precise rules on access to data generated by connected products, the Space Act remains entirely silent on the data-protection issues raised by Earth observation and satellite geolocation.

The omission is the more remarkable in light of contemporary technical capabilities. Maxar's *WorldView Legion* constellation, fully deployed in February 2025 with six satellites in mid-inclination and sun-synchronous orbits, commercialises 30-centimetre-class panchromatic imagery sufficient to identify individual vehicles — and, depending on context, persons — with up to fifteen daily revisits over a given location.[16] Planet Labs operates a fleet capable of imaging the entire emerged surface of the Earth at multi-metre resolution on a daily basis; Chinese providers offer comparable services, occasionally at higher resolutions; new entrants are deploying synthetic-aperture-radar and infrared payloads capable of imaging through cloud cover and darkness.[17] Continuous near-real-time monitoring of the orbital environment is no longer a speculative prospect. The European satellite navigation programme Galileo, for its part, achieves twenty-centimetre accuracy through its *High Accuracy Service*, generating extraordinarily granular geolocation data, whose cross-referencing with social media and communication metadata enables forms of surveillance and profiling that the existing legal framework was not designed to apprehend.

The Commission's official rationale for the omission rests on the principle of normative separation: since the GDPR applies horizontally to any processing of personal data, its provisions need not be reproduced in a sectoral instrument focused on the operational dimensions of space safety. The argument is legally accurate but practically insufficient. The proper conceptual frame is that of *normative coexistence* rather than *integration*: the GDPR retains its function as a horizontal, constitutionally anchored regime, applicable to any processing of personal data regardless of the sector in which it occurs.[18]

[16] Maxar Intelligence, "Maxar Successfully Launches Fifth and Sixth WorldView Legion Satellites", press release, 4 February 2025; ESA Earth Online, *WorldView Legion mission* technical sheet, 2025.

[17] ESA Space Debris Office, *ESA's Annual Space Environment Report* (2025 edition).

[18] D. Jerker B. Svantesson, *Article 3 – Territorial Scope*, in C. Kuner, L. Bygrave, C. Docksey & L. Drechsler (eds.), *The EU General Data Protection Regulation (GDPR): A Commentary*, OUP, 2020, 76. On the most recent doctrinal trajectory see European Data

This architectural choice surfaces a first dimension of the paradox running through the present analysis. The Union may regulate what is *processed* in its market, but it encounters some practical limits in enforcing what is *collected* above its territory. A Chinese or American satellite may image any point of European territory in high resolution, monitor strategic sites, demonstrations, troop movements or population flows, without any operative European instrument capable of opposing it.

From a strictly doctrinal standpoint, the fact that data are collected from beyond the territorial surface of any State does not, in itself, preclude the applicability of the GDPR. Pursuant to Article 3(2) GDPR, the Regulation adopts a functional and effects-based criterion for its territorial scope, extending to processing activities related to the monitoring of the behaviour of individuals "as far as their behaviour takes place within the Union".[19] Consequently, even where data-acquisition infrastructures — including orbital ones — operate outside the territorial boundaries of the Union, the relevant criterion remains the location of the data subjects and the contextual nexus between the processing and behaviour occurring in the Union. Earth observation and analogous space-derived data practices that enable tracking, profiling or behavioural analysis of individuals located within the Union may, on this reading, fall within the GDPR's material and territorial scope.

The extraterritorial dimension of both the infrastructure and the controller or processor raises, however, considerable enforcement challenges. Where the processing is carried out by a non-EU entity using technical means located outside the Union's jurisdiction, supervisory authorities encounter structural limits in the effective exercise of the corrective powers granted under Article 58 GDPR.[20] While the Regulation in principle provides for administrative fines, orders to cease processing or restrictions on data flows, the practical enforceability of such measures depends on the existence of jurisdictional nexuses, mutual cooperation frameworks, or assets located within the Union that may be targeted.[21]

Protection Board & European Data Protection Supervisor, *Joint Opinion 2/2026* on the Digital Omnibus initiative, 10 February 2026.

[19] Article 3(2)(b) GDPR.

[20] EDPB, *Study on the enforcement of GDPR obligations against entities established outside the EEA but falling under Article 3(2) GDPR*, 3 April 2023, 11.

[21] In Case C-645/19, *Facebook Ireland and Others*, the CJEU interpreted Article 58 GDPR as conferring substantive enforcement powers on supervisory authorities — including the ability to initiate judicial proceedings under Article 58(5) — while making clear that, in cross-border contexts, such powers are not autonomous and must be

In the absence of an establishment in the Union or of a designated representative under Article 27 GDPR, enforcement may rely only on indirect — and frequently inefficient — strategies: cooperation with third-country authorities, the adoption of adequacy or transfer-related instruments, or the imposition of duties on intermediaries operating within the Union that facilitate access to or use of the data.

The second major lacuna concerns artificial intelligence and autonomous systems. The space sector is currently undergoing the massive incorporation of AI into critical mission functions: autonomous navigation and collision avoidance, automated processing of satellite imagery, on-board data reduction, optimised constellation management, autonomous in-orbit servicing and orbital manoeuvring. Such systems typically operate in *human-out-of-the-loop* configurations, given communication latency and the practical impossibility of real-time human intervention. They raise liability and safety questions which the Space Act simply does not address.[22]

The Artificial Intelligence Act articulates a detailed risk-based classification, imposes transparency and explainability obligations, mandates human oversight for high-risk systems and outlines a tailored liability regime.[23] The Space Act exhibits no equivalent architecture: no classification of autonomous space systems, no *human-in-the-loop* obligation, no liability regime fitted to algorithmically-driven decisions. The omission is particularly salient in a context marked by an increase in incidents involving autonomous satellites: narrowly avoided collisions, uncoordinated manoeuvres, malfunctions of automatic collision-avoidance systems.

A simple thought experiment illustrates the stakes. Two satellites equipped with autonomous collision-avoidance systems find themselves on a converging trajectory. Each AI independently computes an evasive manoeuvre; by an unfortunate symmetry, both algorithms select the same

exercised within the procedural framework of Articles 56 and 60 GDPR.

[22] Space-activity liability is governed by a multi-level regulatory framework that begins with the UN space treaties, which articulate a regime of strict State liability for damage to third parties. The internal rules of the European Space Agency further provide for mechanisms of solidarity and risk-sharing among Member States; in several Member States, national law fills international gaps by reference to civil codes or special statutes. See S. Zolea, *Errore e responsabilità nel diritto dello spazio in Europa: un sistema multilivello*, in *LCEonline*, 61 (2022).

[23] S. Palmieri & T. Goffin, *Artificial Intelligence and Fundamental Rights Under the European Legal Eye*, in V.L. Raposo (ed.), *The European Artificial Intelligence Act: Promises and Perils?*, Springer, 2025, 60.

direction, *aggravating* rather than reducing the collision probability. The impact occurs, generating thousands of debris fragments that endanger other space assets. Where does liability lie? With operator A, whose AI behaved according to its design specifications and complied with all applicable safety standards? With operator B, identically placed? With both? With the algorithm vendor? With the designer of the interoperability protocol that both AIs implemented?

Current international space law — anchored in the 1972 *Convention on International Liability for Damage Caused by Space Objects* — provides an absolute-liability regime for damage caused on the Earth's surface or to aircraft in flight, and a fault-based regime for damage occurring elsewhere. The framework, conceived when satellites were fully controlled by human operators, offers no satisfactory answer to the questions posed by algorithmic autonomy. The Space Act, which might have filled the gap by establishing clear rules for autonomous space systems operating in or above European space, chooses silence.

4. *The sovereignty paradox: regulating what cannot be controlled*

The core difficulty lies in a fundamental asymmetry between the regulatory ambition of the Space Act and the Union's effective means of enforcing its rules beyond its terrestrial borders. The asymmetry is not novel — the GDPR itself confronts it — but it reaches a particular intensity in the space domain, exposing the structural limits of the European extraterritorial method.[24]

The GDPR's extraterritoriality operates through a relatively straightforward mechanism: any entity wishing to process personal data of individuals located in the Union, or to offer them goods and services, must comply with the Regulation under penalty of administrative fines reaching four per cent of global annual turnover. The mechanism succeeds because the European market — with its roughly four hundred and fifty million high-purchasing-power consumers — represents an economic stake that few global operators can afford to forgo. The major U.S.

[24] Parallel difficulties have emerged in the property-law dimension of space activities: see V. Barela, *Space Colonization: Which Regulations and in Whose Interests?*, in *The Italian Law Journal*, 9 (2023), 728, 735.

technology platforms,[25] initial reluctance notwithstanding, complied with the GDPR precisely because the alternative — exclusion from the Union market — was economically untenable.

The Space Act reproduces the same logic: any space operator providing services or data in the Union must register in URSO, comply with security, resilience and sustainability requirements, and accept the supervision of European authorities.[26] Refusal entails exclusion from the European market. Yet, unlike the GDPR — which regulates processing activities carried out, in large part, on European territory or in identifiable data centres — the Space Act purports to regulate activities that *literally* take place above Europe, in a domain that by definition eludes any territorial sovereignty.

This regulatory impotence is not merely theoretical.[27] American commercial observation satellites routinely image European military bases, nuclear facilities and sensitive research centres; Chinese satellites do likewise. The resulting imagery is then commercialised — not within the Union, in order to avoid the constraints of the Space Act or the GDPR, but to third-party clients, governmental or private, who may exploit it at will.[28] The Union may prohibit the use of such imagery on its own territory, but it cannot prevent its acquisition from orbit, nor its commercialisation elsewhere in the world.

The 1967 *Outer Space Treaty* — the cornerstone of international space law — provides that outer space "shall be free for exploration and use by all States". It prohibits the national appropriation of space, the placement of weapons of mass destruction in orbit and the establishment of military bases on celestial bodies. It does not, however, prohibit Earth observation from space at any resolution.[29] The absence of such a prohibition is not the result of inadvertence: it reflects a deliberate choice by the two superpowers of the era, the United States and the Soviet Union, both of which were already systematically using observation satellites for intelligence purposes

[25] The reference is to the so-called *GAFAM* group (Google, Apple, Facebook, Amazon, Microsoft).

[26] B. Petit, *Proposition de règlement European Space Act: les premiers pas de l'Union dans l'arène du NewSpace*, in *Revue de l'Union européenne*, 2025, 563.

[27] See, for instance, A. Guyomarc'h, *Property on Space Resources: The Search for a Terminology*, in *Market & Innovation*, 2 (2023), 73.

[28] A.S. Courdier & S. Manciaux, *New Space, la nouvelle économie des activités spatiales: enjeux juridiques et éthiques*, LexisNexis, Paris, 2023.

[29] *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies* (Outer Space Treaty), 1967, Articles I–IV.

and had no interest in mutually proscribing the capability.

The corollary principle, often described as the *open skies* doctrine,[30] is that no State may oppose the overflight of its territory by a satellite, regardless of mission, provided that no weapons or hazardous objects are released. The principle was tenable when space capabilities were the monopoly of superpowers and imagery resolution was measured in metres; it has become deeply problematic in an era of commercial sub-metre imagery and continuous constellation-based monitoring of the entire Earth's surface.[31]

Europe thus finds itself in a paradoxical condition. It can adopt the world's most stringent rules on data protection, on the security of information systems, on the environmental sustainability of space activities, and can impose them on every operator seeking access to its market. But it cannot prevent an actor that *chooses* not to access that market from nonetheless freely collecting data on its territory, citizens and infrastructure from orbit. The limit is neither contingent nor technical: it is structural and follows directly from the architecture of international space law and from the physical reality of outer space.

5. *Mega-constellations and the dilution of regulatory control*

This operational limit is significantly compounded by the technological and economic evolution of the sector, encapsulated in the transition from *Old Space* to *New Space*. For decades, space activities were the near-exclusive preserve of States and a limited number of large industrial contractors operating under government contracts; satellites were few, individually expensive and subject to strict national control. The drastic reduction in launch costs, the miniaturisation of satellites and the emergence of private operators — SpaceX, Planet Labs, Capella Space and a growing number of Chinese players — have transformed the sector beyond recognition.

The quantitative shift is staggering. In 2020, the population of

[30] The principle is often understood as reflecting Article I of the Outer Space Treaty, which establishes that outer space "shall be free for exploration and use by all States without discrimination" and "shall be the province of all mankind"; the Treaty itself does not, however, employ the *open skies* terminology.

[31] P.J. Blount, *Developments in Space Security and Their Legal Implications*, in *Law/Technology*, 44 (2011); A. Lee, *The Future of the Law on the Moon*, in *Journal of Air Law and Commerce*, 88 (2023).

active satellites stood at approximately three thousand; according to the European Space Agency, the figure exceeds fifteen thousand in 2025, and projections converge on a possible total of around one hundred thousand orbital assets by 2030, with announcements by private operators cumulatively pointing to as many as four hundred thousand satellites in the longer term.[32] To this quantitative transformation a qualitative one is added: where traditional satellites were unique, highly sophisticated and individually launched assets, the new constellations consist of hundreds or thousands of standardised, mass-produced satellites at comparatively low unit cost.

The Space Act addresses this transformation only imperfectly, as the Starlink case — analysed in Section 6 — demonstrates with particular clarity. At the time of the proposal, in June 2025, SpaceX's constellation already comprised over six thousand satellites in low Earth orbit, with authorisation from the U.S. Federal Communications Commission (FCC) to deploy up to twelve thousand and pending requests extending towards forty-two thousand. By March 2026 the constellation surpassed ten thousand active satellites, representing approximately two-thirds of all operational satellites in orbit.[33] Amazon's *Project Kuiper* envisages over three thousand units, while a Chinese mega-constellation segment — *Guowang*, *Qianfan/Spacesail* — is now under accelerated deployment.

Against this backdrop the Space Act articulates a deceptively simple principle: any operator providing space services in the Union must register in URSO and comply with the regulatory requirements.[34] The principle, however, runs against a substantial practical limit: the Regulation will apply only to satellites launched after 1 January 2030, while the tens of thousands of satellites already in orbit — operational for an additional five to ten years, often longer with mission extensions — fall entirely outside its scope.

The temporal carve-out for existing satellites is not a deliberate political accommodation but a technical and legal necessity. It is materially impossible to impose technical requirements retroactively on objects already in

[32] European Space Agency, *Around 100 000 satellites are expected to be in orbit by 2030*, April 2025; National Space Operations Centre, UK Space Agency and Ministry of Defence, *The Future Space Environment*, 16 May 2024; J.C. Barentine et al., *Satellite Megaconstellations*, online compendium, 2025.

[33] J. McDowell, *Jonathan's Space Report*, statistics as of March–May 2026; *Scientific American*, "SpaceX Reaches Milestone of 10,000 Starlink Satellites in Orbit", 17 March 2026.

[34] J. Tretow, *The EU Space Act: Why the National Security Clause Should Be Removed*, in *Stanford Space Law & Policy Perspectives*, January 2026, 1.

orbit. A satellite cannot be recalled for upgrading as one would a defective automobile; in-orbit modifications are exorbitantly costly, technically demanding and frequently unfeasible. The alternative — mandatory de-orbiting of non-compliant satellites — would be economically catastrophic for operators and legally vulnerable under the international-law principle protecting legitimately undertaken investments.

The operative consequence is the coexistence of two regimes for at least a decade, and most likely longer: *legacy* satellites launched before 2030, operating under the national rules — or the regulatory vacuum — of their registration States, and new satellites subject to the Space Act provided they seek to provide services in the Union. The resulting duality not only entails considerable regulatory complexity but also engenders a risk of competitive distortion, with incumbent operators benefiting de facto from an exemption that may be exploited throughout the operational life of their existing constellations.

More fundamentally still, the multiplication of orbital assets and the attendant regulatory fragmentation raise the question of the long-term sustainability of the orbital environment. The Space Act imposes obligations regarding end-of-life de-orbiting and the limitation of debris generation, but these duties only bind satellites launched after 2030 by operators seeking access to the European market. Tens of thousands of satellites operated by other actors, at other times or for other markets, fall outside its reach.

Yet orbital space — and particularly the low Earth orbit (LEO) regime in which mega-constellations concentrate — is a finite resource. Not in the sense that physical space is lacking — space is by definition vast — but in the sense that the number of stable and useful orbital trajectories is finite, and each additional satellite incrementally raises the probability of collision. The *Kessler syndrome*, formulated by NASA scientist Donald J. Kessler together with Burton G. Cour-Palais in 1978,[35] describes a regime in which the density of objects in LEO becomes such that collisions generate debris at a rate exceeding atmospheric removal, triggering a self-sustaining cascade in which fragments from one collision precipitate further collisions, eventually rendering certain orbital bands unusable for decades or centuries.

The hypothesis is no longer purely theoretical. The 2009 collision between the active U.S. communications satellite *Iridium 33* and the derelict Russian military satellite *Kosmos 2251* — the first accidental

[35] D.J. Kessler & B.G. Cour-Palais, *Collision Frequency of Artificial Satellites: The Creation of a Debris Belt*, in *Journal of Geophysical Research*, 83(A6), 1978, 2637; D.J. Kessler, "The Kessler Syndrome", *Space Safety Magazine*, 2012.

hypervelocity collision between two intact satellites in orbit — generated over two thousand catalogued fragments, many of which will persist in orbit through the end of the century.[36] The 2007 Chinese antisatellite test against *Fengyun-1C* added more than three thousand five hundred trackable fragments; the 2021 Russian antisatellite test against *Kosmos 1408* generated approximately fifteen hundred additional ones and forced the International Space Station to perform an avoidance manoeuvre.[37] With the exponential multiplication of orbital assets, the probability of further such incidents increases mechanically. ESA has accordingly launched its *Zero Debris Charter*, setting 2030 as the target for a debris-neutral orbital environment — a deadline whose feasibility is itself contested.[38]

The Space Act seeks to respond to this challenge through stringent debris-management requirements. Yet its effectiveness is constrained by both its geographical and temporal scope: it can regulate only Union operators and third-country actors seeking market access, and only with respect to satellites launched after 2030. Orbital sustainability, however, requires by its very nature a global and immediate approach. Debris is indifferent to jurisdictional borders: a Chinese satellite fragmenting in orbit threatens European satellites as much as American ones, and the conscientious de-orbiting of a European satellite is futile if hundreds of American or Chinese satellites remain in orbit indefinitely after the end of their operational life.

The sovereignty paradox reappears here from a different angle. The Union may articulate the most stringent rules on space sustainability in the world. It may impose them on its own operators and on third parties seeking access to its market. It cannot, however, constrain the conduct of actors that operate beyond its regulatory reach — and these actors are precisely the ones deploying the largest mega-constellations: SpaceX with Starlink, Amazon with Kuiper, and the Chinese initiatives.

[36] NASA Orbital Debris Program Office, *Orbital Debris Quarterly News*, Iridium 33 / Kosmos 2251 collision (February 2009).

[37] NASA, *Orbital Debris Quarterly News* on the Kosmos 1408 ASAT event, November 2021; for an overview see *Aerospace America*, "Understanding the misunderstood Kessler Syndrome", May 2025.

[38] ESA Space Debris Office, *Zero Debris Charter*, 2024; *SpaceNews*, "The countdown to clean orbits has begun with ESA's Zero Debris Charter", January 2026.

6. *The Starlink case: a paradigmatic stress-test for the European legal arsenal*

No analysis of the Space Act can avoid a close examination of the Starlink case, which constitutes in many respects the most revealing test of the Union's capacity to enforce its rules in the space domain. Starlink is not simply the largest constellation ever deployed: by its scale, deployment rhythm, business model and degree of geopolitical instrumentalisation, it places traditional categories of space law under stress and exposes the relative impotence of European regulation.

As of May 2026, Starlink comprises in excess of ten thousand operational satellites in low Earth orbit, representing approximately two-thirds of all active satellites combined.[39] The company has received FCC authorisation for an initial deployment of twelve thousand units, with subsequent filings extending the figure to thirty thousand or, in some projections, forty-two thousand satellites. The constellation already provides high-speed broadband services in more than one hundred and fifty countries, including most Member States. Millions of users — private citizens, enterprises, vessels, aircraft, and even armed forces — depend daily on Starlink for their connectivity. In Ukraine, Starlink has become an indispensable component of civilian and military communications infrastructure, with several analysts attributing to it a significant role in Ukraine's defensive resilience against the Russian invasion.[40]

What, then, can the Space Act actually do? Formally, the answer is unambiguous: Starlink — like any operator providing space services within the Union — would be required to register in URSO, to comply with the cybersecurity, sustainability and safety requirements of the Regulation, and to accept supervision by European authorities. In the event of refusal, access to the European market would be denied. The formal answer, however, conceals a much more complex reality.

First, the Space Act applies only to satellites launched after 1 January 2030. The Starlink satellites already in orbit in 2025, together with all those launched in the four intervening years, fall entirely outside its scope. While the operational lifespan of an individual Starlink satellite is

[39] SpaceX Inc., Starlink mission reports, 2025–2026; *Scientific American*, "SpaceX Reaches Milestone of 10,000 Starlink Satellites in Orbit", 17 March 2026.

[40] U.S. Agency for International Development – Office of Inspector General, *Ukraine Response: USAID Did Not Fully Mitigate the Risk of Misuse of the Starlink Satellite Terminals It Delivered to Ukraine*, Final Report E-121-25-003-M, 11 August 2025 (concerning 5,175 Starlink terminals, 1,508 funded by USAID and 3,667 donated by SpaceX, delivered to Ukraine's State Service of Special Communications and Information Protection).

approximately five years and replenishment is therefore required, by 2030 Starlink will already have consolidated its dominant position, its revenue base and its user community. The Space Act, in short, risks arriving *after the battle*.

Second — and this is the decisive point — the question must be addressed of what would occur should SpaceX simply elect not to comply. Could the Union credibly exclude Starlink from its market? The question is far from rhetorical. The economic, social and strategic implications are considerable.

In economic terms, Starlink currently provides connectivity in rural and isolated areas where terrestrial operators have neither the means nor the incentive to invest. Farmers, small and medium-sized enterprises, schools, emergency services, ships and aircraft depend on this connectivity. A regulatory ban on Starlink would deprive these users of an essential service, with predictable political and social fallout. The Union, which has elevated universal connectivity to the rank of strategic objective, cannot easily afford to generate digital deserts by regulatory fiat.

Strategically, European dependence on Starlink extends well beyond civilian uses. Several European armed forces employ Starlink for tactical communications, exploiting its resilience, low latency and jamming resistance. During the Russian invasion of Ukraine, Starlink demonstrated itself to be a critical communications infrastructure, performing where terrestrial networks had been destroyed or neutralised; NATO itself regards Starlink as a potential component of its resilient communications architecture. In this context, banning Starlink on grounds of regulatory non-compliance would amount to a voluntary degradation of European defence capabilities.

This strategic dependence is not accidental: it is the consequence of a substantial technological and industrial lag. The Union does not currently dispose of a constellation equivalent to Starlink. The *Infrastructure for Resilience, Interconnectivity and Security by Satellite* — IRIS² — represents the European sovereign response: a multi-orbit constellation of approximately 290 satellites in LEO and MEO, the subject of a twelve-year, €10.6 billion concession contract signed on 16 December 2024 between the European Commission, ESA and the *SpaceRISE* consortium (SES, Eutelsat, Hispasat). Full IRIS² services are scheduled to enter operation by 2030, although the timeline has already been publicly questioned by Member-State military authorities themselves.[41] By that horizon Starlink

[41] ESA & European Commission, *IRIS² programme*, December 2024 launch

may have deployed twenty thousand satellites or more and consolidated a *quasi-monopolistic* position in the segment of low-orbit satellite internet.

The technological asymmetry is compounded by a jurisdictional one. SpaceX is a U.S. company, subject to U.S. law; its satellites are registered in the United States; its ground-control infrastructures are predominantly located on American soil. The Union may attempt to sanction its activities within the European territory, but it possesses no direct hold on its space operations.[42] If SpaceX were to continue providing services in Europe despite a formal ban, how could the Union prevent it? Could it seize the satellites in orbit? Manifestly not. Could it jam Starlink communications? Technically perhaps; legally and politically, certainly not.

7. *High-resolution Earth observation and the spatial dimension of privacy*

Beyond the questions of regulatory compliance and strategic dependence raised by the Starlink case, the multiplication of low-orbit satellites raises a further fundamental issue that the Space Act, by its silence, leaves entirely unresolved: the protection of privacy against Earth observation from orbit.

Contemporary commercial Earth observation reaches resolutions that, twenty years ago, were the exclusive preserve of classified military assets. Maxar Intelligence's *WorldView Legion* constellation, fully deployed in February 2025, commercialises 30-centimetre-class imagery — sufficient, in operational terms, "to identify objects approximately the size of a car hood from orbit"[43] — with up to fifteen daily revisits over given locations.[44] Planet Labs operates a constellation of more than two hundred satellites capable of imaging the entire emerged surface of the Earth on a daily basis at multi-metre resolution; Chinese providers offer comparable or superior services; new entrants are deploying synthetic-aperture-radar

documentation; ESA CSC, *ESA confirms kick-start of IRIS² with European Commission and SpaceRISE*, December 2024; *Space Intel Report*, "French Military: Iris² won't be deployed by 2030", April 2026.

[42] G. De Minico & M.F. De Tullio, *Feedback on the Digital Networks Act*, European Commission, Ref. Ares(2025) 4875093, 19 June 2025.

[43] *SpaceNews*, "SpaceX launches third pair of Maxar's WorldView Legion imaging satellites", 5 February 2025.

[44] Maxar Intelligence press release of 4 February 2025; ESA Earth Online, *WorldView Legion*, 2025.

and infrared sensors capable of imaging through cloud cover and darkness.

These capabilities, combined with artificial intelligence for automated image analysis, produce surveillance possibilities without historical precedent. It is now technically feasible to track in near real-time the movements of specific vehicles, to detect automatically changes in land use, to identify patterns of economic or agricultural activity, to count individuals present at public gatherings, to monitor industrial or military sites, to detect activities such as illegal deforestation or unauthorised construction.

From a public-interest perspective the benefits are undeniable. Earth observation enables the monitoring of climate change, the detection of natural disasters, the coordination of rescue operations, the verification of compliance with international agreements such as the Paris Climate Agreement, and the fight against illegal deforestation and unregulated fishing. The positive applications are innumerable.

From the standpoint of fundamental rights, however, the same capabilities raise vertiginous questions. At what point does a satellite image cease to be a mere territorial representation and become a processing of personal data within the meaning of the GDPR? When the resolution allows identification of a vehicle whose owner may be deduced by cross-referencing with other databases? When it permits the distinction of a natural person, even where their facial features remain unrecognisable? When the analysis of time-series imagery enables the reconstruction of an individual's habitual movements?

Article 4(1) GDPR defines personal data as "any information relating to an identified or identifiable natural person", regarding as identifiable any person who "can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity". The definition is deliberately broad and functional: it does not require that the information, considered in isolation, permit identification, but only that it contributes to identification within an ensemble of means reasonably likely to be used. The Court of Justice of the European Union has consistently endorsed the broad scope of this definition, most recently in *EDPS v SRB*.[45]

Applied to satellite imagery, the definition suggests that an image enabling the identification of a vehicle associated with a person, the

[45] CJEU, Case C-413/23 P, *European Data Protection Supervisor v Single Resolution Board*, judgment of 4 September 2025; commentary in *Privacy World*, "Towards a Contextual Concept of Personal Data Under the GDPR", March 2026.

reconstruction of habitual movement patterns, or the determination of presence at a specific location at a given time would qualify as personal data. The corresponding collection, processing, retention and commercialisation would, in principle, be required to comply with the GDPR's substantive principles: lawfulness, fairness and transparency, purpose limitation, data minimisation, accuracy, storage limitation, integrity and confidentiality.

The application of these principles to space-based image collection is, however, immediately problematic. The GDPR conditions the lawfulness of processing on consent or another legitimate basis under Article 6. How is one to obtain the consent of the millions of individuals whose vehicles, homes and workplaces are imaged daily by orbital sensors, without their knowledge and without any practicable means of objection? The Regulation provides for rights of access, rectification and erasure. How is the right to erasure to be exercised against an archived satellite image showing one's vehicle in the driveway at a particular moment?

These are not purely theoretical concerns. Civil-rights organisations have repeatedly raised concerns regarding the use of high-resolution satellite imagery to monitor demonstrations, identify participants in political gatherings or track the movements of public figures. Investigative journalism has documented that certain intelligence services acquire commercial imagery as a supplement to their own surveillance capabilities, in this way circumventing the legal constraints applicable to their proper means.

The Space Act, which might have seized the opportunity to establish a coherent framework, opts instead for complete silence. It imposes no privacy-specific obligations on operators of observation satellites; it defines no resolution thresholds beyond which specific safeguards would apply; it draws no distinction between observation for commercial, scientific or surveillance purposes; it establishes no prior-authorisation or oversight mechanism for observation activities entailing heightened risks to privacy. Even pre-existing international guidance — most notably the 1986 UN *Principles Relating to Remote Sensing of the Earth from Outer Space*[46] — predates the digital era of high-resolution sensing and receives no normative reinforcement from the proposed Regulation.

The sovereignty paradox surfaces here once again. The Union, which possesses the most demanding data-protection regime in the world, can apply it to any processing of European residents' data — yet it cannot

[46] UN General Assembly, *Principles Relating to Remote Sensing of the Earth from Outer Space*, Resolution 41/65 (1986).

prevent the collection from orbit of imagery of its territory. It may sanction the commercialisation of such imagery within Europe, but it cannot reach its collection, processing, archiving or use elsewhere in the world. The sky is, in a literal sense, the limit of the Union's sovereignty in data protection matters.

The limit is far from trivial. It means that European citizens, whose data are protected by the GDPR when collected on European soil, lose that protection the moment data is collected from orbit. Their movements, activities and properties may be observed, recorded, analysed and commercialised without their consent, without notification and without effective recourse, provided the observation is carried out from a satellite registered in a third State and operated by an entity not seeking to commercialise services within the European market.

The asymmetry is the more problematic in that it does not concern privacy in the narrow sense alone. It also implicates economic and strategic security. Observation satellites can monitor industrial sites, reveal technological innovations, track the movement of goods, detect research and development activities. In an environment of intense economic competition and systematic industrial espionage, the inability to control what is observed from orbit constitutes a major strategic vulnerability.

8. Conclusion: between regulatory ambition and geopolitical realism

The analysis of the EU Space Act reveals a paradoxical legal instrument. At the formal level, it represents a substantial step forward: it establishes the first harmonised regulatory framework for space activities at the supranational level; it articulates robust cybersecurity and sustainability requirements; and it extends to the space domain the regulatory grammar that has come to characterise Union law. It testifies to a legitimate ambition — to prevent the space sector from becoming a regulatory wild west and to impose high standards of safety and environmental responsibility.

At the substantive level, however, the Space Act encounters structural limits that significantly attenuate its effective reach. These limits do not derive from drafting deficiencies or unfortunate political choices that might be remedied through amendment; they flow from the very nature of outer space as a domain that eludes classical territorial sovereignty, and from the relative position of the Union within the international space order.

The first limit — designated in this analysis as the *sovereignty paradox* — consists in the Union's impossibility of controlling what occurs above its territory. The Space Act may regulate access to the European market but cannot prevent an operator that chooses not to access that market from

freely collecting data on European territory, monitoring its infrastructures and observing its citizens from orbit. The limit is not peculiar to the Space Act: it equally affects the GDPR and, more broadly, the entirety of the European regulatory acquis.

The second limit lies in the asymmetry of capabilities between the Union and the principal space powers — primarily the United States and China. Europe does not currently possess the technical instruments to compel a recalcitrant operator to comply with its rules: it cannot seize satellites in orbit; it cannot jam space-based communications without provoking diplomatic and strategic escalation of potentially catastrophic dimensions. It depends for its critical space services on infrastructures controlled by extra-European actors — a dependence that is the consequence of decades of underinvestment in autonomous European space capabilities and which severely reduces the credibility of the sanctions envisaged by the Space Act.

The third limit concerns the substantive lacunae of the text, notably the wholesale absence of provisions on privacy and the insufficient governance of artificial intelligence and autonomous systems. These omissions are not mere oversights: they reflect a choice — conscious or otherwise — to separate the operational dimensions of space safety from the questions of fundamental rights and data protection. The choice is legally defensible by reference to the horizontal scope of the GDPR, but practically inadequate, since the GDPR itself confronts the sovereignty limit and cannot prevent the collection of data from orbit by extra-European actors.

Confronted with these limits, three options present themselves to the Union. They are not mutually exclusive but each entails significant political and budgetary commitments.

The first is the *improvement of the existing regulatory framework*. The Space Act, in its current form, is a proposal subject to the ordinary legislative procedure: the European Parliament and the Council will have the opportunity to amend it substantially before adoption. It would be desirable for those amendments to address — at least in part — the identified lacunae, by introducing clear rules on data protection in the context of space-based observation, by establishing a governance framework for artificial intelligence in space systems, and by reinforcing the coordination mechanisms with third-State legislation in order to prevent jurisdictional conflicts. This avenue, however, does not resolve the deeper problem that relevant actors may simply elect to ignore the European rules.

The second is the *development of autonomous European space capabilities*. If the Union genuinely intends to control what occurs in and from space above its territory, it must possess the technical means to do so. This implies the rapid deployment of sovereign European constellations for communication, observation and navigation — IRIS², Copernicus and Galileo being the first steps of a much wider effort; the strengthening of *space surveillance and tracking* (SST) capabilities to detect and track all objects in orbit; and, politically more sensitive, the eventual acquisition of space-deterrence capabilities — the means to neutralise hostile space systems without necessarily resorting to kinetic destruction. The corresponding investments amount to tens, if not hundreds, of billions of euro, and require a political will currently lacking.

The third option is *space diplomacy*: the construction of a binding international framework capable of imposing common rules on all space actors. The international space-law regime, anchored in the 1967 *Outer Space Treaty* and complemented by the UNOOSA *Guidelines for the Long-Term Sustainability of Outer Space Activities* (LTS Guidelines, 2019),[47] is manifestly insufficient with regard to data protection, artificial intelligence, mega-constellations, debris management, resource exploitation and the progressive militarisation of orbit. A new international convention negotiated under the auspices of the United Nations and signed by all space-faring powers might in principle establish common rules and create the conditions for genuine global space governance. The prospects for such a negotiation are, it must be acknowledged, extremely modest: geopolitical rivalries between the United States, China and Russia, the divergent economic interests of public and private actors, and the absence of any credible enforcement mechanism, render the emergence of substantive consensus highly improbable.

The EU Space Act ultimately exemplifies, in a single instrument, both the *ambition* and the *limits* of European regulatory power. With the GDPR, the Union demonstrated its capacity to impose its standards beyond its borders through the leverage of market access. That mechanism, however, presupposes the locatability of the regulated object — and that condition fails in space. Data can be localised, servers can be seized, undertakings can be compelled; satellites, by contrast, escape by their very nature any seizure, evolving in a domain that belongs to no one and that all may freely use.

[47] UNOOSA, *Guidelines for the Long-Term Sustainability of Outer Space Activities*, 2019, consolidated in *United Nations International Space Law Booklet*, Vienna, 2025.

This realisation does not condemn the Space Act to irrelevance. It will establish important rules for European operators and for those seeking access to the European market. It will create a harmonised and predictable framework that will benefit the development of the European space sector.

Through the exemplarity of its standards, it may, in time, influence the evolution of international space law. It cannot, by itself, protect European citizens from space-based surveillance, guarantee the sustainability of the orbital environment or secure Europe's strategic autonomy in the space domain. Those objectives demand instruments that lie far beyond the law: massive investment in technical capabilities, a sustained political will, and a long-term strategic vision that a fragmented and hesitant Europe has yet to embrace.

Beatrice Ermini

When Regulation Becomes Risk: Mapping Italy's Space Law (No. 89/2025) into Enterprise ESG Architectures

Summary: 1. Introduction – 2. Legal Analysis of Italian Law No. 89/2025 – 3. Methodological Construction – 4. Operational Application within an Enterprise Risk Management Framework – 5. Conclusions.

1. *Introduction*

The progressive consolidation of the space economy as a strategic industrial sector has intensified the need for regulatory frameworks capable of governing the increasingly complex interaction between public authorities, private operators, and technological innovation. Space activities, once predominantly state-driven and exploration-oriented, have progressively evolved into a structured industrial ecosystem in which private entities play a central role in both upstream and downstream segments of the value chain. This transformation has generated a parallel evolution of legal frameworks, aimed at ensuring that the expansion of commercial space activities remains consistent with principles of international responsibility, safety, and sustainability.

In parallel, the increasing centrality of Environmental, Social and Governance (ESG) criteria in corporate governance has profoundly reshaped the conceptualization of enterprise performance and risk. ESG factors are no longer interpreted as ancillary reputational considerations, but rather as structural determinants of financial resilience, regulatory compliance, and long-term value creation. In particular, the integration of ESG principles into Enterprise Risk Management (ERM) systems reflects a broader shift from traditional, financially driven risk models toward more holistic frameworks that incorporate environmental impact, social responsibility, and governance quality as core dimensions of risk exposure.

These two trajectories — the juridification of space activities and the operationalization of sustainability within corporate governance — are increasingly converging. In highly regulated and technologically intensive sectors such as aerospace, legal obligations do not merely function as external compliance constraints but actively contribute to shaping corporate risk architectures. Regulatory requirements influence investment decisions, operational design, supply chain structures, and ultimately the strategic positioning of firms operating in this sector. As a result, law becomes not only a normative framework but also a structural component of enterprise risk formation[1].

Despite this evident convergence, a significant gap persists between legal analysis and risk management practice. Space law scholarship has extensively examined foundational issues such as authorization regimes, state responsibility, liability for damage caused by space objects, and the evolving governance of private space activities.[2] In parallel, management literature has developed sophisticated models for ESG integration within corporate risk management systems, emphasizing the role of sustainability in enhancing organizational resilience.[3] However, relatively limited attention has been devoted to the operational translation of sector-specific legal obligations into structured and measurable ESG risk indicators capable of being directly integrated into Enterprise Risk Management systems.

This gap is particularly relevant in the aerospace industry, where regulatory density, technological complexity, and high-risk operational environments intersect. In this context, legal obligations are not abstract constraints but concrete determinants of operational feasibility and strategic risk exposure. Yet, despite their relevance, these obligations are rarely translated into standardized metrics that can be systematically incorporated into corporate risk models. This limitation reduces the capacity of firms to anticipate regulatory impacts and to integrate legal compliance into strategic decision-making processes.

The present article is positioned within this research gap and is developed in continuity with an ongoing doctoral research project focused on the integration of ESG criteria into Enterprise Risk Management

[1] P. Velte, *Sustainable board governance and environmental performance: European evidence,* in *Business Strategy and the Environment*, 33(4), 2024, pp. 3397–3421.

[2] F. Tronchetti, *Fundamentals of Space Law and Policy,* Springer, New York, 2013; S. Marchisio, *Lezioni di diritto aerospaziale*, Sapienza Università Editrice, Roma, 2021.

[3] R.G. Eccles, I. Ioannou, G. Serafeim, *The Impact of Corporate Sustainability on Organizational Processes and Performance*, in *Management Science*, 60(11), 2014, pp. 2835–2857.

systems in the aerospace sector, with particular reference to industrial operators such as Avio S.p.A. The underlying assumption of the research is that sustainability cannot be treated as an external reporting requirement but must be embedded within the core decision-making architecture of firms through structured, quantitative, and operational tools.

Against this background, the article proposes an integrated analytical framework structured around three interconnected pillars. The first consists of a legal analysis of Italian Law No. 89/2025, interpreted not merely as a regulatory instrument but as a generator of structured risk conditions. The second develops a methodological model for translating legal norms into ESG risk metrics through a structured decomposition and mapping process. The third applies this framework within an Enterprise Risk Management perspective, demonstrating how ESG indicators derived from legal obligations can be operationalized to support strategic decision-making in aerospace organizations.

The central contribution of the article lies in demonstrating that regulatory frameworks in the space sector can be systematically reinterpreted as structured sources of ESG-relevant risk indicators. This transformation enables a shift from a compliance-oriented approach to regulation toward a risk-based governance model in which legal obligations are directly integrated into enterprise risk architecture. In doing so, the article contributes to bridging the conceptual and operational divide between space law, sustainability governance, and corporate risk management in high-technology industries.

2. *Legal Analysis of Italian Law No. 89/2025*

Italian Law No. 89/2025 represents a significant milestone in the evolution of national space governance, marking the transition from a fragmented regulatory landscape to a structured legal framework governing the conduct of space activities by private operators. The law must be interpreted within the broader architecture of international space law, particularly in light of the principles established by the Outer Space Treaty, which attributes international responsibility to States for national activities in outer space, including those carried out by non-governmental entities.

In this context, national legislation plays a crucial role in operationalizing

international obligations by defining procedures, responsibilities, and control mechanisms applicable to private actors. As observed in the literature, the proliferation of domestic space laws reflects the progressive commercialization of outer space and the consequent need to reconcile public oversight with private initiative Italian Law No. 89/2025 is fully embedded within this trend, positioning itself as both an instrument of compliance with international law and a tool for fostering the development of a competitive national space industry.

A more granular reading of the law reveals an articulated normative architecture structured around a set of key provisions that directly shape the operational conditions of space activities. Article 3 defines the territorial and personal scope of the regime, extending its application to space activities conducted on Italian territory as well as to Italian operators acting abroad. Articles 5 and 6 establish the objective and subjective requirements for authorization, including technical reliability, financial capacity, operational safety across all mission phases, infrastructure resilience, cybersecurity standards, environmental sustainability, and the reputational integrity of the applicant.

At the core of the legislative framework lies the authorization regime, which functions as the primary mechanism through which the State exercises control over space activities conducted by private entities. This regime is institutionally centred on the Agenzia Spaziale Italiana (ASI), which performs technical assessments and ongoing supervisory functions, while the formal authorization is issued by the Presidency of the Council of Ministers. This institutional configuration is not merely procedural: it concentrates regulatory discretion within a politically coordinated decision-making structure, thereby introducing an additional layer of governance risk for private operators.

Operators are required to obtain prior authorization by demonstrating compliance with the requirements set out in Articles 5 and 6. Beyond its formal legal function, the authorization regime generates significant implications for corporate governance. Authorization procedures introduce elements of uncertainty, procedural complexity, and timing constraints that directly affect investment decisions and project feasibility. In this sense, authorization can be interpreted not only as a legal prerequisite but also as a source of governance risk, influencing the strategic planning of aerospace firms.

Closely connected to the authorization system is the framework of ongoing supervision established by the law. Italian Law No. 89/2025

adopts a model of continuous oversight, requiring operators to maintain compliance throughout the lifecycle of their activities under the monitoring of competent authorities, primarily the ASI. This shift from ex ante control to continuous supervision reflects the dynamic nature of space operations and their associated risks.[4]

From a governance perspective, compliance becomes an ongoing organizational function requiring internal control systems, reporting procedures, and structured risk monitoring mechanisms. Firms must institutionalize compliance within their organizational structures, integrating regulatory interaction into daily operations. This reinforces the link between legal obligations and governance-related ESG factors, particularly in terms of transparency, accountability, and internal control effectiveness.

Another fundamental pillar of the law is the liability regime applicable to space operators, defined in Articles 18 to 20. The Italian framework introduces a dual-track liability system, distinguishing between strict liability for damage occurring in the atmosphere or on Earth, and fault-based liability for damage occurring in outer space, while the State retains international responsibility and a right of recourse against the operator where liability is engaged.

This dual structure generates a layered exposure profile that must be actively managed by operators. The obligation to secure insurance coverage and implement risk mitigation measures translates into financial risk variables, including insurance costs, provisioning strategies, and contractual allocation of liability. In ESG terms, liability rules operate at the intersection of governance and social dimensions, linking legal accountability to financial resilience, public safety, and stakeholder protection.

The law also incorporates explicit provisions related to safety, environmental protection, and sustainability. Environmental protection is expressly included among the objective requirements for authorization under Article 5, alongside safety and cybersecurity. The framework further addresses space debris mitigation and the sustainable use of outer space, aligning national regulation with international sustainability standards.[5]

These provisions contribute to both environmental and social ESG dimensions, covering emissions, orbital debris management, occupational

[4] F. Tronchetti, *Fundamentals of Space Law and Policy*, Springer, New York, 2013.

[5] S. Di Pippo, *The Contribution of Space for a More Sustainable Earth*, in *Yearbook on Space Policy 2017*, Springer, Wien, 2019, pp. 35–46.

safety, and risk prevention. Sustainability therefore emerges as an embedded regulatory requirement that directly affects operational and strategic decision-making.

A further relevant dimension is introduced in Title V, which establishes industrial policy instruments for the space economy. Article 22 provides for a five-year National Plan for the Space Economy, while Article 23 establishes a multi-annual Fund supporting private investment, with priority for SMEs and innovative start-ups.

These instruments operate simultaneously as opportunity and risk drivers: they enable access to public funding and strategic alignment, but also generate dependency on political priorities, budget cycles, and administrative allocation mechanisms. They therefore represent a relevant component of governance-related ESG risk exposure.

From an enterprise risk perspective, the architecture of Italian Law No. 89/2025 can be further decomposed into a set of distinct risk-generating provisions. Article 3 defines the territorial and personal scope of the regime, extending its reach to space activities conducted by operators of any nationality on Italian territory and by Italian operators abroad. Articles 5 and 6 establish the objective and subjective requirements for authorisation, encompassing safety in all operational phases, infrastructure resilience, cybersecurity, environmental sustainability, and reputational integrity of the applicant. Articles 18 to 20 introduce a dual-track liability regime, distinguishing between strict liability for atmospheric damage and fault-based liability for damage occurring in outer space, with the State retaining a right of recourse against the operator where international responsibility is engaged. Article 22 mandates the adoption of a five-year National Plan for the Space Economy, while Article 23 establishes a multi-annual Fund supporting private investment, with specific provisions favouring SMEs and innovative start-ups.

The Italian Space Agency (ASI) is positioned at the centre of the supervisory architecture, exercising both technical evaluation and ongoing oversight functions, while authorisation is formally issued by the Presidency of the Council of Ministers. This institutional configuration is not a mere procedural detail: it concentrates regulatory discretion within a politically led decision-making centre, generating an additional layer of governance risk that aerospace operators must internalise within their strategic planning. From an enterprise risk perspective, each of these provisions can be reinterpreted as a structured driver of exposure, susceptible of operational translation into ESG-relevant indicators.

The temporal coincidence between the entry into force of Law No. 89/2025 on 25 June 2025 and the publication on the same date of the European Commission's proposal for an EU Space Act[6] is not merely symbolic. It generates a transitional regulatory environment in which national operators are required to comply with the Italian framework while simultaneously anticipating the prospective harmonisation imposed by the European regime. This dual-layer condition produces a specific category of regulatory risk — what may be termed anticipatory compliance risk — whereby firms must align internal control systems and capital allocation strategies with a normative scenario that is not yet fully defined. For aerospace operators with significant European exposure, this risk is structurally embedded in investment decisions and must therefore be incorporated within the ESG governance dimension of any Enterprise Risk Management system.

The Italian law must also be considered within the evolving European regulatory context, particularly in relation to the EU Space Act proposal. This reinforces the transition toward a multi-level regulatory framework in which national and European norms interact in shaping the governance of space activities.

Overall, Italian Law No. 89/2025 can be understood as a comprehensive regulatory framework that not only governs legality but also structures corporate risk exposure across ESG dimensions. Authorization, supervision, liability, sustainability, and industrial policy collectively define a regulatory ecosystem in which legal obligations are directly translated into enterprise risk variables.

3. *Methodological Construction*

Before presenting the analytical framework, it is necessary to clarify the methodological perimeter of the proposal. The model developed in this section is conceptual and ex-ante in nature: it provides a structured logic for translating legal obligations into ESG-relevant risk indicators but does not yet incorporate empirical calibration on real corporate data. Empirical validation is reserved for subsequent stages of the underlying doctoral

[6] European Commission, *Proposal for a Regulation of the European Parliament and of the Council on the Safety, Resilience and Sustainability of Space Activities in the Union* (EU Space Act), COM(2025) 335 final, Brussels, 25 June 2025.

research, which will apply the framework to financial and operational disclosures of a representative sample of European aerospace operators. This methodological caveat is essential to position the contribution as a theoretical scaffolding rather than as a finalised measurement tool.

The methodological framework is based on the assumption that legal norms, particularly in highly regulated sectors such as aerospace, do not merely impose external constraints on corporate behaviour but actively generate structured conditions of risk exposure. Each legal provision embedded in Italian Law No. 89/2025 defines obligations, procedural requirements, and constraints that shape organisational decision-making. In this sense, law is interpreted as an endogenous component of enterprise risk systems[7].

The first step of the model consists in decomposing Law No. 89/2025 into its core regulatory components, including: the scope of application (Art. 3), the authorization and compliance regime (Arts. 5–6), the liability framework (Arts. 18–20), and the strategic instruments introduced by the legislator, namely the National Space Economy Plan (Art. 22) and the Multi-annual Space Fund (Art. 23)[8]. These provisions are then translated into operational variables that reflect their impact on corporate structures and decision-making processes. Authorization rules are interpreted as sources of regulatory dependency affecting investment timing; supervisory obligations translate into compliance intensity; liability provisions generate financial exposure and insurance requirements; environmental clauses affect operational sustainability constraints; and public policy instruments introduce strategic alignment incentives.

These organizational variables are then mapped onto the three ESG dimensions (Environmental, Social, Governance), according to the nature of the risk generated. Governance risks relate to compliance systems, regulatory interaction, and internal control structures; environmental risks concern sustainability requirements and debris mitigation; social risks involve occupational safety, workforce protection, and external stakeholder impacts. The resulting translation is formalised in the synoptic matrix below.

[7] T. Mahler, *Defining Legal Risk in Commercial Contracting for Strategic Advantage: Potentials and Prospects*, Turku University of Applied Sciences, Turku, 2007, pp. 10–31; R. McCormick, C. Stears, *Legal and Conduct Risk in the Financial Markets*, 3rd ed., Oxford University Press, Oxford, 2018.

[8] Legge 13 giugno 2025, n. 89, *Disposizioni in materia di economia dello spazio*, in Gazzetta Ufficiale della Repubblica Italiana, Serie Generale, n. 144 del 24 giugno 2025.

Table 1 — Mapping of Law No. 89/2025 obligations onto ESG-ERM risk dimensions

Legal provision (Law 89/2025)	**Operational variable**	**ESG dimension**	**Candidate risk indicator**
Art. 5 — objective requirements (safety, cybersecurity, environment)	Compliance design and certification	G / E / S	Number of certified processes; cyber-incident rate
Art. 6 — subjective requirements (integrity, conduct)	Reputational and integrity controls	G	Existence of Model 231; integrity audits per year
Arts. 18–20 — dual liability regime	Insurance coverage and provisioning	G / S	Insurance-to-revenue ratio; provisions for litigation
Art. 22 — National Space Economy Plan	Strategic alignment with public investment	G	Share of revenues from Plan-aligned projects
Art. 23 — Multi-annual Space Fund	Access to public co-financing	G / E	Funding obtained for sustainability-linked R&D
Implicit — debris mitigation	End-of-life management of assets	E	Compliance with IADC guidelines; deorbiting plans

The logic underlying the translation presented in Table 1 is not based on a direct one-to-one correspondence between legal provisions and ESG indicators, but rather on a functional decomposition process. Legal norms are interpreted in terms of their operational effects on firms, and these effects are then associated with measurable proxies. Consequently, the indicators included in the table should be understood as analytical representations of risk conditions generated by law, rather than as direct measurements of legal compliance. For instance, certification-related indicators under Article 5 capture the organisational response to safety, cybersecurity, and environmental obligations, rather than the legal rule itself.

A second methodological consideration concerns the multidimensional nature of several provisions. As illustrated in the matrix, certain legal norms generate effects across multiple ESG dimensions simultaneously. The liability regime under Articles 18–20, for example, produces both governance-related impacts (through insurance structures and financial provisioning) and social impacts (through third-party risk exposure). This overlap reflects the intrinsic complexity of regulatory risk in aerospace

contexts and should not be interpreted as a methodological inconsistency. On the contrary, it highlights the need for a framework capable of capturing cross-dimensional interactions between legal obligations and sustainability outcomes.

A third limitation relates to the use of proxy indicators. Many of the variables identified in the table are indirect measures of underlying risk exposure. Indicators such as the number of certifications or the insurance-to-revenue ratio provide observable and comparable signals but cannot fully capture the complexity of the underlying legal and operational risks. Moreover, data availability and disclosure practices may vary across firms, particularly with respect to qualitative governance and compliance information. These limitations do not undermine the analytical value of the framework but rather define its epistemic boundaries.

Finally, to ensure consistency across heterogeneous indicators, the framework incorporates a multi-criteria decision-making structure based on the Analytic Hierarchy Process (AHP)[9]. This approach allows for the hierarchical decomposition of ESG dimensions into risk categories and observable indicators, with weights assigned through structured expert judgment. While this article does not provide empirical calibration of weights, it defines the decision architecture necessary for their future estimation. In this way, the model establishes a bridge between qualitative legal analysis and quantitative risk aggregation, preparing the ground for subsequent empirical applications.

4. *Operational Application within an Enterprise Risk Management Framework*

The operational relevance of the proposed methodological framework emerges most clearly when it is situated within the context of Enterprise Risk Management systems in the aerospace industry. In such environments, characterized by high technological complexity, strict regulatory oversight, and significant exposure to environmental and safety-related risks, the integration of ESG indicators derived from legal obligations becomes not only useful but necessary for effective risk governance.

Enterprise Risk Management in aerospace organizations has progressively evolved from a predominantly financial and operational

[9] T.L. Saaty, *Decision Making with the Analytic Hierarchy Process*, in *International Journal of Services Sciences*, 1(1), 2008, pp. 83–98.

control function toward a more integrated system capable of capturing multidimensional sources of risk. In this evolution, ESG factors have assumed an increasingly central role, particularly in relation to regulatory compliance, reputational exposure, and long-term strategic resilience[10]. Within this context, the translation of legal obligations into structured ESG indicators provides a mechanism to bridge regulatory requirements and managerial decision-making processes.

In the case of aerospace operators such as Avio S.p.A., this integration is particularly relevant due to the strategic importance of launch systems, the complexity of industrial supply chains, and the high level of interaction with national and European regulatory authorities. Space launch activities are inherently exposed to environmental, safety, and governance risks, all of which are directly influenced by regulatory frameworks such as Italian Law No. 89/2025. Consequently, legal compliance cannot be treated as an isolated administrative function but must be embedded within the broader enterprise risk architecture.

4.1 *A mini-case application: Avio S.p.A.*

Applying the decomposition logic developed in Section 3, the principal obligations established by the Italian Space Law map onto Avio's operational reality as follows. The authorisation regime under Article 4, administered by the Italian Space Agency (ASI) under the political authority of the Presidency of the Council of Ministers, generates a category of governance risk that materialises in the form of timing uncertainty and procedural exposure for each new launch service contract. This risk is not merely abstract: any delay in obtaining or renewing operational authorisations directly affects revenue recognition under multi-year ESA contracts and the firm's ability to honour Copernicus-related commitments.[11] The objective and subjective requirements established by Articles 5 and 6 — encompassing operational safety, infrastructure resilience, cybersecurity and environmental sustainability — translate into

[10] P. Velte, *Sustainable Board Governance and Environmental Performance: European Evidence, in Business Strategy and the Environment*, 33(4), 2024, pp. 3397–3421; S. Pizzi, A. Venturelli, F. Caputo, *Restoring Trust in Sustainability Reporting: The Enabling Role of External Assurance*, in *Current Opinion in Environmental Sustainability*, 68, 2024, Article 101437.

[11] Avio S.p.A., *Annual Financial Report 2024 integrated with the Sustainability Statement*, Colleferro, 2025, pp. 18–34.

a measurable compliance design effort, observable through the company's certification portfolio (ISO 9001, ISO 14001, ISO 27001, gender equality), the integration of cybersecurity within enterprise governance, and the disclosure of environmental management systems within the ESRS E1 and E5 perimeters.[12]

The dual-track liability regime introduced by Articles 18 to 20 — strict liability for atmospheric damage, fault-based liability for damage occurring in outer space, with State recourse — is perhaps the legal provision with the most direct quantitative translation into financial risk indicators. For Avio, this regime materialises in the structure of insurance coverage attached to each launch campaign, in the level of provisions for litigation, and in the risk, premium implicitly priced into multi-launch commercial contracts. From a CSRD perspective, the liability profile feeds directly into the disclosures on financial materiality required under ESRS 2, and into the resilience analysis that auditors are required to assure under the new framework.[13] Finally, the provisions on space debris mitigation, although only implicitly addressed by the Italian text, intersect with the company's operational decisions regarding deorbiting, passivation and end-of-life management of upper stages — areas in which Avio has already disclosed its alignment with international guidelines such as the IADC and the COPUOS Long-term Sustainability Guidelines.[14]

Translating this mapping into a synthetic indicator structure produces the matrix presented in Table 2, which extends the general framework of Section 3 with the empirical referents observable in Avio's 2024 disclosures.[15]

[12] Ivi, pp. 104–136; *Commission Delegated Regulation* (EU) 2023/2772 of 31 July 2023 establishing the European Sustainability Reporting Standards (ESRS).

[13] *Directive (EU) 2022/2464* (Corporate Sustainability Reporting Directive – CSRD); Legislative Decree No. 125 of 6 September 2024.

[14] United Nations Committee on the Peaceful Uses of Outer Space (COPUOS), *Guidelines for the Long-term Sustainability of Outer Space Activities*, Vienna, 2018; Inter-Agency Space Debris Coordination Committee (IADC), *Space Debris Mitigation Guidelines*, Revision 2, 2020.

[15] Avio S.p.A., *Annual Financial Report 2024 integrated with the Sustainability Statement*, cit.; Avio S.p.A., FY 2024 Results, Press Release, 13 March 2025.

Table 2 — Legal-to-ESG mapping applied to Avio S.p.A. (FY2024)

Legal provision (Law 89/2025)	**Operational variable (Avio)**	**ESG dimension**	**Observable indicator (FY2024)**
Art. 4 — authorisation regime	Time-to-authorisation for ESA/Copernicus contracts	G	Number of launch services contracts under active authorisation cycle
Art. 5 — objective requirements (safety, cyber, environment)	Certification portfolio and management systems	G / E / S	ISO 9001/14001/27001 coverage; cyber-incident rate disclosed under ESRS S1
Art. 6 — subjective requirements (integrity, conduct)	Reputational and integrity governance	G	Membership of Business Integrity Forum (Transparency); Model 231 in force
Artt. 18-20 — dual liability regime	Insurance coverage and provisions	G / S	Insurance-to-launch-revenue ratio; provisions for litigation in Notes
Art. 22 — National Space Economy Plan	Strategic alignment with ESA / Copernicus / defence	G	Share of backlog from public/ institutional customers (≈100% in 2024)
Art. 23 — Multi-annual Space Fund	Access to public co-financing for sustainability-linked R&D	G / E	Funding obtained for Vega-E and M10 (methane-oxygen) development
Implicit — debris mitigation	End-of-life management of upper stages	E	Deorbiting and passivation procedures disclosed in Sustainability Statement
Cross-cutting	ESG-linked executive remuneration	G	Weight of ESG KPIs in variable pay (20% in 2024, up from 16% in 2023)

Read through the lens of CSRD double materiality, the indicators in Table 2 reveal an interesting analytical property: the same legal obligation simultaneously generates an outside-in financial risk — for example, the cost of insurance under the dual liability regime — and an inside-out impact dimension — the externalities associated with launch

operations and orbital debris. This dual nature is precisely what justifies the integration of legal-derived indicators within the ESRS reporting perimeter, rather than within a separate compliance silo.[16] In other words, Law No. 89/2025 does not produce risks that are external to the sustainability framework: it produces risks that are constitutive of the materiality assessment that Avio is now required to perform every year.

From an Enterprise Risk Management perspective, the value added by this framework becomes visible when one considers how Avio could use the composite ESG-legal index in scenario analysis. Three illustrative scenarios merit attention. In a regulatory tightening scenario — for example, the entry into force of the EU Space Act with stricter cybersecurity and debris mitigation requirements — the index would re-weight the environmental and governance components upwards, signalling the need for additional capital expenditure on compliance infrastructure.[17]

In a liability shock scenario — a launch failure with damage to third parties — the financial materiality channel would translate into a measurable increase in the governance-financial sub-index, with direct implications for credit risk and insurance pricing. In a strategic alignment scenario — a new National Space Economy Plan emphasising sustainable launch technologies — the framework would highlight the positive correlation between Avio's M10 engine development and the public funding pipeline established by Article 23, supporting the case for accelerated R&D investment.[18]

The case of Avio thus illustrates how the conceptual model developed in Section 3 ceases to be a purely theoretical exercise and becomes operationally relevant precisely when applied to a firm that already operates at the intersection of three regulatory regimes: national space law, EU sustainability reporting, and the prospective European space framework. It also reveals a more general analytical insight. In firms of this kind, ESG indicators are no longer an external sustainability narrative grafted onto financial reporting, but the natural output of a legal-operational architecture that the firm is required to build in any case. The contribution of the proposed framework is, therefore, less to "create"

[16] EFRAG, European Sustainability Reporting Standards (ESRS) – Set 1, Brussels, 2023; EFRAG, IG 1 – *Materiality Assessment. Implementation Guidance*, Brussels, 2025.

[17] European Commission, *Proposal for a Regulation of the European Parliament and of the Council on the safety, resilience and sustainability of space activities in the Union* (EU Space Act), COM (2025) 335 final, Brussels, 25 June 2025.

[18] Avio S.p.A., FY 2024 Results, cit., pp. 2–5; European Space Agency (ESA), Vega-E *Programme Documentation and Industrial Development Framework*, 2024.

indicators than to organise them — to provide a structured rationale linking specific legal obligations to specific ESG metrics, in a way that is both theoretically defensible and operationally usable within an Enterprise Risk Management system.

It remains essential to acknowledge the limits of this illustrative application. The mapping presented above is ex-ante and qualitative: it shows how the framework can be populated with publicly available data but does not yet provide a calibrated index. A full empirical validation would require access to internal risk-register data and to the AHP weighting workshops that the methodology presupposes — a step reserved for the subsequent stages of the underlying doctoral research. Nonetheless, even at this illustrative level, the Avio mini case demonstrates that the translation from legal obligation to ESG indicator is neither artificial nor purely doctrinal: it traces a path that aerospace operators are already, de facto, walking under the combined pressure of Law No. 89/2025 and the CSRD framework. The contribution of the present article is to render this path explicit, structured, and analytically tractable.

5. *Conclusions*

This article has explored the intersection between space regulation, sustainability governance, and enterprise risk management, proposing a structured methodological framework for translating legal obligations into operational ESG risk indicators within the aerospace industry. Starting from Italian Law No. 89/2025, the analysis has demonstrated how contemporary space regulation does not merely function as a set of compliance requirements but increasingly operates as a generator of structured risk conditions affecting corporate decision-making processes.

The first contribution of the paper lies in its legal analysis of the regulatory framework governing space activities. The study has shown that authorization regimes, continuous supervisory mechanisms, liability structures, and safety and sustainability obligations collectively define a complex normative architecture that directly shapes corporate risk exposure. Rather than being interpreted solely as external constraints, these legal provisions can be understood as embedded sources of governance, environmental, and social risks, particularly relevant in highly regulated and technologically advanced sectors such as aerospace.

The second contribution consists in the development of a methodological model capable of translating legal norms into ESG risk indicators. By decomposing legal obligations into operational variables and mapping them onto Environmental, Social, and Governance dimensions, the proposed framework provides a structured approach to bridging the gap between legal analysis and risk management. The integration of multi-criteria decision-making techniques, in particular the Analytic Hierarchy Process, enables the transformation of heterogeneous qualitative and quantitative inputs into a coherent analytical structure suitable for decision-support purposes.[19]

The third contribution relates to the operational integration of the proposed framework within Enterprise Risk Management systems. The analysis has shown that ESG indicators derived from legal obligations can be effectively embedded within corporate risk architectures, enabling firms to move from static compliance approaches toward dynamic and anticipatory risk management practices. In this perspective, the ESG risk index functions not merely as a reporting instrument but as a strategic tool supporting scenario analysis, regulatory forecasting, and decision-making processes.

From a broader theoretical perspective, the findings of this study contribute to three distinct but interconnected streams of literature. First, they extend space law scholarship by reframing regulatory instruments as sources of structured risk rather than purely normative constraints. Second, they contribute to ESG literature by providing a more operational and legally grounded understanding of sustainability indicators. Third, they enrich Enterprise Risk Management research by demonstrating how regulatory frameworks can be systematically integrated into risk modeling architectures.[20]

From a managerial perspective, the proposed framework offers practical implications for aerospace operators operating in highly regulated environments. In particular, it supports the integration of compliance, sustainability, and risk management into a unified decision-making structure, thereby enhancing organizational resilience and strategic adaptability. This is particularly relevant in the context of rapidly evolving

[19] T.L. Saaty, *Decision Making with the Analytic Hierarchy Process*, in *International Journal of Services Sciences*, 1(1), 2008, pp. 83–98.

[20] R. McCormick, C. Stears, *Legal and Conduct Risk in the Financial Markets*, 3rd ed., Oxford University Press, Oxford, 2018; T. Mahler, *Defining Legal Risk*, in *Commercial Contracting for Strategic Advantage: Potentials and Prospects*, Turku University of Applied Sciences, Turku, 2007, pp. 10–31.

regulatory frameworks at both European and international levels, where the ability to anticipate and quantify regulatory impacts represents a critical competitive advantage.

Nevertheless, the study is not without limitations. The proposed framework remains conceptual and requires further empirical validation through application to real industrial datasets. Moreover, while the focus on the aerospace sector ensures analytical depth, future research could extend the model to other highly regulated industries characterized by complex ESG-risk interactions.

A further methodological limitation deserves explicit acknowledgement. The framework proposed in this article relies on an interpretative translation of legal obligations into risk indicators and therefore presupposes a degree of normative discretion in the selection and weighting of variables. While the Analytic Hierarchy Process mitigates this concern by structuring expert judgement, it does not eliminate the inherent subjectivity of the legal-to-managerial translation process.[21] Future research should therefore address two complementary questions: first, the sensitivity of the composite ESG index to alternative weighting schemes; and second, the convergence (or divergence) between the indicators derived from the proposed framework and those reported under the European Sustainability Reporting Standards (ESRS) within the CSRD perimeter. This second line of inquiry would allow the framework to be tested against an externally validated reporting infrastructure, strengthening its operational relevance for aerospace operators subject to mandatory sustainability disclosure.[22]

In conclusion, the article argues that space regulation should no longer be interpreted solely as a compliance framework, but rather as a structural driver of ESG risk formation. By translating legal obligations into measurable risk indicators, the proposed approach contributes to a more integrated understanding of how law, sustainability, and risk management interact in shaping the governance of high-technology industries.

In this sense, compliance evolves from a static legal requirement into a dynamic strategic capability, essential for long-term resilience and sustainable industrial development.

[21] T.L. Saaty, *Decision Making with the Analytic Hierarchy Process*, cit.

[22] EFRAG, European Sustainability Reporting Standards (ESRS) – Set 1, Brussels, 2023; EFRAG, IG 1: Materiality Assessment – Implementation Guidance, Brussels, 2025.

Frans von der Dunk

International third-party liability, national space law & EU space law

Summary: 1. Introduction – 2. International third-party liability under space law and national space law in the EU – 3. The draft EU Space Act and private sector space activities – 4. EU Member State space laws and the scope of the authorization regime – 5. EU Member State space laws and the substance of international third-party liability – 6. Concluding remarks.

1. *Introduction*

In the summer of 2025, the EU Commission proposed a Regulation on the safety, resilience and sustainability of space activities in the Union, also known for short as the Draft EU Space Act[1] since it purports to generally address the same issues that national acts, statutes and laws on space in a number of countries, including many EU Member States, address: namely, the engagement of the private sector in space activities.[2]

While the Draft EU Space Act is as of this writing going through the EU process for creating legislation[3], it has raised discussion on many issues in this context of private sector participation in space activities, of which the current paper will address only one: that of international third-party liability[4].

[1] Proposal for a Regulation of the European Parliament and of the Council on the safety, resilience and sustainability of space activities in the Union (hereafter Draft EU Space Act), Brussels, 25 June 2025, COM(2025) 335 final.

[2] See on this in general, I. Marboe, *National space law*, in *Handbook of Space Law* (Eds. F.G. von der Dunk & F. Tronchetti)(2015), 127-204; F.G. von der Dunk, *Advanced Introduction to Space Law* (2020), 18, 115-25.

[3] Notably as per Art. 294, Treaty establishing the European Community as amended by the Treaty of Lisbon amending the Treaty on European Union and the Treaty establishing the European Community (hereafter Treaty on the Functioning of the European Union), Lisbon, done 13 December 2007, entered into force 1 December 2009; OJ C 326/47 (2012).

[4] 'Liability' here is generally defined as 'the legal accountability of persons or entities for

2. *International third-party liability under space law and national space law in the EU*

Ever since the engagement of the private sector[5] became a matter of substance at least in some sectors of the human space endeavour – roughly since the 1980s, especially in the context of satellite communications – the issue of (international) third-party liability for damage resulting from space activities has been a major part of the discussion. Following Article VII of the 1967 Outer Space Treaty[6] as further elaborated by the 1972 Liability Convention[7] namely, States are internationally held liable for damage caused by space objects also if these are manufactured, launched, owned and/or operated by private entities, as long as such States qualify as 'launching States'[8].

Further to Article VI of the Outer Space Treaty requiring "authorization and continuing supervision by the appropriate State Party to the Treaty" of "national activities in outer space (…) carried on (…) by non-governmental entities", many States have indeed established national space legislation to regulate such national space activities and the third-party liabilities that might result therefrom.

damage caused to other persons or entities', in accordance with applicable law and the conditions and parameters it applies; 'third-party liability' then is to be defined as 'liability for damage caused to third parties', that is to anyone not directly involved in the activities causing the damage in question. See further already F.G. von der Dunk, *Commercial Space Activities: An Inventory of Liability – An Inventory of Problems*, in *Proceedings of the Thirty-Seventh Colloquium on the Law of Outer Space* (AIAA)(1995), 165-6.

[5] The term 'private' is here used as the opposite of 'public', meaning 'governmental' as well as 'intergovernmental', even though this is clearly an oversimplification as in particular in the world of space activities many activities are one way or another of a hybrid character, comprising private and public actors in various complicated relationships.

[6] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies (hereafter Outer Space Treaty), London/Moscow/Washington, done 27 January 1967, entered into force 10 October 1967; 610 UNTS 205; TIAS 6347; 18 UST 2410; UKTS 1968 No. 10; Cmnd. 3198; ATS 1967 No. 24; 6 ILM 386 (1967).

[7] Convention on International Liability for Damage Caused by Space Objects (hereafter Liability Convention), London/Moscow/Washington, done 29 March 1972, entered into force 1 September 1972; 961 UNTS 187; TIAS 7762; 24 UST 2389; UKTS 1974 No. 16; Cmnd. 5068; ATS 1975 No. 5; 10 ILM 965 (1971).

[8] See Art. I(c), Liability Convention (*supra* n. 7). This clause defines the 'launching State' as the State launching the space object causing the damage, the State procuring that launch, the State whose territory is used for that launch and/or the State whose facilities are used for that launch.

Such national space law regimes addressing third-party liability in particular would serve to impose, as part of the authorization required to become active in outer space, conditions and limitations so as to minimize the chances of damage being caused in the first place, and second, in case third-party damage giving rise to international liability claims nevertheless occurs, obligations to reimburse the State – since the *State* would be required to compensate such damage under the Outer Space Treaty and the Liability Convention.

As for the European Union, currently thirteen States have adopted such national space legislation with a comprehensive approach to, in principle, *all* private sector space activities – in chronological order: Sweden (1986)[9], Belgium (2005)[10], Netherlands (2007)[11], France (2008)[12], Austria (2011)[13], Denmark (2016)[14], Greece (2017)[15], Finland (2018)[16], Portugal

[9] *Lag om rymdverksamhet* (hereafter Swedish Act on Space Activities), 1982: 963, 18 November 1982; *National Space Legislation of the World*, Vol. I (2001), at 398; *Space Law – Basic Legal Documents*, E.II.1; 36 *Zeitschrift für Luft- und Weltraumrecht* (1987), at 11.

[10] Law on the Activities of Launching, Flight Operations or Guidance of Space Objects (hereafter Belgian Space Law), of 17 September 2005 (adopted 28 June 2005); *Nationales Weltraumrecht / National Space Law* (2008), at 183; United Nations/Nigeria Workshop on Space Law, National Legislation and Policy – Selected Texts (2005), at 92.

[11] *Wet houdende regels omtrent ruimtevaartactiviteiten en de instelling van een register van ruimtevoorwerpen (Wet ruimtevaartactiviteiten)* (hereafter Dutch Space Law), 24 January 2007; 80 *Staatsblad* (2007); *Nationales Weltraumrecht / National Space Law* (2008), at 201.

[12] *Loi relative aux opérations spatiales* (hereafter French Law on Space Operations); *Loi n° 2008-518 du 3 juin 2008*; unofficial English translation 34 *Journal of Space Law* (2008), 453.

[13] Austrian Federal Law on the Authorisation of Space Activities and the Establishment of a National Space Registry (*Bundesgesetz über die Genehmigung von Weltraumaktivitäten und die Einrichtung eines Weltraumregisters (Weltraumgesetz)*(hereafter Austrian Outer Space Act), as adopted by Parliament on 6 December 2011; Federal Law Gazette I No. 132/2011 of 27 December 2011; 61 *Zeitschrift für Luft- und Weltraumrecht* (2012), 37-42, 56-61.

[14] *Lov om aktiviteter i det ydre rum* (hereafter Danish Law on Activities in Outer Space), Act. No. 409, passed by Parliament with the third treatment, 11 May 2016, entered into force 1 July 2016; *Parliament Gazette*, 2015-17, No. L 128.

[15] Law # 4508; Licensing of space activities – Registration in the National Register of Space Objects – Establishment of Hellenic Space Agency and other provisions; Government Gazette of the Hellenic Republic (hereafter Greek National Space Law), 22 December 2017, Issue # A 200.

[16] *Laki avaruustoiminnasta* (hereafter Finnish Act on Space Activities), No. 63/2018, of 23 January 2018.

(2019)[17], Luxembourg (2020)[18], Slovenia (2022)[19], Slovakia (2024)[20] and Italy (2025)[21]. In addition, Germany has a national law dealing only with satellite remote sensing[22] and Spain a law on space object registration[23], while Liechtenstein, a country not legally a Member State of the European Union yet for most practical purposes part thereof, has a space law of general scope in place[24], but those will not be further addressed here.

The thirteen national EU Member State laws that provide the focus of this paper all provide for a rather comprehensive regime of authorizing and monitoring private space operations under the respective jurisdictions, and if necessary sanctioning the operators, and they all consequently also deal with third-party liability.

[17] Decree-Law No. 16/2019 (hereafter Portuguese Law on Space Activities), of 22 January 2019; *Official Journal, 1st Series – No. 15*, of 22 January 2019.

[18] *Loi portant sur les activités spatiales et modifiant: 1. la loi modifiée du 9 juillet 1937 sur l'impôt sur les assurances dite "Versicherungsgesetz" ; 2. la loi modifiée du 4 décembre 1967 concernant l'impôt sur le revenu*)(hereafter Luxembourgish Law on Space Activities); of 15 December 2020, published 28 December 2020. Note that Luxembourg had already in 2017 adopted an Act specifically addressing private space mining activities: Law on the exploration and utilization of space resources (*Loi du 20 juillet 2017 sur l'exploration et l'utilisation des ressources de l'espace*)(hereafter Luxembourgish Space Resources Law); of 20 July 2017, published 28 July 2017.

[19] *Zakon o Vesoljskih Dejavnostih (ZVDej)* (hereafter Slovenian Space Activities Act); No. 003-02-1/2022-78.

[20] Act No. 378 of 28 November 2024 on the regulation of space activities and on the amendment to Act of the National Council of the Slovak Republic No. 145/1995 Coll. on administrative fees, as amended (hereafter Slovak Space Act).

[21] *Legge 'Disposizioni in materia di economia dello spazio'* (Law with Provisions on the space economy)(hereafter Italian Law on the Space Economy), Nr. 89, 13 June 2025; *Gazzetta Ufficiale* Nr. 144 of 24 June 2025.

[22] *Gesetz zum Schutz vor Gefährdung der Sicherheit der Bundesrepublik Deutschland durch das Verbreiten von hochwertigen Erdfernerkundungsdaten (Satellitendatensicherheitsgesetz)*, 23 November 2007, effective 1 December 2007; *Bundesgesetzblatt (2007)*, Part I No. 58 of 28 November 2007, 2590.

[23] Royal Decree No. 278/1995 establishing in the Kingdom of Spain of the Registry foreseen in the Convention adopted by the United Nations General Assembly on 2nd November 1974, 24 February 1995, Prime Minister's Chancellery; United Nations/ Nigeria Workshop on Space Law, National Legislation and Policy – Selected Texts (2005), at 201.

[24] Act on the Authorization of Space Activities and the Registration of Space Objects, of 5 October 2023; Liechtenstein Law Gazette Vol. 2023 No. 443 of 6 December 2023.

3. *The draft EU Space Act and private sector space activities*

It is in this legal environment, of thirteen EU Member States having a full-fledged national space law and authorization regime while the fourteen other EU Member States have either no space law at all or one which only addresses a certain part of the field, that the draft EU Space Act has entered the discussion.

All proposed EU law needs to comply with the three ruling principles of conferral, subsidiarity and proportionality in order to maximize its chances at becoming adopted. Conferral means that "the Union shall act only within the limits of the competences conferred upon it by the Member States in the Treaties to attain the objectives set out therein. Competences not conferred upon the Union in the Treaties remain with the Member States".[25] Subsidiarity means that

> in areas which do not fall within its exclusive competence, the Union shall act only if and in so far as the objectives of the proposed action cannot be sufficiently achieved by the Member States, either at central level or at regional and local level, but can rather, by reason of the scale or effects of the proposed action, be better achieved at Union level.[26]

Finally, proportionality means that "the content and form of Union action shall not exceed what is necessary to achieve the objectives of the Treaties".[27]

In terms of conferral specifically, the Draft EU Space Act bases the competence of the Commission to propose a Regulation like the one at issue on Article 114 of the Treaty on the Functioning of the European Union, following which

> [t]he European Parliament and the Council shall, acting in accordance with the ordinary legislative procedure and after consulting the Economic and Social Committee, adopt the measures for the approximation of the provisions laid down by law, regulation or administrative action in Member States which have as their object the *establishment and functioning of the internal market*.[28]

[25] Art. 5(2), Treaty on the Functioning of the European Union (*supra* n. 3).

[26] Art. 5(3), Treaty on the Functioning of the European Union (*supra* n. 3).

[27] Art. 5(4), Treaty on the Functioning of the European Union (*supra* n. 3).

[28] Art. 114(1), Treaty on the Functioning of the European Union (*supra* n. 3); italics added.

The Explanatory Memorandum to the draft EU Space Act in this regard claims:

> The lack of coordination between the different regulatory approaches has led to a fragmented regulatory landscape whereby emerging obstacles may impede the functioning of the internal market in space services and space-based data in the Union.
>
> The proposed EU Space Act harmonises the legal framework across the Union, integrating requirements laid down by national space legislations to avoid overlap, duplication and conflicts and to improve the functioning of the internal market.[29]

The fact, however, that thirteen EU Member States have already legislated on matters of space, and in particular also on the private sector involvement therein and the requirement to authorize and continuously supervise them pursuant to Article VI of the Outer Space Treaty, also brings into the discussion Article 189 of the Treaty on the Functioning of the European Union, which provides that while "the European Parliament and the Council, acting in accordance with the ordinary legislative procedure, shall establish the necessary measures, which may take the form of a European space programme", this has to occur while "*excluding any harmonisation of the laws and regulations of the Member States*".[30]

This then raises the question whether the Commission is entitled to use Article 114 in order to circumvent the principled prohibition of harmonization which Article 189(2) provides for. Ultimately, the answer to this question will depend upon whether the Member States, who in Council and together with the European Parliament pursuant to the ordinary legislative procedure will have to approve the Draft EU Space Act before it would enter into force, find such a circumvention acceptable in light of overriding interests in harmonizing national requirements of the thirteen EU Member States that already have a national space law and at the same time applying them to the fourteen that do not.

In this regard, the Explanatory Memorandum, arguing subsidiarity and proportionality, broadly claims that analyses and the results of stakeholder engagement confirm that discrepancies between, *casu quo*

[29] Draft EU Space Act (*supra* n. 1), at p. 2.

[30] Art. 189(2), Treaty on the Functioning of the European Union (*supra* n. 3); italics added. See further on this *e.g.* F.G. von der Dunk, *The EU Space Competence as per the Treaty of Lisbon: Sea Change or Empty Shell?*, in *Proceedings of the International Institute of Space Law 2011* (IISL)(2012), 382-92.

absence of, EU Member State national space laws and regulations provide substantial obstacles to growth for the EU space industry.[31]

Interestingly enough, while third-party liability arrangements including reimbursement and insurance obligations generally are a subject of major interest to the private space sector, given that the assumed risks can be in the range of many millions of euros[32] and the related insurance fees would be commensurately large, the Draft EU Space Act does not mention this issue at all.[33]

The question this paper attempts to answer then is: would that claim – of substantial obstacles arising from the current fragmented legal situation throughout the European Union – make sense as far as third-party liability is concerned? And would the answer to that question perhaps, one way or another, explain why the Draft EU Space Act does *not* attempt to address (at least on the face of it[34]) the issue of national space law clauses on international third-party liabilities?

The issue of international third-party liability and how it is handled by national space law, including that of EU Member States, concerns many different questions; in view of constraints of size the current paper however will only address two of those.

4. *EU Member State space laws and the scope of the authorization regime*

The first question concerns the scope of the respective national space laws, with a view to who would be subject to the applicable substantive rules imposed regarding the handling of international third-party liability. This goes back essentially to the issue of Article VI of the Outer Space Treaty, which requires States to authorize and supervise "national activities in outer space" without further specifying what the term 'national' refers

[31] See Draft EU Space Act (*supra* n. 1), esp. at pp. 4–6.

[32] *Cf. e.g.* the US approach, where under the Commercial Space Launch Act launch operators are required to accept, and take out insurance alternatively accept financial responsibility for, third-party liabilities up to caps of US$ 500,000,000; § 50914, 51 U.S.C. 509.

[33] It only refers in very general terms to the liability applicable to space activities pursuant to the Outer Space Treaty; see preambular para. (6), Draft EU Space Act (*supra* n. 1).

[34] Note, that the general reference to harmonization for the purposes of safety might at some point in the future perhaps come to be referred to as a baseline from which to also address any disparate national legislation on the issue of liability.

to – how have the Member States at issue interpreted this term, read to what categories of private actors have they applied their respective authorization regimes? Do they apply a territorial criterion – all operators on national territory require an authorization – or a nationality criterion – all operators with the relevant nationality require an authorization – or both? Are also operators undertaking their space activities from nationally registered vessels, aircraft or other facilities included in the regime?[35]

The results of a summary analysis here would be as follows. In the following countries the following categories of activities undertaken by private actors require authorization by the relevant national authorities.

- Sweden: activities undertaken from national territory or by natural or juridical persons with Swedish nationality[36];
- Belgium: activities undertaken from zones or installations under Belgian jurisdiction or control; as well as potentially those undertaken by natural or legal persons with Belgian nationality[37];
- Netherlands: activities undertaken from Dutch territory, ships or aircraft; as well as potentially those undertaken by natural or juridical persons with Dutch nationality or organized from Dutch territory[38];
- France: launch activities undertaken from French territory, means or facilities or by operators with French nationality; other space activities only if undertaken by operators with French nationality[39];
- Austria: activities undertaken from Austrian territory, vessels or airplanes, or by natural or legal persons with Austrian nationality[40];
- Denmark: activities undertaken from Danish territory, craft or

[35] Note, that the Recommendations on national legislation relevant to the peaceful exploration and use of outer space, UNGA Res. 68/74, of 16 December 2013; UN Doc. A/RES/68/74; at para. 2 provides: "The State, taking into account its obligations as a launching State and as a State responsible for national activities in outer space under the United Nations treaties on outer space, should ascertain national jurisdiction over space activities carried out from territory under its jurisdiction and/or control; likewise, it should issue authorizations for and ensure supervision over space activities carried out elsewhere by its citizens and/or legal persons established, registered or seated in territory under its jurisdiction and/or control (…)".

[36] See Sec. 2, Swedish Act on Space Activities (*supra* n. 9).

[37] See Art. 2, Belgian Space Law (*supra* n. 10).

[38] See Sec. 2, Dutch Space Law (*supra* n. 11).

[39] See Art. 2, French Law on Space Operations (*supra* n. 12).

[40] See Sec. 1(1), Austrian Outer Space Act (*supra* n. 13).

facilities, or by operators with Danish nationality[41];
- Greece: activities undertaken from Greek territory or facilities, or by natural or legal persons with Greek nationality, or in any other scenarios making Greece a launching State[42];
- Finland: activities undertaken from Finnish territory, vessels or aircraft, or by citizens or legal persons with Finnish nationality[43];
- Portugal: activities undertaken from Portuguese territory, vessels, aircraft or facilities, or by operators with Portuguese nationality[44];
- Luxembourg: activities undertaken from Luxembourgish territory or installations, or by natural or legal persons with Luxembourgish nationality[45];
- Slovenia: activities undertaken from Slovenian territory, vessels or aircraft, or with space objects registered in Slovenia, or by citizens or legal persons with Slovenian nationality[46];
- Slovakia: "regulated space activities", suggesting potentially all activities within national jurisdiction[47]; and
- Italy: activities undertaken from Italian territory or by operators with Italian nationality[48].

In other words: all States concerned require authorizations for activities conducted from national territory or by national persons or entities, although sometimes it is made dependent upon certain scenarios (such as in Belgium, the Netherlands and France). Many, but not all, in addition one way or another refer to vessels/ships, aircraft and/or installations or facilities registered nationally, and one (the Slovenian one) explicitly even includes nationally registered space objects. In sum: these national acts still show a sizeable measure of variation in the application of the scope of their respective authorization regimes also as relevant for international third-party liability related obligations, certainly beyond the territorial application of jurisdiction for such purposes.

It might already be noted that the Draft EU Space Act defines its scope

[41] See Art. 2, Danish Law on Space Activities (*supra* n. 14).

[42] See Art. 1(2), Greek National Space Law (*supra* n. 15). This, of course, refers back to the definition of the 'launching State' as per Art. I(c), Liability Convention (*supra* n. 7).

[43] See Sec. 1, Finnish Act on Space Activities (*supra* n. 16).

[44] See Art. 2(1), Portuguese Law on Space Activities (*supra* n. 17).

[45] See Art. 1, Luxembourgish Law on Space Activities (*supra* n. 18).

[46] See Art. 2, Slovenian Space Activities Act (*supra* n. 19).

[47] Sec. 1(a), Slovak Space Act (*supra* n. 20).

[48] See Art. 3, Italian Law on the Space Economy (*supra* n. 21).

in terms of the categories of private operators subject to its regime rather broadly, that is to all relevant categories of space activities "carried out by space services providers *established in the Union*" as well as those "carried out by international organisations and space services providers established in third countries when providing space-based data or space services in the Union".[49] Its application would, therefore, indeed result in the need for considerable amendment of various EU Member State laws in their current state, as its own terms are only generally aligned with those of most but certainly not all of the national laws assessed above.

5. *EU Member State space laws and the substance of international third-party liability*

The other question concerns the core matter of substance of any obligation of authorized operators to reimburse the State authorizing them in case of an international third-party liability claim pursuant to Article VII of the Outer Space Treaty and/or the Liability Convention. Have the Member States at issue required full, comprehensive reimbursement, with or without exceptions, or have they imposed limits upon any reimbursement obligation, either in general terms or precisely indicated?

The results of a summary analysis here would be as follows. In the following countries the following approach to reimbursement of State-paid international third-party liability claims by, respectively recourse thereof to, the authorized operator has been taken.

- Sweden: reimbursement is mandatory and comprehensive, "unless special reasons tell against this"[50];
- Belgium: the government has a right of recourse, which may however be limited by further legislation or regulation[51];
- Netherlands: the government has a right of recourse up to the mandatory insurance cover which is not further specified[52];
- France: the government has a right of recourse limited to a sum between € 50 and € 70 million per launch[53];

[49] Art. 1(2.a), Draft EU Space Act (*supra* n. 3); italics added.

[50] Sec. 6, Swedish Act on Space Activities (*supra* n. 9).

[51] See Art. 15, Belgian Space Law (*supra* n. 10).

[52] See Sec. 12, Dutch Space Law (*supra* n. 11).

[53] Further to Arts. 14, 15, French Law on Space Operations (*supra* n. 12), this has so far

- Austria: the government has a right of recourse limited to € 60 million for damage on Earth[54];
- Denmark: the government has a right of recourse, but further regulations may provide limits thereto[55];
- Greece: the government has a right of recourse limited to € 60 million or less[56];
- Finland: the government has a right of recourse subject to the national tort liability act as well as limited to € 60 million[57];
- Portugal: the government has a right of recourse, to be limited as per further regulation[58];
- Luxembourg: unlimited liability applies, which presumably means that the recourse obligation would also be unlimited[59];
- Slovenia: the government has a right of recourse limited to € 60 million[60];
- Slovakia: the government has a right of recourse limited to € 60 million [61]; and
- Italy: the government will automatically assert a claim for reimbursement, to be limited by a decree to no less than € 50 million (respectively € 20 million for exclusively-research activities) for damage on Earth[62].

In other words: almost all States *allow* for recourse, whereas in some cases recourse is applied automatically; a considerable number of States use a limit of € 60 million while a few use other limits and yet others defer limits to further regulation. Only two States (Sweden and Luxenbourg) proceed on the assumption that there would be full reimbursement. In sum: these national acts still show considerable variation already in the mere obligations to reimburse international third-party liability related claims – noting that, as said, the Draft EU Space Act actually does not address the issue at all.

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always been set at € 60 million.

[54] See Secs. 4(4), 11, Austrian Outer Space Act (*supra* n. 13).

[55] See Art. 12, Danish Law on Space Activities (*supra* n. 14).

[56] See Art. 11, Greek National Space Law (*supra* n. 15).

[57] See Sec. 7, Finnish Act on Space Activities (*supra* n. 16).

[58] See Art. 18, Portuguese Law on Space Activities (*supra* n. 17).

[59] See Art. 4, Luxembourgish Law on Space Activities (*supra* n. 18).

[60] See Art. 6, Slovenian Space Activities Act (*supra* n. 19).

[61] See Sec. 11, Slovak Space Act (*supra* n. 20).

[62] See Arts. 19, 21, Italian Law on the Space Economy (*supra* n. 21).

6. *Concluding remarks*

The next, resulting final question on this issue then is: even ignoring any variation in other key aspects of the national authorization regimes pertaining to third-party liability (the set of actual space activities *ratione materiae* subjected to the relevant obligation; the imposition or not of third-party liability insurance, *et cetera*): is this variety a real problem?

It should be noted that at least US companies have for many years made a strong push for more support from the US government in the realm of third-party liability, meaning for example a long-duration commitment to limitation of recourse and a revision of the calculation of the Maximum Probable Loss which determines the limitation for every individual launch operation.[63]

This suggests that it might be worthwhile to consider whether the final EU Space Act should not (also) address international third-party liability as the most directly and most quantifiable, money-related aspect of an Internal Market for safe space services. At the same time, given its obvious quantifiability, it might suggest that for political and strategic reasons the Commission has so far refrained from what would otherwise have been an obvious push for limiting third-party liability for operators to a sum of € 60,000,000 per launch – and for harmonizing existing national space law in an unequivocal, direct and directly visible manner, in contradiction to what Article 189(2) of the Treaty on the Functioning of the European Union seems to require.

Evidently, the last word in this respect will be for the EU Member States in Council and the individual Members of the European Parliament, as they will have to ultimately agree one way or another, in accordance with the ordinary legislative procedure of Article 294 of the Treaty on the Functioning of the European Union.

[63] See *e.g.* Commercial Space Launch Insurance: FAA Needs to Fully Address Mandated Requirements; GAO-18-57, of 16 January 2018.

Rafael Moro Aguilar

Some Reflections on the Proposed EU Space Act

Summary: 1. Overview – 2. Potential Repercussions – 3. The EU draft Space Code of Conduct – 4. Dark and Quiet Skies in the EUSA – 5. Conclusions.

1. *Overview*

The European Commission presented the initial draft EU Space Act (EUSA) on 25 June 2025. The act has adopted the form of a EU Regulation, which means that once it enters into force, it will apply directly in all 27 EU Member States. The proposed Regulation is aimed to establish a common and transparent legal framework for many aspects of space operations and to create a single market for space services within the EU. It also intends to make the EU space sector more competitive, innovative, resilient, and environmentally friendly.[1]

The three key "pillars" that compose this draft Regulation are safety, resilience, and sustainability:

- Safety refers to the protection of space assets from physical threats, such as space debris, collisions, etc. The Act introduces robust rules for tracking space objects and mitigating space debris, preserving Europe's secure and uninterrupted access to space.
- Resilience refers to the protection of space assets and infrastructure from cyberthreats and from other kinds of attacks. Tailored cybersecurity requirements will strengthen protection of European space infrastructure and ensure business continuity.
- Sustainability means the protection of the natural environment (both terrestrial and space). Operators will need to assess and reduce the environmental impact of their space activities, while

[1] European Commission, Directorate General of Defence, Industry and Space, "EU Space Act": https://defence-industry-space.ec.europa.eu/eu-space-act_en. See also for an overview: https://www.european-space-act.com/.

benefiting from support for innovation in emerging technologies like in-orbit servicing and debris removal.[2]

The new rules will apply to both EU and non-EU operators providing space services in Europe.

The EUSA has a long text encompassing a number of different aspects of space activities, and for that reason it looks more like a code than a space act. If we compare its extension to national space acts, the original EUSA had 119 articles while the United Kingdom Outer Space Act[3] has 15 sections, the United States Commercial Space Launch Act[4] has 25 sections, the French Space Operations Act[5] has 30 articles, and the Italian Space Act[6] has 31 articles.

The list of definitions is particularly extensive. This section of the EUSA reads almost like a space dictionary, particularly when compared to the short lists of definitions that are usually contained in ordinary space acts. The original EUSA contained as many as 73 definitions (56 can be counted in the most recent version) as opposed to e.g. 17 definitions contained in the recent Italian Space Law.

A public consultation process followed the presentation of the draft EUSA in the Fall of 2025.[7] the European Commission assessed the many comments and responses received, and the original draft text of the Space Act has undergone a complete revision. Many changes have been introduced into the text, and already two new versions of the text have been circulated. The latest available version is the updated "compromise" proposal issued by the Cypriot Presidency of the EU on March 30, 2026.[8]

The proposal (referenced as 2025/0335(COD)) is currently under

[2] *Ibidem*.

[3] UK, Outer Space Act, 1986: https://www.legislation.gov.uk/ukpga/1986/38/introduction.

[4] USA, Commercial Space Launch Act, 1984: https://www.congress.gov/bill/98th-congress/house-bill/3942/text.

[5] France, Space Operations Act, 2008: https://www.legifrance.gouv.fr/loda/id/JORFTEXT000018931380?init=true&page=1&query=loi+des+operations+spatiales&searchField=ALL&tab_selection=all.

[6] Italy, Law on Space Economy, 2025: https://www.outerspacelawsapienza.it/2025/06/25/italy-new-law-on-space-economy/.

[7] The public consultation was open from 15 July 2025 to 7 November 2025 and is currently closed. Feedback and comments can be found here: https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/13971-EU-Space-Act-new-rules-for-safe-resilient-and-sustainable-space-activities_en.

[8] This latest version of the EUSA is still not publicly available.

negotiation among the EU Member States. The European Commission will carry out a last revision of the text before presenting a final version to the European Parliament and the Council of the European Union, following the ordinary legislative procedure of the EU. Final passage would follow some time in 2027, after which there would be a transition period that would last a couple of years.[9]

2. *Potential repercussions*

This is an ambitious initiative that, if successful, could indeed have an influence in the global regulation of outer space. This initiative could position Europe as a global standard-setter on a number of space matters. This has happened before in other sectors. For instance, the 2018 General Data Protection Regulation (GDPR)[10] had repercussions beyond the European borders with regard to the protection of personal data.

This is even more likely because, as already noted, the EUSA rules will apply not only to European space operators coming from the EU's 27 countries, but also to any space operators from outside the EU who intend to offer in-orbit services and space-based data in Europe.

As a result, the EUSA could have an international impact that could be similar to the "California effect" that has been observed in North America.[11]

Inspired by the notion of the "California effect," a "Brussels effect" has also been identified as a phenomenon where the EU's strict regulatory standards for consumer health, environmental protection, and data privacy (GDPR) become *de facto* global standards adopted by multinational corporations.

According to this thesis that was first developed by American legal scholar Anu Bradford, the EU is already acting in practice as the world's leading regulator in many areas, creating a "Europeanization" of global,

[9] Council of the European Union, Progress Report on the EU Space Act, 8 May 2026: https://data.consilium.europa.eu/doc/document/ST-8861-2026-INIT/en/pdf.

[10] Available at https://gdpr-info.eu/.

[11] Coined by UC Berkeley professor David Vogel in the 1970s, the term "California effect" highlights how jurisdictions with high standards can drive regulatory shifts upwards. The strictest regulations for many industries enacted by the State of California has exactly this kind of positive effect within the United States market.

digital, and environmental standards.[12]

Although Prof. Bradford makes a compelling case on behalf of the EU regulatory powers, perhaps this view is too optimistic. In the case of the EUSA in particular, it is too early to discern what its true transnational effect may be.

3. *The EU draft Space Code of Conduct*

In order to gain context, it is useful to move one step back in history. The EUSA is not the first initiative of the EU in the areas of space regulation and space governance.

Between 2010 and 2014, the EU prepared several versions of an International Code of Conduct for Outer Space Activities. This draft Code of Conduct (CoC) was presented in COPUOS and other international fora, and its adoption was proposed to all spacefaring nations at the time.[13]

A number of themes that are now found in the EUSA were already included in the proposed CoC. They relate mostly to the first and the third pillar of the proposed EUSA Regulation, i.e. safety and sustainability.

Already the Preamble of the Code made clear that sustainability was one of the main considerations of the initiative: "In order to safeguard the continued peaceful and sustainable use of outer space for current and future generations."[14]

And in terms of both safety and sustainability, the Preamble further stated: "Taking into account that space debris affects the sustainable use of outer space, constitutes a hazard to outer space activities and potentially limits the effective deployment and utilisation of associated outer space capabilities".[15]

For that reason, the EU was "[C]onvinced that a multilateral code of conduct aimed at enhancing the safety, security, and sustainability of outer space activities could become a useful complement to international law as

[12] https://geopolitique.eu/en/articles/the-european-union-in-a-globalised-world-the-brussels-effect/.

[13] The last version of the EU Space Code of Conduct was presented in 2014. It can be found at: https://www.eeas.europa.eu/sites/default/files/space_code_conduct_draft_vers_31-march-2014_en.pdf.

[14] CoC, Preamble at 1.

[15] CoC, Preamble at 8.

it applies to outer space."[16] The text further stated: "The purpose of this Code is to enhance the safety, security, and sustainability of all outer space activities pertaining to space objects, as well as the space environment."[17] In order to secure these three goals, "[T]he Subscribing States resolve to establish and implement policies and procedures to minimise the risk of accidents in space, collisions between space objects, or any form of harmful interference with another State's peaceful exploration, and use, of outer space."[18]

It is no coincidence that, according to its own words, the proposed CoC gravitated around the aforementioned three objectives, since those objectives largely correspond to the three "pillars" that compose the current draft EUSA: safety, resilience (which has largely replaced security, although arguably the two still have some overlapping aspects), and sustainability.

Of course, the CoC included a number of other aspects that are not repeated in the current draft EUSA, because the EUSA has a very different nature. The EUSA is not conceived as an international non-binding legal instrument, as the CoC was, but instead it is proposed as a binding piece of regulation for the internal EU market.

For instance, the Code established the adoption by States of transparency and confidence-building measures "with the aim of enhancing mutual understanding and trust, helping to prevent confrontation and foster national, regional and global security and stability."[19] Transparency and confidence-building measures are a typical instrument of international diplomacy that have no meaning in a national or, in this case, a supranational piece of legislation. In the context of the EU CoC, transparency and confidence-building measures were meant as a multilateral way to address the issues of trust and security of space activities, and to ensure that outer space would remain as a domain for exclusively peaceful activities.

Another goal of the CoC as stated in its text was to promote the universal adoption and adherence to the main legal and political regulatory instruments concerning outer space activities, such as the UN outer space treaties and the ITU regulations. This aspect is also not repeated now, as the EUSA is an internal EU Regulation and does not encroach on the

[16] CoC, Preamble at 10.

[17] CoC, Purpose and Scope at 19.

[18] CoC, Measures on Outer Space Operations and Space Debris Mitigation at 49.

[19] CoC, Purpose and Scope at 21.

foreign policy of the EU Member States.

Still, due to its particular importance, the 1967 Outer Space Treaty[20] is specifically mentioned in the Explanatory Memorandum that accompanies the text of the EUSA. It is relevant noting here that at present, after the accession of Latvia to the Treaty in May 2025, all 27 EU Member States are parties to the Outer Space Treaty, which is generally considered the cornerstone of international space law.[21]

Eventually, the CoC presented by the European Union did not receive enough international support and was shelved. But even though it was not adopted, it still constitutes a clear precedent when analyzing the current EUSA.

4. *Dark and Quiet Skies in the EUSA*

What has been the impact so far of the proposed initiative of the EUSA, and what will be the foreseeable repercussions? One can try to find out by selecting one particular aspect of the draft EUSA, and examining in detail how it has fared so far. The particular aspect selected for this purpose has been included in the Act as part of the sustainability pillar, and it pertains to a very recent problem that has arisen due to the use of very large constellations of satellites, or "mega-constellations". We are referring to the light pollution and radio pollution that is caused by megaconstellations, an unexpected impact that has brought about the novel issue of the protection of dark and quiet skies.[22]

A number of people, including the present author, were consulted by the EU Commission back in 2023 through personal interviews that were conducted remotely. As a result of those consultations, the Commission

[20] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies, opened for signature in January 1967, entered into force in October 1967. https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introouterspacetreaty.html.

[21] UNOOSA, Status of International Agreements relating to activities in outer space as at 1 January 2026, UN Doc. A/AC.105/C.2/2026/CRP.9/Rev.1, available at: https://www.unoosa.org/res/oosadoc/data/documents/2026/aac_105c_22026crp/aac_105c_22026crp_9rev_1_0_html/AC105_C2_2026_CRP09Rev01E.pdf.

[22] On the issue of dark and quiet skies there is already an abundant literature. See e.g. the Report of the IISL Working Goup on Light Pollution of the Night Sky from a Space Law Perspective, which was coordinated and edited by the present author in 2023. Available at: https://iisl.space/iisl-working-group-reports/.

was recommended by many interviewees to include the issue of the protection of dark and quiet skies in the EUSA proposal.

A preambular paragraph was therefore included in the preliminary section of the text: "(65) Due to increased orbital traffic, astronomers encounter light and radio frequency disruptions in their astronomical campaigns. Such disruptions have a direct impact on research and planetary defence capabilities. As a result, mitigation measures should be developed to protect the dark and quiet sky."[23]

And then a whole provision was included in the text, originally numbered as Article 72:

> "1. Union spacecraft operators shall establish a plan containing measures that are adequate to limit light and radio pollution in accordance with paragraph 2.
>
> 2. The visual magnitude for spacecraft during the entire lifetime, including the design requirements on low reflectivity coating or shielding, shall be at least 7 magnitude.[24]
>
> The plan referred to in paragraph 1 shall include all of the following elements:
>
> (a) a description of the technical and operational measures implemented by the Union spacecraft operator to reduce the visible brightness of the spacecraft and to minimise the impact of satellites on astronomical observations;
>
> (b) a description of the technical and operational measures implemented by the Union spacecraft operator to limit disruptions for radio astronomy observatories and to minimise the impact of satellites on astronomical observations."[25]

During the public consultation period that was opened in 2025 when the European Commission requested for public feedback on the draft EU Space Act, there were many different reactions to many aspects of the text. The reactions to the proposed rules on dark and quiet skies were typical or representative of the reactions elicited by the text as a whole.

The American Astronomical Society showed an enthusiastic support for

[23] https://eur-lex.europa.eu/resource.html?uri=cellar:c7906dd7-5283-11f0-a9d0-01aa75ed71a1.0001.02/DOC_1&format=PDF.

[24] This paragraph is actually an erroneous formulation; it should say instead "shall not exceed the 7th magnitude." This mistake was spotted and pointed out by many commentators during the public consultation process.

[25] https://eur-lex.europa.eu/resource.html?uri=cellar:c7906dd7-5283-11f0-a9d0-01aa75ed71a1.0001.02/DOC_1&format=PDF.

EUSA: "On behalf of over 8500 members of the American Astronomical Society, we are writing to express our strong support for provisions in the EU Space Act related to the protection of dark and radio-quiet skies."

The comment from the AAS, dated in Washington, D.C. in July 2025, further stated that: "We are encouraged by this legislation's recognition of the growing challenges that satellites pose for the natural night sky and for astronomical observations… The protections for dark and quiet skies within the EU Space Act represent a critical step forward in balancing commercial innovation with responsible stewardship of our shared space environment."[26]

The US Government, however, adopted a very different, much more critical attitude. The State Department's Comments on the Proposed EU Space Act document were released on 4 November 2025. The document was prepared jointly by the Departments of State and Commerce with the input of more than 70 US space companies and trade associations that have a commercial interest in the European market.[27]

Much of the US Government's criticism centered around the addition of new regulatory requirements, particularly in the area of environmental protection, which it warned would "drive up costs" and therefore would be detrimental to business. It also accused the proposal of being a "regulatory overreach" and further stated: "As a general matter, the United States expresses deep concern regarding measures in the proposed Act that would impose unacceptable regulatory burdens on U.S. providers of space services to European customers."[28]

On the dark and quiet skies issue, the US Government said: "Some of the proposed obligations, such as the requirement to limit the visual brightness of spacecraft to 7th magnitude at all times, are technically impossible to implement consistent with other important mission objectives (despite ongoing industry efforts to mitigate satellite reflectivity)."

It also said that "The United States firmly believes that regulation will not spark the innovation needed to solve the reflectivity concern. Technology neutral rules will support innovative solutions and commercial industry better than costly compliance measures to include certain technological requirements like the reflectivity thresholds for satellites."

[26] https://compasse.aas.org/aas-comment-on-the-proposed-european-union-space-act/.

[27] https://space.commerce.gov/eu-space-act-update-april-2026/.

[28] See the full text of the US comments at: https://cdn.table.media/assets/europe/090166e52487d15d.pdf.

The comments added that "[S]uch a regulation could also disproportionately affect U.S. companies who operate at lower altitudes as compared to the major EU-based companies who tend to operate at higher altitudes." The US constellation *Starlink* in particular operates at very low earth orbit and thus would be particularly affected by the brightness limit requirement.

The submission, therefore, requested that the future Act recognize US licensing and avoid "burdensome" duplication of requirements. If a constellation has already received authorization from the US government, this license should be automatically recognized by third parties such as the EU. Otherwise, the EU legislator will be exceeding its territorial limits and attempting to project its influence on other jurisdictions by imposing extra requirements.

In brief, a number of rules included in the EUSA, including in particular the requirements proposed for the protection of dark and quiet skies, have been considered a) Impossible to implement; b) Inefficient and costly; c) A protectionist measure; and d) A regulatory overreach.

But is truly the EUSA a regulatory overreach? It has been noted elsewhere that this extraterritorial effect of national space laws is not unusual. For instance, in the US, the Federal Communications Commission applies many of its rules both to American companies and to foreign companies seeking access to the US market. This included the FCC's decision in 2022 to reduce the time licensees would have to deorbit their satellites after the end of their missions from 25 years to just 5 years.[29]

In any case, in part due to the criticism above, new versions of the draft EUSA have modified significantly the original draft rules, and in particular those rules that were intended to protect dark and quiet skies. The issue has not disappeared from the draft, but the rules that were initially proposed have now been made more "light-touch" than the original ones.

In the case of former Article 72, the 7th magnitude limit has been eliminated, because of its disproportionate impact on US constellations and its supposed impossibility to maintain. The other requirement that was imposed on all constellations still remains in the proposal: This is the need to present a plan detailing the measures adopted to protect dark and

[29] Jeff Foust, "The long arm of a European space law", The Space Review, December 8, 2025. https://www.thespacereview.com/article/5117/1.

quiet skies.[30]

If the EUSA is adopted as it is now, the proposed rule on dark and quiet skies would align EU legislation with the requirements that are already in place and currently met by US operators of large satellite constellations. The FCC is requesting that each US operator reach a coordination agreement with the scientific community aimed at mitigating the impact of the respective satellite constellation to ground-based astronomical observatories.

5. *Conclusions*

The road to safety, resilience, and sustainability in outer space is paved with obstacles.

One lesson that can be learned from these two experiences: the international CoC proposed by the EU 15 years ago, and the current pitfalls encountered by the proposed EUSA, is that outer space is an international area, and as such, it is better regulated with the agreement and acquiescence of all the nations and stakeholders involved in its exploration and utilization.

Another important lesson that can be extracted is that regulating space matters is always going to be hard, and that in order to succeed, one must be patient and persistent. The EU is to be commended for its persistence in this effort.

Finally, it is fortunate that the protection of dark and quiet skies has not disappeared from the EUSA. Far from being just an excuse for protectionism, or a case of overreach, this is a real problem and a legitimate concern that will require the adoption of international regulation at some point.

[30] European Parliament, Legislative Train, May 2026: "As for balancing sustainability with proportionality, the draft report provides for an amendment of Article 72 on light and radio pollution that would keep the objective and the technical description required to the applicant, but without the obligation consisting in guaranteeing that 'the visual magnitude for spacecraft during the entire lifetime, including the design requirements on low reflectivity coating or shielding, should be at least 7 magnitude'." Available at: https://www.europarl.europa.eu/legislative-train/carriage/eu-space-law/report?sid=10301.

Alfonso Giordano

Orbital Geographies of EU Space Governance: Single Market, Strategic Autonomy and a Core–Periphery Landscape

Summary: 1. Introduction – 2. EU Space Law, the Draft EU Space Act and the "Single Market for Space" – 3. EU Space Policy, Strategic Autonomy and Industrial Policy – 4. New Space, Commercial Regulation and Orbital Sustainability – 5. Applying the Quadruple Filter to EU Space Governance – 6. Conclusion.

1. *Introduction*

1.1 *Research Questions and Thesis: Orbit as a Spatial and Governance Problem*

This chapter approaches European Union space policy from the standpoint of an economic-political geographer working on outer space, contributing to a law-and-policy volume on space governance.[1] Legal instruments - the Treaty on the Functioning of the European Union, the draft EU Space Act, the Regulation establishing IRIS², the Joint Communication on a European space strategy for security and defence - are treated throughout as materials for analysing governance trajectories, territorial and industrial geographies, and an emerging "space society" in orbit, rather than as the primary object of doctrinal legal analysis.[2] Where legal debates are unavoidable - competence under Articles 4(3), 114 and 189 TFEU, or the extraterritorial reach of the draft Space Act - they are presented concisely and functionally, in service of a broader argument about where orbit sits within the EU's institutional and territorial map,

[1] A. Giordano, *The New Space Age. Integrating Social Sciences and Humanities in Policy-Making*, LUISS CISS-SPHERE Policy Report, No. 1, Rome, 2025.

[2] Treaty on the Functioning of the European Union (TFEU); EU Space Act proposal, COM(2025) 335 final; Regulation (EU) 2023/588; JOIN(2023) 9 final.

and who gains or loses a voice in governing it.

The starting observation is that the EU now pursues two governance rationales for space that pull in different directions. The first is a single-market rationale: the draft EU Space Act proposes to harmonise the patchwork of national space laws that Member States have developed, relocating orbit into the institutional geography of the internal market.[3] The second is a strategic-autonomy rationale: EU space policy has progressively absorbed security and defence framing, from the Union Space Programme through to the 2023 Space Strategy for Security and Defence, treating orbital infrastructure - Galileo, Copernicus, and now IRIS² - as an instrument of European sovereignty rather than only of service delivery.[4] These two rationales imply different institutional geometries: Brussels internal-market procedures on one hand, and a security-institutional layer drawing in defence ministries and the External Action Service on the other. Both are superimposed on a pre-existing governance triangle of the EU, the European Space Agency, and the Member States that was designed for neither.[5]

A third element belongs to this picture from the outset, not as a later addition: orbit itself has changed. Falling launch costs and the entry of dominant commercial actors have densified low Earth orbit with smallsats and constellations at a pace and scale that European governance instruments did not anticipate.[6] Both the single-market template and the strategic-autonomy agenda are therefore being constructed for - and, in the case of IRIS², adding to - an orbital environment whose basic parameters of density and occupation are no longer ones the EU controls or can easily reshape. Framing orbit as a "contested Global Space" - one with non-exclusive access, escalating rival uses, and governance still in formation - captures this structural condition more precisely than the language of

[3] Cfr. Section 2 below; B. Bartóki-Gönczy, K. Malinowska, *Paradigm Shift in the European Union's Space Policy: Institutional Restructuring and Its Possible Consequences for the CEE Region*, in *Frontiers in Political Science*, 7, 2025, art. 1536170.

[4] Regulation (EU) 2021/696; JOIN(2023) 9 final; cfr. G.G. Müller, P. De Man, *Incremental Shifts, Strategic Orbits: The Evolution of EU Space Policy Through Gradual Security Linkages*, in *Global Policy*, 29 January 2026, pp. 1-13.

[5] Cfr. P. De Man, J. Wouters, *EU Space Governance at the Threshold of a New Era*, in *Global Policy*, 28 May 2025.

[6] E. Marcucci, V. Gatta, C. Ciucci, D. Cucina, *When Launch Costs Fall, Do Satellites Shrink and Constellations Grow? The Role of SpaceX in the SmallSat Revolution*, in *Research in Transportation Business & Management*, 67, 2026, 101663.

commons alone.[7] This chapter asks how the EU attempts to reconcile a single-market template for space with a strategic-autonomy trajectory, in an orbital environment increasingly crowded and commercially dominated by non-EU actors, and what tensions this reconciliation generates, for whom, and with what prospects for resolution. In doing so, it situates an economic-political geography of orbit within the broader comparative space-law framework of the volume, treating space governance as a problem of both institutional design and spatial inequality.

The thesis advanced here is that EU space governance is not a coherent strategy converging on a settled endpoint, but an unresolved compromise between these rationales, pursued through overlapping instruments and an institutional architecture inherited from an earlier, less contested period of European space cooperation.[8] This compromise is most visible along three fault lines that recur throughout the chapter and structure its conclusion: a competence-and-legitimacy fault line, concerning where authority over orbit resides among EU institutions, ESA and Member States; a core-periphery fault line, concerning who benefits from - and who bears the adjustment costs of - a single market and an autonomy agenda whose industrial geography remains concentrated; and an autonomy-versus-sustainability fault line, concerning whether EU-controlled infrastructure deployed in pursuit of autonomy is compatible with the EU's own commitments to a sustainable orbital environment.[9] None of these fault lines is new in the sense of being unrecognised in the literature; what this chapter offers is a way of holding them together, systematically, through the analytical lens introduced below.

1.2 *The Quadruple Filter, the Geographies of Outer Space and the "Space Society"*

To examine these tensions without collapsing them into a single verdict – "the Space Act is a good idea", "strategic autonomy is unrealistic" - this chapter adopts as its lens the "quadruple filter" developed within the CISS-SPHERE (Space Policies, Humanities and Exogeographical Research Ecosystem) Unit (LUISS) and presented at the 1st SPHERE

[7] S. Gstöhl, J. Larik, *The European Union and the Governance of Contested Global Spaces*, in *Journal of European Integration*, 45, n. 8, 2023, pp. 1107-1124, which identifies the EU as a latecomer in governing spaces of this kind.

[8] Cfr. De Man, Wouters, *EU Space Governance*, cit.

[9] Cfr. Section 5.5 and Section 6 below.

Space Policy Forum.[10] The filter poses four distinct questions about any space-policy option: what is technically feasible, given existing industrial, institutional and orbital constraints; what is socially desirable, and for whom; what is legally and ethically legitimate, in the sense of whether the claim to regulate is recognised as well-founded by the relevant actors; and what is systemically sustainable over time, given orbital density and institutional coordination demands.[11]

Two features distinguish this filter from a purely legal or purely industrial-policy reading, and both matter for how the chapter proceeds. First, it keeps the four questions analytically separate. A policy can be feasible and desirable while raising serious legitimacy concerns; it can be feasible, desirable and legally unproblematic while still being unsustainable. The draft EU Space Act and IRIS² - the two instruments around which much of the later analysis turns - are precisely cases where this separation does real work: the Space Act's harmonisation logic may be institutionally feasible while its legal basis remains contested (Sections 5.1 and 5.3), and IRIS² may be both feasible and desirable as a connectivity programme while sitting in tension with orbital-sustainability commitments (Section 5.4).[12] Second, the filter is applied here not to legal texts in isolation, but to a governance architecture operating in a populated orbital environment - the EU Space Act, the strategic-autonomy turn, the EU-ESA-Member States triangle, IRIS², the EU's space-situational-awareness capabilities, and the extension of horizontal cybersecurity instruments to space assets - considered together as a system that must function within, not above, the densifying orbit described in 1.1.[13]

This is where the chapter's second conceptual debt becomes important. The analysis here sits within the geographies of outer space, the field that over the past decade has established orbit and the extra-terrestrial as legitimate objects of geographical enquiry, and within their orbital register in particular: orbit treated as a space of settlement, infrastructure and governance, not merely as a technical domain to be regulated from a distance.[14] These orbital geographies form one pole of a wider Earth–

[10] Giordano, *The New Space Age*, cit., ch. 3.

[11] Ivi.

[12] Cfr. Sections 5.1, 5.3 and 5.4 below.

[13] Cfr. Sections 2-4 below.

[14] O. Dunnett, A.S. Maclaren, J.M. Klinger, K.M.D. Lane, D.J. Sage, *Geographies of Outer Space: Progress and New Opportunities*, in *Progress in Human Geography*, 43, n. 2, 2019, pp. 314-336, as a key survey of human-geography work on outer space, including orbit and extra-terrestrial environments, over the past decade.

orbit–planet continuum; the present chapter works on this orbital pole, leaving the continuum's extraterrestrial pole to exogeography, as developed elsewhere.[15]

Read through this lens, the social-desirability question can be reframed in terms of an uneven orbital ecumene: as constellations, debris-mitigation regimes and STM-/SSA-mediated traffic multiply, the capacity to operate safely and with adequate standing in orbit becomes itself an unevenly distributed good, independent of formal market-access rules, shaping how Section 5.2 examines who is positioned to participate in an increasingly occupied orbit and how Section 5.4 examines whether that occupation can be sustained. Exogeography – developed as the study of the continuum's extra-terrestrial pole, its planetary environments, habitable interfaces and prospective populations[16] – is invoked here only as a parallel frame at the points where orbital infrastructures act as interfaces toward that pole, most clearly at the "high boundary" examined in Section 4.2, where work on orbital thresholds situates itself between the two registers.

A third conceptual resource - comparative space law - enters not as an internal EU story but as a way of widening the frame against which the EU's single-market template is read. The proliferation of national space laws across Europe and beyond represents not regulatory disorder awaiting EU correction, but a set of genuinely different institutional models for organising the relationship between public authority, private operators, and an orbital environment that no national legal system fully controls.[17] Zolea's edited volume on comparative visions in space law, and Benvenuti

[15] G. Casagrande, *Luoghi nello spazio: alcune prime sfide dell'esogeografia di fronte alla presenza e alle attività umane al di fuori della Terra*, in *Semestrale di Studi e Ricerche di Geografia*, 32, n.2, 2021, pp. 37-50, as an early programmatic systematisation of exogeography; A. Giordano, *Orbital Thresholds: Exogeography and the Evolving Geopolitics of the High Boundary*, in *Geopolitica. Journal of Geopolitics and Related Matters*, XIV, n. 1, 2025, pp. 119-142, which reads the "high boundary" as a dynamic interface between Earth, orbit and the extra-terrestrial along an Earth–orbit–planet continuum.

[16] G. Casagrande, *Looking out of the Cradle. Possible Geographical Perspectives on the Beginning of Human Expansion in Outer Space*, in G. Casagrande, A. D'Ascenzo (eds.), *Oltre la Terra. Verso l'esogeografia*, Società Geografica Italiana, Rome, 2025, as an early geographical framing of the initial phases of human expansion beyond Earth; A. Giordano, *Human Migration, Settlements and Life in Outer Space: Some Initial Reflections on an Imagined Population Exogeography*, in G. Casagrande, A. D'Ascenzo (eds.), *Oltre la Terra, verso l'Esogeografia*, Società Geografica Italiana, Rome 2025, pp. 121-165, as an initial attempt to extend the categories of population geography to the extra-terrestrial pole of that continuum.

[17] Cfr. Section 2.2 below.

and Zolea's argument that legal comparison can help sketch new rules and institutions for an emerging space society, treating this diversity as a resource: a set of alternative legal models and institutional imaginaries against which any single template, the EU's included, can be evaluated rather than assumed to be the self-evident endpoint of harmonisation.[18] This framing matters directly for Section 5.3, where the legitimacy filter asks not whether the EU Space Act is doctrinally sound in isolation, but whether the EU's proposed template is likely to be recognised as legitimate against this wider landscape of models for an emerging space society.

1.3 *How the Chapter Proceeds*

The chapter moves from mapping legal and institutional geographies, through the strategic-autonomy and industrial-policy layer, to the New Space and infrastructural layers within which this governance architecture must now operate, before applying the quadruple filter explicitly to the whole.

Section 2 examines the draft EU Space Act as the centrepiece of the single-market rationale: how the choice of Article 114 TFEU relocates orbit institutionally, how the Act's harmonisation template relates to the diversity of existing national space laws, how its extraterritorial reach overlays a market-access geography onto the registration, and launch-based geographies of international space law, and how its harmonisation logic distributes costs and benefits across Member States with mature and emerging space sectors.[19] Section 3 turns to the strategic-autonomy rationale: the shift from civilian programmes toward a security-oriented agenda, the Draghi report's industrial-policy framing and the debate over a European Space Law, the evolving EU-ESA-Member States governance triangle, and the industrial and external geography of flagship infrastructures, including their extension through partnerships with emerging economies.[20]

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[18] S. Zolea (ed.), *Comparative Visions in Space Law: Legal Models, Private Law and Governance Beyond Earth*, Roma Tre Press, Rome, 2024; S. Benvenuti, S. Zolea, *Legal Comparison as a Tool to Sketch New Rules and Institutions for a Space Society*, in *International Journal for the Semiotics of Law*, 16 April (online), 2026.

[19] Cfr. Section 2 below.

[20] Cfr. Section 3 below; A. Giordano, *From Access to Agency: Digitalisation and Space Technology Partnerships between Europe and Emerging Economies for Sustainable Development*, in W. Reiners, S. Lucatello (eds.), *Digitalisation Towards Sustainable*

Section 4 addresses the orbital environment these rationales must operate within: the densification of orbit driven by falling launch costs and New Space actors, IRIS²'s position at the intersection of digital autonomy and orbital sustainability, the governance of the "high boundary" between airspace and outer space through Space and Air Traffic Management, and the extension of horizontal EU cybersecurity and critical-infrastructure instruments to space assets.[21] Section 5 then applies the quadruple filter to this material as a whole - feasibility, social desirability, legal and ethical legitimacy, systemic sustainability - before a cross-filter synthesis identifies the three fault lines that Section 6 takes up as the chapter's principal findings and open questions.[22] Throughout, the vantage point remains that of an economic-political geography of an emerging space society: orbit not as an abstract legal or technical space, but as a space of settlement, infrastructure and governance whose occupation is being negotiated, unevenly and incompletely, through the instruments examined here.

2. *EU Space Law, the Draft EU Space Act and the "Single Market for Space"*

2.1 *Orbit Inside the Internal Market: Competence and Institutional Geography*

From the standpoint of economic-political geography, the relevant question is less what the Treaties say about space competence than where this places orbit within the institutional map of the Union. Article 4(3) TFEU assigns research, technological development and space to what the literature describes as a "parallel" competence: the EU may act, especially through programmes, without preventing Member States from continuing to act on their own.[23] This formula codifies a pre-existing territorial arrangement rather than creating a new one, national agencies, national launch and ground infrastructure, and the intergovernmental European Space Agency, whose membership only partly coincides with

Development: Comparative Perspectives from Emerging Economies and Europe, Taylor and Francis, London (forthcoming, 2026).

[21] Cfr. Section 4 below.

[22] Cfr. Section 5 below.

[23] Treaty on the Functioning of the European Union (TFEU), Art. 4(3).

that of the EU.[24] Article 189 TFEU, the Treaty's dedicated space clause, sits comfortably within this arrangement: it speaks of "joint initiatives", research support and industrial competitiveness, but its second paragraph rules out "any harmonisation of the laws and regulations of the Member States", in effect fencing space off from the regulatory instruments through which the EU has integrated other sectors.[25]

The draft EU Space Act's reliance on Article 114 TFEU - the general internal-market harmonisation clause - therefore does more than swap one legal basis for another; it relocates orbit from one institutional geography to a different one. As De Man and Wouters note, the Commission had reached the limits of what programmatic action under Article 189 could deliver: the Union Space Programme and EUSPA, established by Regulation (EU) 2021/696, remain instruments that Member States and operators essentially opt into. It therefore turned to the instrument that has historically absorbed sectors such as telecommunications, financial services and product safety into the single market, a binding, horizontally applicable regulatory framework.[26] Bartóki-Gönczy and Malinowska read this as part of a wider "paradigm shift" in EU space governance, from space as a domain of coordinated national programmes to space as a domain of regulated economic activity, governed by the same internal-market logic that already applies to goods, services and data.[27]

The consequences of this shift are institutional before they are legal. Internal-market legislation under Article 114 is shaped through a specific configuration of actors: Commission directorates-general responsible for the single market and industry, the European Parliament and Council as co-legislators, and ultimately the Court of Justice as arbiter. Within this configuration, actors experienced in shaping EU internal-market rules - EU-level industry associations, Member States with strong Brussels-facing lobbying capacity - tend to carry more weight than the specialised, agency-to-agency diplomacy that has historically characterised ESA-centred cooperation.[28] Pulling orbit into the internal market is thus also a redistribution of regulatory voice: it changes who is well positioned to influence the rules governing an increasingly commercial and crowded orbital environment, independently of whether those rules

[24] Cfr. De Man, Wouters, *EU Space Governance,* cit.

[25] Treaty on the Functioning of the European Union, Art. 189(2).

[26] De Man, Wouters, *EU Space Governance*, cit.; Regulation (EU) 2021/696.

[27] Bartóki-Gönczy, Malinowska, *Paradigm Shift*, cit.

[28] Ibidem; Treaty on the Functioning of the European Union (TFEU), Art. 114(1) and (3).

are, in substance, good ones. Whether this reallocation is desirable, and for whom, is a question deferred to Section 5.2; what matters here is that the choice of Article 114 is itself a spatial and institutional choice, not merely a technical one.

This choice also leaves an underlying ambiguity unresolved, one that runs through the rest of the section. Article 114's harmonisation logic and Article 189(2)'s explicit exclusion of harmonisation point toward two different ways of conceiving orbit: as an extension of the internal market - a space of economic activity subject, like any other, to approximation of national laws - or as a strategic territory whose occupation and use Member States have so far been reluctant to fully delegate.[29] The draft EU Space Act does not choose between these framings; it attempts to hold both together, and it is in this attempted synthesis, as the next subsection shows, that the chapter's central tensions begin to take shape.

2.2 *National Models, Market Access and the Jurisdictional Geography of Orbit*

The draft Act is generally organised around three pillars - safety (authorisation, collision avoidance, end-of-life management of space objects), resilience and cybersecurity (protection of space systems and ground infrastructure), and environmental sustainability (the orbital footprint of missions and the use of orbital "slots" as a finite resource) - with the overarching aim of aligning or replacing the national authorisation regimes that a growing number of Member States have adopted.[30] For an economic-political geographer working on outer space, the interesting question is not whether these pillars are technically sound in isolation, but what kind of single template they propose for a domain that has, until now, been governed through a genuinely plural set of national models, models differing not in degree but in how each organises the relationship between national territory, jurisdiction and orbital activity.[31]

These national models embody distinct logics. France's framework was developed alongside a state that retains direct operational involvement in launch from its own territory (the Guiana Space Centre) and has long

[29] De Man, Wouters, *EU Space Governance*, cit.

[30] Cfr. Bartóki-Gönczy, Malinowska, *Paradigm Shift*, cit.; I. Bratu, F. von der Dunk, *The EU Space Act Proposal and Its Impact on the Dutch Space Ecosystem*, SSRN working paper, version 11 August 2025.

[31] Zolea (ed.), *Comparative Visions in Space Law,* cit.

treated space capability as a component of national strategic autonomy; its law reflects a model in which the state's territorial reach extends quite literally into the act of launching.[32] Luxembourg's legislation, by contrast, was designed to attract internationally mobile commercial activity - initially space resource utilisation - through a predictable, operator-friendly registration regime; it functions less as territorial control over launch than as a jurisdictional "anchor", in a manner reminiscent of flag registries in maritime law.[33] The Netherlands and other smaller Member States with national laws but limited domestic launch capability represent a third model, governing primarily the licensing of operators established on national territory but whose activities and value chains extend well beyond it.[34] Italy's more recent move (Law No. 89/2025) adds a further variant: a Member State with an established upstream industrial base formalising a national framework relatively late, in parallel with the emergence of an EU-level one.[35]

The comparative-law literature treats this diversity not as a problem awaiting correction but as evidence of legal systems experimenting with different answers to a shared question: how to draw the boundary between public authority and private operators, and between national legal order and an activity that is inherently transnational and infrastructurally mediated.[36] Benvenuti and Zolea go further, suggesting that the proliferation of models reflects an ongoing search for institutional forms adequate to an emerging "space society", one that does not map neatly onto inherited categories of public/private or domestic/international law.[37] The comparative private-law perspective underpinning the present volume is developed at monographic length by its editor.[38]

[32] Bratu, von der Dunk, *The EU Space Act Proposal*, cit.

[33] Zolea (ed.), *Comparative Visions in Space Law*, cit. On the characterisation of national space objects as "islands of sovereignty" extended into orbit - and of mega-constellations as "archipelagos of sovereignty" occupying considerable swathes of the space commons - see A. Marino, T. Cheney, *Centring Environmentalism in Space Governance: Interrogating Dominance and Authority Through a Critical Legal Geography of Outer Space*, in *Space Policy*, 63, 2023, 101521.

[34] Bratu, von der Dunk, *The EU Space Act Proposal*, cit.

[35] Cfr. T. Graziani, S. Zolea (eds.), *Verso una legge italiana sullo spazio*, Callive/Mediabooks, Rome, 2025; Legge 13 giugno 2025, n. 89. Disposizioni in materia di economia dello spazio (25G00095).

[36] Zolea (ed.), *Comparative Visions in Space Law*, cit.

[37] Benvenuti, Zolea, *Legal Comparison as a Tool to Sketch New Rules,* cit.

[38] S. Zolea, *Comparative Space Law: The Space Frontier from a Private Law Perspective*, in

Seen in this light, the EU Space Act's harmonisation pillar is best understood not as the imposition of order on an untidy legal landscape, but as the proposal of one governance model among several for a shared and increasingly occupied orbit: a model whose default orientation - common substantive standards, a harmonised authorisation procedure, mutual recognition across Member States - is that of the single market.[39] The outer space geographical stakes are considerable. At the very moment when orbit is becoming more occupied - by satellites, constellations, debris and dependent infrastructures - the Act proposes a single template for how European legal systems will relate to that occupation, applied across a France oriented toward sovereign launch, a Luxembourg oriented toward jurisdictional attractiveness, a Netherlands oriented toward operator licensing, and an Italy formalising its own framework in parallel. Whether one template can accommodate this range of territorial relationships to orbit, without simply privileging the model closest to the EU's own internal-market default, is a question the chapter carries forward into Section 5.

The draft Act's envisaged application to operators not established in the EU - but providing satellite communications, Earth observation or positioning services to EU-based customers - extends the question of "who relates to orbit, and how" beyond the Union's territory, into the global geography of the space economy.[40] Baumann and Prall situate this within the now-familiar pattern of the "Brussels effect": the tendency of EU market-access conditions, given the size of the EU market, to function as de facto global standards adopted by non-EU operators even for activities registered or conducted elsewhere, as already seen in data protection and digital services regulation.[41]

What is distinctive about applying this pattern to space is the way it overlays a third spatial logic onto an already layered map of orbital jurisdiction. The 1967 Outer Space Treaty and the 1975 Registration Convention allocate jurisdiction over space objects, and international responsibility for space activities, according to where an object is registered and which state authorises or supervises its launch, categories anchored to

Studies in Space Law, 25, Brill Nijhoff, Leiden, 2025.

[39] Bartóki-Gönczy, Malinowska, *Paradigm Shift*, cit.; Bratu, von der Dunk, *The EU Space Act Proposal*, cit.

[40] I. Baumann, K. Prall, *The EU Space Act's Global Reach: Legal Considerations on the Envisaged Application to Non-EU Operators*, in *Global Policy*, 2025, pp. 1-8.

[41] A. Bradford, *The Brussels Effect: How the European Union Rules the World*, Oxford University Press, Oxford, 2020.

specific national territories and launch infrastructures.[42] An EU market-access condition follows a different geography altogether: it is anchored to where services are consumed, independently of where the underlying object is registered or launched.[43] For a non-EU constellation registered and launched outside the Union but serving European customers, the geography of compliance (where it sells services) no longer coincides with the geography of registration and launch (where it is built, registered and operated), three overlapping but non-identical maps of the same orbital activity.[44]

Crucially, this is not only a "non-EU operator" issue. Bratu and von der Dunk's analysis of the Dutch ecosystem shows that EU-based operators are themselves frequently nodes within transnational value chains, with satellites manufactured, launched, insured and operated across multiple jurisdictions and the EU-based entity representing one point in a wider international structure.[45] For such operators, an EU framework that adds market-access requirements on top of the registration- and launch-based obligations they already meet elsewhere does not simplify their position but adds a further layer to an already multi-jurisdictional geography of compliance.[46] This layering - registration geography, launch geography, and now market-access geography, each with its own logic and each potentially generating its own obligations - is precisely what Section 5.3 examines, in asking whether the EU's attempt to project its standards along the geography of market access can be absorbed by the broader system of orbital governance without generating duplicative or conflicting requirements for the operators who, in practice, build, launch and service the infrastructure that populates orbit.

2.3 *Core, Periphery and Uneven Access to Orbit*

The Act's harmonisation logic also has an internal geography, one that maps onto - and may reshape - existing core-periphery patterns within the EU's space economy. On one side are Member States with mature space

[42] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (Outer Space Treaty), 1967; Convention on Registration of Objects Launched into Outer Space, 1975.

[43] Baumann, Prall, *The EU Space Act's Global Reach*, cit.

[44] Ibid.

[45] Bratu, von der Dunk, *The EU Space Act Proposal*, cit.

[46] Bartóki-Gönczy, Malinowska, *Paradigm Shift*, cit.

sectors and, in several cases, established national space laws: France, with sovereign launch infrastructure and a large industrial base; Germany and Italy, with significant upstream manufacturing capacity; the Netherlands, with a more service and operator-oriented sector; Luxembourg, with its jurisdictionally specialised registration model. These are the Member States whose regulators, industries and legal traditions have the longest track record of engagement with the questions the Act now addresses at EU level, and whose operational realities are, almost by default, most likely to be reflected in the substantive content of the resulting EU-wide standards.

On the other side are the CEE Member States and other relative latecomers, whose space sectors - where they exist - tend to be smaller, more recently established, and oriented less toward upstream capacity (launch, large-scale manufacturing) than toward participation in EU-level programmes: Copernicus and Galileo downstream applications, Horizon Europe research funding, Cohesion Policy support. Bartóki-Gönczy and Malinowska frame the position of these states in genuinely two-sided terms. A harmonised EU framework could lower their barriers to entry by removing the need to build costly national licensing regimes from scratch, allowing smaller administrations to operate directly under an EU-level framework rather than developing specialised legal and technical capacity domestically. But if that framework is substantively shaped by the realities and industrial interests of Member States with mature, upstream-oriented sectors, it risks fitting awkwardly with the downstream-oriented, applications-focused profile that characterises much of the CEE space economy, reproducing in regulatory form a core-periphery pattern that already exists in the industrial geography of European space activity.[47]

This is, at root, a distributive question, about who benefits from, and who bears the adjustment costs of, a single regulatory template applied across structurally different national space ecosystems, and about whether the resulting distribution reinforces or helps to rebalance existing patterns of core and periphery.[48] It is also, from the standpoint of economic-political geography, a question about participation in an increasingly populated orbit, and thus about the emerging social geography of a "space society" rather than market access alone.[49] The legal-institutional architecture traced across this section - the relocation of orbit into the internal market

[47] Ibid.

[48] Cfr. De Man, Wouters, *EU Space Governance*, cit.

[49] Cfr. Giordano, *Orbital Thresholds*, cit.; Casagrande, *Looking out of the Cradle,* cit.

via Article 114 (§2.1), the EU template proposed as one governance model among several (§2.2), the layered geographies of registration, launch and market access brought into contact by the Act's extraterritorial reach (§2.2), and the core-periphery distribution this generates (§2.3) - provides the material that Section 5 revisits through the quadruple filter, from the standpoint of an economic-political geography of orbit as a space of settlement, infrastructure and governance.

3. *EU Space Policy, Strategic Autonomy and Industrial Policy*

3.1 *From Civilian Programmes to Strategic Autonomy: Securitisation and the Industrial-Policy Turn*

If Section 2 traced how the EU Space Act relocates orbit into the institutional geography of the internal market, this section traces a parallel - and not entirely compatible - relocation: the gradual incorporation of space into the EU's security and defence agenda. For most of its history, EU-level space activity was organised around civilian, scientific and applications-oriented programmes - Galileo for positioning, Copernicus for Earth observation - whose governance, while never entirely apolitical, was framed in terms of public services, scientific cooperation and industrial competitiveness rather than security.[50] Regulation (EU) 2021/696, establishing the Union Space Programme and EUSPA, consolidated this programme-based architecture under a single regulatory umbrella while still framing the Union's role primarily as that of a programme owner and funder operating alongside Member States and ESA.[51]

The Joint Communication on a "European Union Space Strategy for Security and Defence" (JOIN(2023) 9 final) marks a more explicit shift.[52] Müller and De Man describe this not as a sudden rupture but, in the language of their article's title, as the product of gradual security linkages: a process by which security considerations, initially attached to civilian programmes as secondary objectives (the resilience of Galileo's signal, dualuse applications of Copernicus data), progressively become organising principles in their own right, reshaping how new initiatives are

[50] Regulation (EU) 2021/696.
[51] Regulation (EU) 2021/696.
[52] JOIN(2023) 9 final.

framed from the outset.[53] Cellerino's analysis of the "security and defence dimension of the Union in space" situates this within a broader pattern in EU integration, in which competences initially developed for civilian purposes are incrementally extended into security-relevant domains, often without a parallel extension or clarification of the EU's formal competence to act in defence matters.[54]

For an economic-political geographer, what matters about this shift is less its normative framing than its spatial and institutional implications. A space policy organised around civilian programmes operates largely through the institutional geography already familiar from Section 2: EUSPA, the Commission's space directorates-general, and a programmatic relationship with ESA and national agencies.[55] A space policy organised around security and defence draws in a different set of actors - the European External Action Service, Council formations dealing with foreign affairs and defence, and national defence ministries - whose involvement in space governance was historically marginal.[56] The strategic-autonomy turn is therefore not simply a change in the goals of EU space policy; it changes who governs orbit from Brussels, adding a security-institutional layer onto the internal-market layer of Section 2[57], with consequences for both the coherence and the long-term sustainability of the resulting architecture that Section 5.4 takes up directly.

The Draghi report's recommendations on defence and space, as analysed by Miglio and Nota, gave this security-oriented turn a more explicit competitiveness framing. European strategic autonomy in space is presented not only as a matter of security in the narrow sense, but as a question of whether European industry can remain a credible, autonomous supplier of space capabilities - launch, satellites, ground infrastructure - in a global market increasingly dominated by non-European, and particularly US, commercial actors.[58] Miglio and Nota read the report as calling for a more centralised, EU-level approach to

[53] Müller, De Man, *Incremental Shifts, Strategic Orbits,* cit.

[54] C. Cellerino, *EU Space Policy and Strategic Autonomy: Tackling Legal Complexities in the Enhancement of the 'Security and Defence Dimension of the Union in Space'*, in *European Papers*, 8, n. 2, 2023, pp. 487-501.

[55] Cfr. Regulation (EU) 2021/696.

[56] Cfr. JOIN(2023) 9 final; Müller, De Man, *Incremental Shifts, Strategic Orbits*, cit.

[57] De Man and Wouters describe this as a transition "from a regulatory power in space to a strategic space actor"; cfr. De Man, Wouters, *EU Space Governance*, cit.

[58] A. Miglio, C. Nota, *Implementing the Draghi Report's Recommendations on Defence and Space*, Policy Paper n. 64, Centro Studi sul Federalismo, Torino, 2024.

space industrial policy, capable of mobilising investment, consolidating fragmented national industrial bases, and supporting capabilities - heavy launch, secure connectivity, in-orbit servicing - that no single Member State's industrial base can sustain alone.[59]

The subsequent debate over a "European Space Law" - distinct from, but related to, the EU Space Act of Section 2 - should be read in this light: as a proposal to extend the EU's regulatory and industrial-policy reach into areas (defence-related space capabilities, dual-use technologies, critical space infrastructure) that the internal-market-oriented Space Act, anchored in Article 114 TFEU, was not designed to address directly.[60] Dalledonne, Dousset and Aliberti frame this as a response to a structural problem in the geography of European space industry: production capacity, supply chains and critical know-how remain distributed across national industrial bases of very different sizes, with limited EU-level mechanisms beyond programme funding to coordinate or consolidate this capacity in line with strategic-autonomy objectives.[61]

From a territorial perspective, the report's framing and the European Space Law debate it has fed represent an attempt to add a further institutional layer onto the architecture mapped so far: alongside the internal-market layer (Section 2) and the emerging security-institutional layer just described, a more explicitly industrial-policy-oriented layer aimed at consolidating European industrial capacity in ways that national industrial policies - however well-resourced in France, Germany or Italy - cannot achieve alone.[62] Hainaut's analysis of "European autonomy in orbit" suggests that the credibility of this layer depends on whether it can mobilise genuinely EU-level resources and decision-making, rather than functioning as a coordinating label for what remain, in practice, nationally funded and nationally controlled industrial programmes, a question taken up when assessing technical feasibility in Section 5.1.[63]

[59] Ibid.

[60] Cfr. De Man, Wouters, *EU Space Governance,* cit.

[61] S. Dalledonne, J. Dousset, M. Aliberti, *New Challenges for European Space Industrial Policy*, in *Global Policy*, 2 June 2025.

[62] Miglio, Nota, *Implementing the Draghi Report's Recommendations*, cit.; Dalledonne, Dousset, Aliberti, *New Challenges*, cit.

[63] B. Hainaut, *European Autonomy in Orbit*, International Centre for Defence and Security (ICDS), EU Defence Series n. 5, Tallinn, 2026.

3.2 *The EU–ESA–Member States Governance Triangle*

The layering described in §3.1 - internal market, security institutions, industrial policy - does not occur on a blank canvas. It is superimposed onto a pre-existing governance triangle formed by the European Union, the European Space Agency, and the Member States, each retaining distinct roles, instruments and constituencies, and each occupying a different position in the institutional geography of "Brussels, ESA, national capitals" that has long characterised European space governance.[64]

The three corners of this triangle operate by different logics. ESA, an intergovernmental organisation predating the EU's own space competence and with a membership extending beyond the EU (including the United Kingdom, Norway and Switzerland), works through optional programmes to which Member States subscribe individually, with funding and decision-making weighted by subscription level, a geometry that does not map onto EU procedures and that gives ESA's larger contributors (notably France, Germany and Italy) influence independent of EU-level mechanisms. The EU, through EUSPA and the Commission, brings a different logic: EU-level funding (the Union Space Programme budget, Horizon Europe, and, for IRIS², the Union Secure Connectivity Programme established by Regulation (EU) 2023/588) tied to EU decision-making procedures, in which all Member States participate as co-legislators regardless of their level of space-industrial development.[65] Member States, finally, continue to run national programmes, national agencies and - as discussed in Section 2 - in several cases national space laws, retaining a layer of activity that neither ESA nor the EU fully absorbs.[66]

De Man and Wouters discuss this as a 'governance triangle' between the EU, ESA and the Member States, whose internal balance is shifting unevenly. EU-level instruments are expanding in scope, from programme funding toward regulation (as the EU Space Act exemplifies) and toward security framing (as JOIN(2023) 9 exemplifies). ESA, meanwhile, retains functions, particularly those requiring optional, variable-geometry funding that EU procedures do not easily replicate, such as exploration or capability-development programmes that not all Member States wish

[64] De Man, Wouters, *EU Space Governance*, cit.

[65] Regulation (EU) 2021/696; Regulation (EU) 2023/588 of the European Parliament and of the Council of 15 March 2023 establishing the Union Secure Connectivity Programme for the period 2023-2027.

[66] Cfr. Section 2 above, and sources cited therein.

to fund equally, that an EU-centred architecture would struggle to absorb without significant institutional redesign.[67] Müller and De Man's account of "gradual security linkages" can be read, in this light, as also describing a gradual redistribution within the triangle: as EU-level instruments increasingly carry security framing, the EU's institutional weight grows relative to ESA's, even where the underlying capabilities - launchers, satellites, ground infrastructure - continue to be developed through industrial bases and ESA programmes that predate this shift.[68]

This redistribution connects directly to Section 2. The choice of Article 114 as the legal basis for the EU Space Act (§2.1) and the EU's expanding role within the governance triangle are two facets of the same movement: the Brussels institutions acquiring a more central position in the governance of orbit, relative to both ESA's intergovernmental geometry and Member States' national frameworks.[69] Whether this more centralised configuration is institutionally sustainable - whether it can function coherently given ESA's continued existence, its non-EU membership, and Member States' continued operation of national programmes and laws - is a question deferred to Section 5.4, where the systemic-sustainability filter is applied not only to the orbital environment but to the governance architecture meant to manage it.

3.3 *Strategic Autonomy, Flagship Infrastructures and the External Geography of EU Space Policy*

The strategic-autonomy and industrial-policy turn finds its clearest material expression in the EU's flagship space infrastructures. Galileo and Copernicus, initially framed as civilian programmes providing positioning and Earth observation services, are increasingly described - including in JOIN(2023) 9 - as components of European digital sovereignty and resilience, whose continued autonomous operation, independent of non-EU positioning and observation systems, is treated as a strategic objective in its own right.[70] The geopolitical geographies of outer space, read from a specifically European vantage point, frame these infrastructures

[67] De Man, Wouters, *EU Space Governance*, cit.

[68] Müller, De Man, *Incremental Shifts, Strategic Orbits*, cit.

[69] Cfr. Section 2.1 above.

[70] JOIN(2023) 9 final; Regulation (EU) 2021/696.

as spatial configurations of power as much as of service provision.[71] IRIS², established by Regulation (EU) 2023/588 as the Union Secure Connectivity Programme for 2023–2027, extends this logic into a newer domain - secure satellite connectivity - explicitly framed as a response to the EU's dependence on non-EU satellite connectivity infrastructure, a dependence that the launch-cost-driven proliferation of constellations discussed in Section 4 has made increasingly visible.[72]

These flagship infrastructures are not evenly distributed achievements of "European" capability in any simple sense. They are assembled through supply chains and industrial consortia in which a relatively small number of Member States - the "core" of the EU's space economy identified in Section 2.3, notably France, Germany and Italy - host the prime contractors, system integrators and much of the upstream manufacturing capacity, while other Member States participate primarily through downstream applications, ground-segment contributions or EU-level funding.[73] Dalledonne, Dousset and Aliberti suggest that the industrial-policy dimension of strategic autonomy, insofar as it seeks to consolidate this base, risks reinforcing rather than rebalancing this distribution, since consolidation tends in practice to favour already-established prime contractors and supply chains over the development of new ones in less central Member States.[74] This is precisely the tension the social-desirability filter in Section 5.2 examines: strategic autonomy and industrial consolidation may be desirable for European competitiveness as a whole while simultaneously reinforcing the core-periphery pattern of §2.3, a tension between two scales at which "desirability" can be assessed, the EU-level and the intra-EU distributive.

The external dimension of this infrastructure adds a further layer of geography, extending beyond the EU's territory into its relationships with third countries, and particularly with emerging economies, a dimension that tests what "autonomy" can mean once EU infrastructure is embedded in global networks. Copernicus and Galileo are not purely

[71] On the geopolitical geographies of outer space from a specifically European vantage point, see B. Doboš, *Geopolitics of the Outer Space: A European Perspective*, Springer, Cham, 2019.

[72] Regulation (EU) 2023/588; cfr. B. Akcali Gur, J. Kulesza, *The EU's Dual Policy Dilemma: Orbital Sustainability and Digital Autonomy Policies Intertwined*, in *Space Policy*, 76, 2026, 101731.

[73] Cfr. Dalledonne, Dousset, Aliberti, *New Challenges*, cit.; Hainaut, *European Autonomy in Orbit*, cit.

[74] Dalledonne, Dousset, Aliberti, *New Challenges*, cit.

"internal" infrastructures: their data, services and underlying technologies are made available, through various cooperation and capacity-building arrangements, to partner countries outside the EU, and the digitalisation of these services creates openings for technology partnerships that extend the reach of EU space infrastructure into global value chains.[75] Giordano's analysis of digitalisation and space technology partnerships between Europe and emerging economies frames this external dimension as integral to how EU space infrastructures operate in practice: their value, and their claim to "autonomy", depends not only on the EU's internal industrial base but on the networks of data-sharing, capacity-building and technology transfer through which their outputs circulate globally, including toward countries that are themselves emerging space actors.[76]

Read together, the flagship infrastructures occupy a position at the intersection of the internal and external geographies this chapter traces. Internally, they are central to the strategic-autonomy and industrial-consolidation agenda, embedded in the core-periphery industrial geography of Section 2; externally, they are nodes in a wider global geography of space-technology cooperation, whose "autonomy" is therefore always relational, defined not only by what the EU can do without relying on others, but by the position EU infrastructures occupy within networks of partnership and dependency that extend well beyond the Union's borders.[77] This relational character of autonomy - internally distributive, externally networked - is one that each of the four filters in Section 5 puts to the test.

4. *New Space, Commercial Regulation and Orbital Sustainability*

4.1 *New Space, Falling Launch Costs and the IRIS² Dilemma in a Crowded Orbit*

The governance architecture traced in Sections 2 and 3 - the EU Space Act's internal-market template, the strategic-autonomy turn, the EU-ESA-Member States triangle, and the flagship infrastructures of Galileo, Copernicus and IRIS² - does not operate in a static orbital environment.

[75] Cfr. Giordano, *From Access to Agency,* cit.

[76] Ibid.

[77] Cfr. Ibid.; Akcali Gur, Kulesza, *The EU's Dual Policy Dilemma*, cit.

It operates in one transformed, over a short period, by a sharp reduction in launch costs and the entry of new commercial actors, above all SpaceX, into the launch market.[78] Marcucci and colleagues document the structural consequences: as the marginal cost of placing mass into orbit has fallen, the economics of satellite design have changed accordingly, favouring smaller, cheaper, more numerous satellites - "smallsats" - deployed in large constellations rather than the smaller numbers of large, expensive satellites that characterised earlier decades. The result is not simply more satellites but a qualitative change in the density and composition of orbital occupation: low Earth orbit in particular now hosts constellations numbering in the thousands, operated by a small number of dominant commercial actors and composed of satellites whose individual lifespans are shorter, and replacement cycles faster, than those of traditional institutional satellites.[79]

This shift carries several implications for the governance geography of Sections 2 and 3. First, it changes the baseline against which the EU Space Act's safety and environmental-sustainability pillars (§2.2) must be assessed: a regulatory template designed around the authorisation of individual missions or operators sits differently in an environment where a handful of constellation operators account for a large and growing share of all active satellites.[80] Second, it sharpens the core-periphery dynamics of §2.3 and §3.3: the commercial entities best positioned to operate at the scale and cost structure that falling launch costs enable are concentrated in a small number of jurisdictions - overwhelmingly the United States, in the case of the dominant constellation operators - while European actors, including those in the EU's "core" space economies, participate primarily as launch-service customers, component suppliers, or operators of smaller European constellations rather than at comparable scale.[81] The strategic salience of specific orbital locations - choke points, transfer corridors, the orbital "high ground", the Lagrange points - has entered even popular

[78] Marcucci et al., *When Launch Costs Fall,* cit.

[79] Ivi.

[80] Cfr. Section 2.2 above, and sources cited therein. For a useful taxonomy of the interdependent challenges involved, see G. Pavesi, J. Wouters, *The Final Frontier? The European Union and the Governance of Outer Space*, in *Journal of European Integration*, 45, n. 8, 2023, pp. 1199-1217, who describe the fundamental dimensions at stake as S3 (Safety, Security and Sustainability) and the converging drivers as C4 (Congestion, Competition, Contestation and Convergence).

[81] Marcucci et al., *When Launch Costs Fall*, cit.; cfr. Hainaut, *European Autonomy in Orbit*, cit.

geopolitical writing, a sign of how widely the "geography of space" is now read as a determinant of power and security.[82]

Third, and most directly relevant to the chapter's outer space geographical framing, the proliferation of smallsats and constellations represents a material densification of the population of orbit. This is not reducible to an industrial-policy question of who builds and operates these satellites; it concerns the changing physical and infrastructural character of the orbital environment itself, an environment that EU policy, through both the Space Act and IRIS², must now navigate as a given condition rather than a blank space awaiting European occupation.[83]

IRIS², established as the Union Secure Connectivity Programme for 2023–2027 by Regulation (EU) 2023/588, occupies a dual position that mirrors the broader tension this chapter has traced since Section 1. It is presented simultaneously as a commercial connectivity service - competing with, or complementing, existing satellite broadband providers - and as an instrument of digital and strategic autonomy, reducing the EU's dependence on non-EU-controlled satellite connectivity infrastructure of the kind discussed above.[84] This dual framing is not unique to IRIS²; it echoes the framing of Galileo and Copernicus as both service-providing programmes and instruments of "digital sovereignty" noted in §3.3. But IRIS² makes the tension unusually explicit, because the connectivity market it seeks to enter is one in which the scale economics described above already favour incumbents operating very large constellations.[85]

Akcali Gur and Kulesza identify the core of this tension in what they term the EU's "dual policy dilemma". Orbital sustainability and digital autonomy, two objectives the EU pursues simultaneously and presents as mutually reinforcing, are in practice in some tension: the pursuit of digital autonomy through EU-controlled constellations implies adding to the satellite population of an orbital environment whose sustainability - in terms of debris risk, collision probability and long-term usability - depends on constraining that population, or at least managing its growth carefully.[86]

[82] In a popular-geopolitics register, see T. Marshall, *The Future of Geography: How Power and Politics in Space Will Change Our World*, Elliott & Thompson, London, 2023, on orbital choke points and the Lagrange points as strategic positions. The work is cited here as an instance of "pop" astropolitics - symptomatic of the mainstreaming of spatial readings of outer space - rather than as a theoretical reference.

[83] Cfr. Giordano, *Orbital Thresholds,* cit.

[84] Regulation (EU) 2023/588.

[85] Cfr. Section 3.3 above; Akcali Gur, Kulesza, *The EU's Dual Policy Dilemma,* cit.

[86] Akcali Gur, Kulesza, *The EU's Dual Policy Dilemma*, cit.

IRIS² does not cause this tension, which is structural to any strategy seeking autonomy through EU-controlled space infrastructure in an already congested orbit; but it makes the tension concrete and policy-relevant in a way that more abstract discussions of "strategic autonomy" do not.[87]

IRIS² thus sits differently under each of the four questions, feasible and, for some actors, desirable, yet directly in tension with orbital sustainability, since an additional EU constellation is also an additional contribution to a density whose management already strains EU space-surveillance capacity across all operators, EU and non-EU alike.[88] These tensions are taken up explicitly through the quadruple filter in Section 5.[89]

4.2 *Launch Licensing, Space Traffic Management and the Governance of Near-Earth Space*

The densification of orbit described in §4.1, and the EU's own contribution to it through IRIS² and other constellations, raise a governance question the Space Act's safety pillar (§2.2) addresses only partially: how is traffic - as opposed to individual launches or individual satellites - managed in an increasingly occupied near-Earth environment, and how does this management interact with the pre-existing governance of airspace, from which outer space is conventionally, but not always clearly, distinguished?[90]

Giordano's analysis of "orbital thresholds" frames this in geographical terms. The boundary between airspace, governed by an established international regime, the Chicago Convention system, and by national sovereignty over the airspace above national territory - and outer space - governed by the very different regime of the 1967 Outer Space Treaty, premised on non-appropriation and freedom of access, is not a sharp physical line but a "high boundary": a zone of increasing technical and legal ambiguity, made more consequential as suborbital and very-low-orbit activities multiply and the operational distinction between "aviation" and "space" becomes less clear-cut.[91] Space Traffic Management (STM), the set of practices and increasingly rules governing the trajectories, manoeuvres

[87] Ibid.

[88] P. Faucher, A. Gillet, *European Union Space Surveillance and Tracking: State of Affairs in 2025 and Strategic Vision*, in *Space Policy*, 74, 2025, 101699.

[89] Akcali Gur, Kulesza, *The EU's Dual Policy Dilemma*, cit.

[90] Cfr. Section 2.2 above.

[91] Giordano, *Orbital Thresholds*, cit.

and conjunctions of objects in orbit, and Air Traffic Management (ATM), the long-established system governing aircraft movements, have historically operated as largely separate regimes, separate institutions, separate data systems, separate allocations of responsibility between states, operators and infrastructure providers. Increasing traffic through the "high boundary" zone - from reusable launch vehicles, suborbital flights, or satellites in decaying orbits re-entering through airspace - makes this separateness increasingly difficult to sustain in practice, even where it remains formally intact in law.[92]

From a governance perspective, STM and ATM are not merely technical coordination problems; they are mechanisms that distribute responsibility and risk among states, operators and infrastructure providers, and that determine in practice who has the operational and informational capacity to access particular orbital regimes safely. An operator's ability to obtain a launch licence, and then to operate within a given orbital regime without generating conjunction risks that others must manage, depends on access to space-situational-awareness data, to coordination mechanisms, and to the administrative capacity to engage with both STM and ATM authorities, capacities themselves unevenly distributed across Member States and across operators of different sizes, echoing the core-periphery and large/small-operator distinctions of Sections 2 and 3.[93] Faucher and Gillet's account of the EU's space surveillance and tracking capabilities (EUSST) situates the EU's own contribution here: EUSST is an attempt to build EU-level space-situational-awareness capacity that could, in principle, support both STM coordination and the conjunction-risk management an increasingly dense orbit requires, but its development to date reflects the same uneven distribution of sensor infrastructure and national capabilities that characterises the EU's space sector more broadly.[94]

The governance of the "high boundary" - and the question of who can access which parts of near-Earth space safely, and on what informational basis - is therefore not a narrowly technical matter to be settled once by a sufficiently detailed STM regime. It has the same distributive structure as the questions raised in Sections 2 and 3: it determines which operators, and indirectly which Member States and regions, can participate in an increasingly crowded orbit without bearing disproportionate compliance costs or disproportionate exposure to risks generated by others. Section

[92] Ibid.

[93] Cfr. Sections 2.3 and 3.3 above.

[94] Faucher, Gillet, *European Union Space Surveillance and Tracking*, cit.

5.4 revisits this material when assessing systemic sustainability, not only of the orbital environment in the narrow physical sense, but of a governance architecture for the "high boundary" that currently divides responsibility between regimes (STM, ATM) not designed with each other, or with current traffic density, in mind.

4.3 *Cybersecurity, Critical Infrastructure and the Uneven Vulnerability of an Orbital "Space Society"*

The final governance layer this chapter examines concerns the application of the EU's horizontal cybersecurity and critical-infrastructure instruments - the NIS2 Directive (Directive (EU) 2022/2555) and the Critical Entities Resilience Directive (Directive (EU) 2022/2557) - to space-related assets: satellites themselves, but also the ground stations, data links and terrestrial infrastructure on which satellite services depend.[95] Cappelletti and Papakonstantinou frame this extension as a "question of strategic legislation". These instruments were designed primarily with terrestrial critical infrastructure - energy grids, financial systems, digital infrastructure providers - in mind, and their application to space raises the question of whether categories and thresholds developed for terrestrial infrastructure (which "essential" or "important" entities are covered, what counts as a reportable incident, what resilience measures are proportionate) translate adequately to an environment where the infrastructure is, in part, physically located in orbit and operationally dependent on space-to-ground links with no direct terrestrial analogue.[96]

This extension intersects with the dual-use and AI dimensions of the defence-space nexus discussed by León-Serrano and colleagues, who document the growing overlap between civilian and military applications of space-based AI systems, in the processing of Earth-observation data, the operation of autonomous satellite systems, or the analysis of space-situational-awareness data discussed in §4.2.[97] As satellites and ground infra-

[95] Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 (NIS2 Directive); Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the Resilience of Critical Entities and Repealing Council Directive 2008/114/EC.

[96] F. Cappelletti, V. Papakonstantinou, *A Question of Strategic Legislation: Can the EU Deal with Cybersecurity Issues in Space?*, in *Telecommunications Policy*, 49, 2025, 102954.

[97] G. León Serrano, F. Pérez Martínez, J.C. De La Fuente Chacón, *The Dual Use of Artificial Intelligence: Analysis of Trends and Policies in the Defence-Space Sector*, in

structure become both more numerous (§4.1) and more central to EU strategic-autonomy objectives (§3.3, §4.1), the question of which actors are subject to - and capable of complying with - cybersecurity and resilience obligations acquires direct bearing on the EU's own autonomy objectives: a constellation or ground-segment provider that falls outside the practical reach of NIS2/CER compliance, whether because of its size, its jurisdiction, or the ambiguity of its classification under instruments designed for terrestrial infrastructure, represents a potential vulnerability in precisely the infrastructure that strategic-autonomy policy treats as critical.[98]

From the perspective of industrial and territorial geography, the practical burden of compliance is unlikely to fall evenly. Larger operators - typically those based in the EU's "core" space economies of §2.3 and §3.3, or large non-EU constellation operators with EU-facing compliance functions - are more likely to meet cybersecurity and resilience requirements as a matter of course, in some cases as an extension of obligations they already meet in other regulated sectors.[99] Smaller operators, including those based in CEE Member States or operating in niche downstream-applications segments of the kind discussed in §2.3, may face a proportionally larger compliance burden relative to their size, a dynamic familiar from horizontal EU regulation in other sectors, but one whose consequences in the space domain extend beyond compliance costs to the question of who can participate, securely and sustainably, in an EU space economy that is simultaneously becoming more commercially dense (§4.1), more autonomy-oriented (§4.1) and more dependent on contested near-Earth traffic-management arrangements (§4.2).[100] Read across the terrestrial–orbital continuum, these inequalities of capacity connect to wider questions of extraction, supply chains and waste in outer space geographies.[101]

This unevenness is the final element the chapter carries forward into Section 5. The "space society" whose emergence this chapter has traced

International Journal of Engineering Business Management, 17, 2025, pp. 1-25.

[98] Cfr. Cappelletti, Papakonstantinou, *A Question of Strategic Legislation*, cit.; León-Serrano et al., *The Dual Use of Artificial Intelligence*, cit.

[99] Cfr. Sections 2.3 and 3.3 above.

[100] Ibid.

[101] On extraction, supply chains and waste as a continuum linking terrestrial and extra-terrestrial geographies, and on the environmental-justice inequalities this produces, see J.M. Klinger et al., *Wasting and Wanting: An Extractive Supply Chain Approach to Outer Space Geographies*, in *Environment and Planning D: Society and Space*, 43, n. 2, 2024, pp. 823-840.

- across the internal-market template of Section 2, the strategic-autonomy and governance-triangle dynamics of Section 3, and the New Space densification and infrastructural extension of this section - is not a uniform space in which all participants face the same opportunities, costs and vulnerabilities. Section 5.2 examines the social-desirability implications of this unevenness directly; Section 5.3 examines whether the EU's instruments, including the horizontal cybersecurity framework discussed here, are adequately calibrated to a space domain whose actors and assets differ so substantially; and Section 5.4 draws together the orbital- and institutional-sustainability threads - debris and congestion (§4.1), the governance of the high boundary (§4.2), and the resilience of an unevenly secured infrastructure base (§4.3) - into a single assessment of whether the architecture this chapter has mapped is sustainable, in the systemic sense the quadruple filter requires, over the medium to long term.

5. *Applying the Quadruple Filter to EU Space Governance*

The preceding sections have mapped a governance architecture that is, in important respects, internally coherent: each instrument discussed - the EU Space Act, the strategic-autonomy turn, IRIS², the EU's emerging STM/SSA capabilities, the extension of NIS2/CER to space - responds to a recognisable policy problem and is defensible on its own terms. The quadruple filter introduced in Section 1.2, drawn from the SPHERE framework, exists precisely because individual defensibility does not guarantee systemic coherence.[102] This section applies the filter's four questions - what is technically feasible, what is socially desirable and for whom, what is legally and ethically legitimate, and what is systemically sustainable - to the architecture as a whole, treating Sections 2 to 4 as the material from which the filter draws rather than re-describing them.

5.1 *Technical Feasibility: An Architecture Built Faster Than Its Capacities*

Asked simply, "what can the EU do?", the architecture mapped in Sections 2 to 4 yields an uneven answer. At the level of regulatory feasibility, the EU Space Act is feasible in the narrow sense that the Union

[102] Giordano, *The New Space Age*, cit., ch. 3; cfr. Section 1.2 above.

has repeatedly produced Article 114-based harmonisation instruments of comparable legal complexity in other sectors (§2.1); the legislative machinery exists, even if its application to space introduces the layering of registration, launch and market-access geographies discussed in §2.2.[103] At the level of industrial feasibility, the picture is more constrained. Strategic autonomy in connectivity, as pursued through IRIS², presupposes an EU launch and manufacturing base capable of deploying and sustaining a constellation at a scale that delivers genuine autonomy rather than a token presence alongside dominant non-EU operators. The New Space landscape of §4.1 - in which falling launch costs have been exploited overwhelmingly by non-EU actors operating at far greater scale - suggests that this presupposition is, at best, only partially met.[104] The EU's industrial base remains real and in some segments competitive, but it is distributed unevenly across a small number of Member States (§2.3, §3.3), and "EU capacity" in the aggregate does not translate straightforwardly into the rapid, large-scale deployment capability the IRIS² timeline implies.[105]

A second feasibility constraint concerns institutional rather than industrial capacity. The governance triangle of §3.2 - EU, ESA, Member States - was not designed for the pace or the type of decision-making that an autonomy-oriented, security-linked space policy now requires. ESA's variable-geometry funding model and the EU's co-legislative procedures operate on different timelines and different logics, and the gradual security linkages Müller and De Man describe have so far layered new objectives onto this triangle without resolving the coordination costs this generates.[106] Similarly, the EU's space-situational-awareness capacity (EUSST), while expanding, remains built on unevenly distributed national sensor infrastructure (§4.2), which constrains the EU's ability to operate the kind of STM coordination role that an increasingly dense orbit - to which IRIS² itself contributes - would require.[107]

The orbital-geographies framing sharpens this assessment in one specific way. A purely legal or institutional reading might treat "feasibility" as a question of whether the EU can pass the relevant legislation, and on that narrow reading, feasibility is largely uncontroversial. Reading orbit

[103] Cfr. Sections 2.1 and 2.2 above.

[104] Cfr. Section 4.1 above; Marcucci et al., *When Launch Costs Fall,* cit.

[105] Cfr. Dalledonne et al., *New Challenges for European Space Industrial Policy*, cit.

[106] Cfr. Section 3.2 above; Müller, De Man, *Incremental Shifts, Strategic Orbits,* cit.

[107] Cfr. Section 4.2 above; Faucher, Gillet, *European Union Space Surveillance and Tracking,* cit.

as an already-occupied, densifying physical environment (§4.1) reframes feasibility as a question of whether the EU can act effectively within an environment whose basic parameters - density, dominant actors, available orbital "real estate" - are no longer ones it controls or can easily reshape. On this reading, the architecture's feasibility gap is less about whether the EU can legislate than about whether legislation, however well-designed, can deliver autonomy outcomes in an orbit whose occupation has already been substantially determined by others.

5.2 *Social Desirability: Whose Single Market, Whose Orbit?*

The desirability filter asks not whether the architecture's objectives are desirable in the abstract - reducing dependence on non-EU infrastructure, harmonising fragmented rules, improving orbital safety are all widely shared goals - but who benefits, and who bears the costs, of the specific instruments through which these objectives are pursued.

Three distributive patterns identified across Sections 2 to 4 recur here. First, the core-periphery pattern in the EU's space economy (§2.3, §3.3): a harmonised EU template (the Space Act), an industrial-policy agenda shaped substantially by the Draghi report's competitiveness framing (§3.1), and flagship infrastructures whose supply chains concentrate in France, Germany and Italy (§3.3) together suggest that the desirability of "strategic autonomy" is more directly experienced by Member States already positioned as upstream industrial actors than by CEE and other latecomers, whose participation remains predominantly downstream and programme-funding-dependent.[108] A single market for space and a strategic-autonomy agenda pursued through industrial consolidation can both be desirable for "Europe" in aggregate while distributing that desirability unevenly within Europe. Second, the large-operator/small-operator pattern running through §4.1 and §4.3: New Space economics favour large constellation operators, and the EU's own responses - IRIS² as a large-scale programme, NIS2/CER obligations calibrated implicitly toward entities with the administrative capacity to meet them - risk reinforcing rather than counterbalancing this concentration, potentially at the expense of the smaller European New Space actors that an internal-market logic might otherwise be expected to support.[109]

[108] Cfr. Sections 2.3 and 3.3 above; Miglio, Nota, *Implementing the Draghi Report's Recommendations,* cit.

[109] Cfr. Sections 4.1 and 4.3 above.

Before turning to the third pattern, where the orbital-geographies framing most substantially alters the question posed to EU space law, it is useful to make explicit the distributive geography that has been unfolding in the background. Table 1 draws together five arguments developed across the chapter – on industrial capacity, flagship-infrastructure supply chains, New Space scale economics, regulatory compliance, and access to a crowded orbit – to show that they trace the same underlying core–periphery pattern.

Table 1. A Recurring Core–Periphery Pattern in EU Space Governance

Axis	Core (examples)	Periphery (examples)	Section
Industrial capacity and launch infrastructure	France (sovereign launch infrastructure), Germany and Italy (upstream manufacturing capacity), Netherlands (service/operator-oriented sector), Luxembourg (jurisdictionally specialised registration model)	CEE Member States and other latecomers: smaller, more recently established sectors oriented toward downstream applications and EU-level programme participation	§2.3
Flagship-infrastructure supply chains	France, Germany, Italy: prime contractors, system integrators, upstream manufacturing capacity for Galileo/Copernicus/IRIS²	Other Member States: downstream applications, ground-segment contributions, EU-level funding only	§3.3
New Space scale economics	Dominant constellation operators (overwhelmingly US-based); EU "core" space economies as launch-service customers/component suppliers	European actors operating smaller constellations, without comparable scale	§4.1
NIS2/CER compliance burden	Larger operators based in the EU's "core" space economies, or large non-EU constellation operators with EU-facing compliance functions	Smaller operators, including those in CEE Member States or in niche downstream-applications segments	§4.3

Axis	Core (examples)	Periphery (examples)	Section
Access/standing in a crowded orbit	Larger, better-resourced operators and Member States, informationally/technically/administratively equipped	Smaller, less-equipped operators and Member States, despite formally open market access	§5.2

This table (author's own elaboration) draws together five distinct arguments developed across the chapter - on industrial capacity, infrastructure supply chains, New Space economics, regulatory compliance, and orbital access - to show that they trace the same underlying distributive geography. The recurrence of this core–periphery pattern, rather than its presence in any single instrument, is what the chapter's social-desirability filter (§5.2) ultimately puts to the test.

The third pattern is where the orbital-geographies framing changes the question most substantially. Desirability cannot be assessed solely in terms of who builds and operates infrastructure, but also in terms of who is positioned within an increasingly populated orbital environment, and on what terms. The categories of population geography - place, access, composition, vulnerability - that population exogeography turns toward imagined extra-terrestrial settlement can be mobilised here only analogically, to think about uneven capacities to participate in an occupied orbit.[110] As orbit becomes more densely occupied by constellations, debris-mitigation obligations, and STM/SSA-mediated traffic (§4.1-4.2), the capacity to operate safely and with adequate situational awareness within this environment becomes itself a distributed good, one that larger, better-resourced operators and Member States can access more readily than smaller ones, independently of formal market-access rules.[111] Space debris, in particular, functions as what Clormann and Klimburg-Witjes call a "boundary infrastructure": material remains that link the orbital environment to terrestrial concerns and enact shared - but unevenly distributed - responsibilities across operators.[112] A "single market for

[110] In Giordano, *Human Migration, Settlements and Life in Outer Space,* cit. Population exogeography concerns prospective extra-terrestrial settlement rather than orbital operators; the concept is used here only by analogy, to frame uneven capacities to operate within an already-occupied orbit.

[111] Ibid.; cfr. Section 4.2 above.

[112] M. Clormann, N. Klimburg-Witjes, *Troubled Orbits and Earthly Concerns: Space Debris as a Boundary Infrastructure*, in *Science, Technology, & Human Values*, 47, n. 5,

space" that is formally open to all EU operators may nonetheless be substantively more navigable for those already equipped - informationally, technically, administratively - to operate within a crowded orbit. This is a desirability question that a purely legal or industrial policy reading of Sections 2 to 4 would not, on its own, surface.

5.3 *Legal and Ethical Legitimacy: Whose Geography Gets to Govern?*

Section 2 traced how the EU Space Act relocates orbit into the institutional geography of the internal market via Article 114 TFEU, against the backdrop of Article 189(2)'s exclusion of harmonisation (§2.1); how its extraterritorial ambitions overlay a market-access geography onto the registration and launch-based geographies of international space law (§2.2); and Section 4.3 how horizontal instruments designed for terrestrial critical infrastructure (NIS2, CER) are extended to space assets. The legitimacy filter does not ask whether these moves are lawful in a narrow doctrinal sense - a question consistently set aside here as outside the chapter's disciplinary scope - but whether the EU's claim to govern orbit along these particular geographies is likely to be recognised as legitimate by the range of actors whose cooperation, compliance or acquiescence the architecture depends upon: Member States, ESA, non-EU partners, and operators themselves.[113]

On the competence question (§2.1), the legitimacy stakes are primarily intra-EU and intergovernmental. Member States that have invested in national space laws (§2.2) and occupy central positions in ESA's variable-geometry programmes (§3.2) may regard an Article 114-based harmonisation of matters Article 189(2) appears to reserve to them either as a legitimate use of internal-market competence, or as an institutional overreach that uses the internal market as a vehicle for objectives the Treaty's space-specific provision was designed to keep at Member State and ESA level.[114] This chapter cannot resolve that question, but its answer shapes whether Member States implement the resulting framework cooperatively or grudgingly, which in turn feeds back into the feasibility assessment of §5.1.

On extraterritorial reach (§2.2), the legitimacy stakes are more clearly

2022, pp. 960-985.

[113] Cfr. Section 1.2 above on the SPHERE legal-ethical legitimacy filter.

[114] Cfr. Section 2.1 above; De Man, Wouters, *EU Space Governance,* cit.

external. Baumann and Prall's "Brussels effect" framing describes a pattern in which EU market-access conditions become global standards because non-EU operators find compliance cheaper than non-compliance. But this pattern depends on the EU market being sufficiently attractive, and on non-EU operators and their home jurisdictions accepting - even reluctantly - that EU market access can legitimately be conditioned on compliance with EU-defined standards for activities registered and licensed elsewhere.[115] Whether this acceptance materialises, or whether non-EU operators and jurisdictions instead treat EU market-access conditions as an illegitimate extension of EU regulatory reach into matters governed by registration- and launching-state jurisdiction under the Outer Space Treaty and Registration Convention, is precisely the kind of question that a "Brussels effect" analysis transplanted from digital regulation to space may not adequately anticipate, given how much more developed and institutionally entrenched the international jurisdictional framework for space activities is, compared to the relatively thin framework for digital services.[116] The status of outer space as res communis, and its comparison with other commons such as Antarctica and the deep seabed, has long been a theme of critical legal geography.[117] On the extension of horizontal instruments (§4.3), the legitimacy question is more operator-facing: NIS2 and CER were designed, and their thresholds calibrated, for terrestrial critical-infrastructure operators, and their application to space raises the question of whether operators newly brought within scope perceive this as a coherent and proportionate application of an existing, legitimate framework, or as a regulatory reach exercised through instruments not designed with their sector in mind, applied because no space-specific equivalent yet exists.[118]

Reading orbit as a governed, populated space adds one further

[115] Baumann, Prall, *The EU Space Act's Global Reach,* cit.

[116] Ibid.; cfr. Section 2.2 above.

[117] On outer space as res communis and the comparison between extra-atmospheric space, Antarctica and the deep seabed from the standpoint of critical legal geography, see C. Collis, *Res Communis? A Critical Legal Geography of Outer Space, Antarctica, and the Deep Seabed*, in J.S. Ormrod, P. Dickens (eds.), *The Palgrave Handbook of Society, Culture and Outer Space*, Palgrave Macmillan, London 2016, pp. 270-293. On the shift from a 'global commons' to a 'contested Global Spaces' framing, see also Gstöhl, Larik, *The European Union and the Governance of Contested Global Spaces*, cit.

[118] Cfr. Section 4.3 above; Cappelletti, Papakonstantinou, *A Question of Strategic Legislation,* cit.

dimension a purely legal reading would not capture.[119] Legitimacy claims over orbit are made by an actor - the EU - whose own internal space governance, as Sections 2 to 4 have shown, remains a layered and partially unresolved architecture (internal market, security, industrial policy, each with a different institutional geometry). An EU that has not fully settled where, internally, the authority to govern orbit resides - Brussels institutions, ESA, or Member States, in what combination - may find its external legitimacy claims, whether over non-EU operators, in STM coordination, or in international fora on orbital sustainability, weakened by this internal unsettledness. Legitimacy, on this reading, is not only about whether a given EU claim to regulate is well-founded in isolation, but about whether the EU, as a composite actor occupying a still-evolving position in the governance triangle of §3.2, is positioned to make such claims with the institutional coherence that external recognition typically requires.

5.4 *Systemic Sustainability: Orbital and Institutional Sustainability as a Single Problem*

The sustainability filter, as formulated in the SPHERE framework, asks not whether a measure is feasible or desirable at a given moment, but whether it can be continued over time, a question with both a physical and an institutional dimension, whose two sides, in this case, are closely intertwined.[120]

The physical dimension is the orbital-sustainability picture assembled across §4.1–4.2: a densifying low Earth orbit, shaped predominantly by non-EU constellation operators exploiting falling launch costs; an EU contribution to this density through IRIS² and other constellations, justified on autonomy grounds but in tension with the EU's own debris-mitigation and environmental-sustainability commitments under the Space Act's third pillar (§2.2, §4.1); and a governance regime for the "high boundary" (§4.2) still divided between STM and ATM frameworks not designed for current traffic levels, supported by an

[119] For a critical legal geography reading of how authority in outer space is not given but produced and contested - and how the legal instruments available can reinforce some actors' authority while marginalising others - see Marino, Cheney, *Centring Environmentalism in Space Governance*, cit.

[120] Giordano, *The New Space Age*, cit., ch. 3.

EUSST capability constrained by uneven national sensor infrastructure.[121] On this dimension alone, the filter raises a direct question about IRIS² specifically: a programme that may score acceptably on feasibility (§5.1, with caveats) and on desirability (§5.2, for some actors) nonetheless adds to a form of orbital density whose long-term sustainability the EU's own policy framework treats as a serious concern, precisely the "dual policy dilemma" identified by Akcali Gur and Kulesza.[122]

The institutional dimension concerns whether the architecture itself - the layering of internal-market (§2.1), security-oriented (§3.1), industrial-policy (§3.1) and horizontal-regulatory (§4.3) instruments onto the pre-existing EU-ESA-Member States triangle (§3.2) - can persist and function coherently over the medium to long term, or whether it represents an accumulation of partially overlapping mandates whose coordination costs grow as each layer is further developed.[123] De Man and Wouters's account of the governance triangle "at the threshold of a new era" can be read as posing exactly this question: the threshold is being crossed by adding new EU-level instruments and framings, but without a corresponding settlement of how ESA's role, Member States' national frameworks, and the EU's expanding regulatory and security ambitions are meant to fit together over time.[124]

Reading orbit as an occupied environment, rather than as a backdrop to EU institutions, brings these two dimensions into a single frame in a way a separate treatment of "orbital environment" and "EU institutions" would not. An institutional architecture that is itself unsettled - in competence (§5.3), in the EU-ESA-Member States balance (§3.2), in the coherence between autonomy and sustainability objectives (§4.1) - is poorly positioned to manage a physical environment whose sustainability increasingly depends on coordinated, sustained governance. Debris mitigation, STM coordination and SSA-sharing are not one-off regulatory acts but ongoing functions requiring institutional continuity and a clear allocation of responsibility over time.[125] The systemic-sustainability question this chapter poses is therefore not "is orbit sustainable" and "is the EU's governance architecture sustainable" as two separate questions,

[121] Cfr. Sections 4.1-4.2 above.

[122] Akcali Gur, Kulesza, *The EU's Dual Policy Dilemma,* cit.

[123] Cfr. Section 3.2 above.

[124] De Man, Wouters, *EU Space Governance*, cit.

[125] Cfr. Faucher, Gillet, *European Union Space Surveillance and Tracking*, cit.

but whether an institutional architecture this provisional can sustain the kind of ongoing governance functions that an orbital environment this densely occupied now requires.

5.5 *Cross-Filter Synthesis: Three Fault Lines*

Drawing the four filters together, three fault lines emerge as the points where the EU's combined single-market and strategic-autonomy project appears most fragile, and where the tensions traced individually across Sections 2 to 4 converge most clearly.

The first is competence-and-legitimacy. The Article 114/189 ambiguity (§2.1, §5.3) is not merely a legal technicality but the foundation on which the entire harmonisation edifice rests. If that foundation is contested - by Member States, by the Court of Justice, or simply through uneven and reluctant implementation - the feasibility of the architecture as a whole (§5.1) is correspondingly weakened, since an EU Space Act whose legal basis is in question cannot deliver the harmonised "single template" (§2.2) on which downstream objectives, including industrial consolidation (§3.1) and IRIS²'s integration into a coherent regulatory environment (§4.1), depend.

The second is core-periphery distribution. The same instruments that score well on aggregate feasibility and desirability - industrial consolidation under a European Space Law (§3.1), flagship infrastructure supply chains (§3.3), New Space economics favouring scale (§4.1) - generate a distribution of benefits and costs across Member States, and across upstream/downstream and large/small operators, that the desirability filter (§5.2) shows to be persistently uneven. This fault line is less likely than the first to produce an acute crisis, but it represents a slow erosion of the "single market" framing's claim to be a neutral, EU-wide good if its benefits continue to concentrate in a few already-central Member States and operators.

The third is autonomy-versus-sustainability. Most sharply visible in IRIS² (§4.1, §5.4), it is present wherever the architecture's autonomy-oriented instruments - constellations, EUSST capacity-building, industrial consolidation for launch and manufacturing - interact with its sustainability-oriented commitments, debris mitigation, orbital-slot management, the Space Act's environmental pillar. This fault line is structurally built into any strategy that pursues autonomy through additional EU-controlled infrastructure in an orbit whose sustainability

depends on restraint; it is the fault line for which the quadruple filter's explicit separation of "desirable" from "sustainable" is most directly necessary, since a purely desirability-oriented assessment of IRIS² would not, on its own, surface this tension.

These three fault lines are not independent of one another. An unsettled competence basis (first) makes it harder to address core-periphery distribution through EU-level redistributive mechanisms (second), and an EU whose internal governance triangle remains unsettled (§3.2, §5.3) is less able to manage the autonomy-sustainability trade-off (third) through coordinated, sustained institutional action. Section 6 takes up these three fault lines as the chapter's principal findings and as an agenda of open questions for the EU's evolving space governance.

6. *Conclusion*

This chapter has argued that EU space governance is best understood not as a coherent strategy moving toward a settled endpoint, but as an unresolved compromise between two rationales - a single-market template, anchored in the internal-market logic of Article 114 TFEU, and a strategic-autonomy trajectory, anchored in an increasingly security-linked reading of EU competence - pursued through overlapping instruments and an institutional architecture that predates both in their current form.[126] Applying the quadruple filter developed by the CISS-SPHERE Unit, and reading the result through the lens of an emerging "space society" in which orbit is an increasingly occupied and stratified environment, has made visible three fault lines along which this compromise appears most fragile.

The first fault line - competence and legitimacy - runs through the chapter's account of how orbit has been relocated into the internal market via Article 114 TFEU, against the backdrop of Article 189(2)'s exclusion of harmonisation, and of how this relocation interacts with a governance triangle whose internal balance the strategic-autonomy turn has begun to shift without resolving.[127] This is not a peripheral legal question to be settled once and set aside: it is the foundation on which the Space Act's harmonised template, the extraterritorial reach of §2.2, and the broader "European Space Law" debate of §3.1 all rest. The open question it leaves

[126] Cfr. Section 1.1 above.

[127] Cfr. Sections 2.1, 3.2 and 5.3 above.

is whether the architecture can function coherently while this foundation remains unsettled and, more specifically, whether an EU that has not internally resolved how authority over orbit is distributed among Brussels institutions, ESA and Member States is well placed to make external legitimacy claims, whether toward non-EU operators under the "Brussels effect" logic of §2.2 or within international fora on orbital sustainability.

The second fault line - core-periphery distribution - runs through the chapter's recurring observation that instruments framed as benefiting "Europe" as a whole (the single-market template, industrial consolidation under the Draghi report's recommendations, flagship infrastructure supply chains) distribute their benefits unevenly across Member States with mature versus emerging space sectors, across upstream and downstream segments, and across large and small operators.[128] This unevenness is not incidental to the strategic-autonomy project but is, to some extent, built into how that project is pursued, through consolidation logics that tend to reinforce existing industrial centres rather than redistribute capacity toward CEE and other latecomer ecosystems. The open question it leaves is whether the EU's space-governance instruments can incorporate redistributive mechanisms - comparable to those in other areas of EU industrial and cohesion policy - without undermining the competitiveness rationale on which the autonomy agenda rests, or whether core-periphery patterns in the EU's space economy will simply persist as an accepted, if unaddressed, feature of the "single market for space".

The third fault line - autonomy versus sustainability - runs through the chapter's treatment of IRIS² as the clearest case in which an instrument that performs reasonably on feasibility and desirability nonetheless sits in tension with the EU's own orbital-sustainability commitments, by adding to the density of an already crowded near-Earth environment.[129] This tension is structural rather than incidental: any strategy pursuing autonomy through additional EU-controlled orbital infrastructure, in an environment whose long-term usability depends on restraint, will reproduce some version of it. The open question it leaves is whether EU space policy can develop mechanisms - institutional, regulatory or diplomatic - that allow autonomy-oriented deployment decisions to be weighed explicitly against their contribution to orbital density, rather than assessed, as they currently tend to be, primarily against feasibility

[128] Cfr. Sections 2.3, 3.3 and 5.2 above.

[129] Cfr. Sections 4.1, 5.4 and 5.5 above.

and desirability criteria that do not by themselves capture this trade-off.

These three fault lines are where the quadruple filter has done the most analytical work. Its contribution has been to keep feasibility, desirability, legitimacy and sustainability analytically separate, so that a measure's strength on one dimension - the competence question, core-periphery distribution, IRIS²'s contribution to orbital density - does not obscure its weakness on another, as a purely legal, industrial-policy or technical reading would each, in its own way, allow. The geographies of outer space and "space society" lens has done complementary work, anchoring the analysis in the spatial distribution of access, voice and risk across an orbital environment that is itself a governed, increasingly populated space, shifting attention from "what can the EU legislate" to "who can operate, safely and with adequate standing, in the orbit that EU policy is helping to shape".

Several lines of further work follow. A closer integration of SSA/STM operational practice with governance analysis would allow the institutional-sustainability questions of §5.4 to be examined with the empirical specificity this chapter has applied to legal and industrial-policy instruments. A more detailed mapping of core-periphery dynamics within the EU space economy - at the level of firms, regions and specific programme allocations - would give the second fault line an empirical foundation beyond the directional argument made here. And engagement with non-EU actors and frameworks in shaping orbital-sustainability regimes would help test whether the legitimacy questions of §5.3 are best addressed through unilateral EU instruments or through the kind of multilateral coordination that an inherently shared orbital environment may ultimately require. The SPHERE quadruple filter and the economic-political geography of outer space developed here offer one way of organising such work, not as a substitute for legal, technical or industrial expertise, but as a means of keeping their respective findings in productive tension, in a domain where no single discipline's criteria can, on their own, capture what is at stake in the governance of an increasingly shared and occupied orbit.

Dario Elia Tosi

The governance of space sector: first remarks on the model outlined by the EU draft Regulation on space activities

Summary: 1. Introductory remarks – 2. The international governance of space activities. – 3 The advent of the European Union's interest in the space sector and the spawn of a hazy and approximate governance – 4. The proposal for the Regulation of space activities and the new European governance of the space sector – 5. The consequences of the Regulation on national governance systems: the cases of France and Italy – 6. Conclusive remarks.

1. *Introductory remarks*

Outer space has always been perceived as a field of research strictly connected to engineering, physics, and, broadly speaking, the domain of hard sciences. Historically, the first subject of interest from the legal perspective came up in the WWII aftermath, it being the space race one of the fields of competition of the two main actors of the global geopolitical framework. Over the last two decades, the landscape has undergone a deep-rooted change. Scientific research and technological innovation prompted the advent of new players. Private actors appeared and started struggling for the promotion and protection of their interests. Therefore, alongside scientific and geopolitical questions, commercial issues have come up. Beyond that, the most recent events have conferred on space activities an ever-growing relevance regarding security and defence interests on a global scale.

Within this framework, the European Union has progressively consolidated its activity in the spatial sector in the attempt to establish and consolidate a pivotal role as a global actor, so as to ease the development of the space economy at European level. The proposal COM(2025)335 (commonly quoted as EU Space Act) published on 25th June 2025 strives to accomplish such a task by offering a thorough regulation of space activities. Among the different aspects dealt with by the EU Commission

in the draft text, it can be observed the attempt to establish a governance system of the public actors concerned. Given the complexity of the context, the draft regulation provides for a multilevel and multifaceted structure of government.

At first sight, the attempt represents an innovation compared to the current global regulatory framework. Indeed, looking at the *corpus iuris spatialis* made up of the 5 Treaties adopted by the international community, no specific regulation on space sector governance can be identified. Likewise, at the European level, the proposal represents the first attempt to regulate the field, the activity of EU institutions having till now focused on specific space programmes.

Nevertheless, upon deeper inspection, both international tools signed up by national governments and space activities performed at EU level encompass provisions related to the governance of the system. In this context, after a brief review of the current provisions related to the governance of space activities at the international level, the present contribution aims at depicting the evolution and outlining the current governance of the space sector at the EU level. Based on that, the study wants to sketch out the most important features of the governance model provided for in the draft regulation submitted by the EU Commission to the Parliament and the Council last year and the possible consequences on the system adopted by some countries.

2. *The international governance of space activities*

The pace of the evolution of the space sector, boosted by the advent of the space economy and the more and more urgent issues related to security and defence, reveals the inability of the current *corpus iuris spatialis* to rule the system. Among the different negative remarks expressed by the scientific community, it can be identified the lack of an effective system of governance. Indeed, at first glance, the 5 Treaties currently in force are characterized by provisions that aim at codifying principles and fundamental rules of space activities: the inner focus of the Treaties insists on duties and rights of the different actors involved in the sector. So, aside from the role played by UN institutions, like the General Assembly and the COPUOS, which focus on regulatory activity, no concrete structure of bodies and organisms in charge of the management of the space sector

can be identified[1].

However, alongside rules and provisions addressed to the subjects involved in space activities, from the analysis of the Treaties some provisions aimed at establishing an essential system of governance can be detected. Looking at the 1967 OST, it can be observed how the Treaty identifies two main actors responsible for the overall governance of the system.

The first and most important subjects charged with the duty to manage the system are the national states. Indeed, by pursuing military or scientific purposes, in that period all space activities were carried out by national governments. Therefore, states were the main recipients of the Treaty provisions, they being asked to act by the rules. Nevertheless, the OST gave them some organizational roles. According to Art. VI, besides the overall responsibility for national space activities, states were conferred the power to authorize all the non-governmental activities and, second, establish a system of continuing supervision. In addition, though not providing for an organic regulation of the topic, Art. VIII asks to states that are parties to the Treaty to set up a registry where all the information about an object launched into outer space should be recorded.

Alongside the role of national governments, within the aim of pursuing effective coordination and avoiding tension between countries, the Treaty identifies the UN Secretary General as the main recipient of all information related to space activities. In particular, Art. XI asks states parties to address to the Secretary General information, "to the great extent feasible and practicable", about the exploration activities carried out and the related results. Second, Art. V further affirms that states parties have to inform the UN organism about any phenomena that can constitute a danger to the life or health of astronauts.

A similar solution can be found even in the following 1968 Treaty on the Rescue of Astronauts. Art. 1 affirms that states parties have to inform the UN Secretary General regarding any accident occurred or conditions of distress or even emergency landing of astronauts in foreign countries.

A third element of the international governance of space activities has been outlined within the framework of the the 1972 Treaty on International Liability for Damages Caused by Space Objects. By reaffirming the accountability of the states parties for damages caused in space activities, the Treaty deals with the case of the inability of the states concerned to agree on a compensation. According to Article XIV

[1] On the structure of the international governance, among many others, see e.g. F. Von der Dunk, *International Space Law*, *Handbook of Space Law*, (F. Von der Dunk, F. Tronchetti (eds.), 2015); S. Marchisio, *The law of outer space activities* (2022)

and the following provisions, in case of no friendly settlement, national governments may establish a Claim Commission. Such an organism, made up of three members appointed by the parties[2], decides on the merit of the dispute and adopt a decision that is "final and binding if the parties have so agreed; otherwise the Commission shall render a final and recommendatory award, which the parties shall consider in good faith" (Art. XIX).

More specific provisions on the organizational issue may be found in the following 1975 Treaty on the Registration of Objects Launched into Outer Space. Expressly recalling what had already been outlined in the OST and the subsequent tools, the Art. II compels launching states to establish and maintain an appropriate registry where they have to register any object launched into Earth orbit or beyond. Alongside, Art. III provides that the UN Secretary General creates and maintains a further register in which the information delivered by the states has to be recorded. Differently from what was expressed with regard to the national registries, there shall be full and open access to the information reported in this register.

Likewise even the 1979 Treaty on the Moon and Other Celestial Bodies identifies the UN Secretary General as the overall caretaker of the system. Indeed, regarding the compliance of the provisions on the exploration and exploitation of the celestial bodies, Art. 15 affirms that if a state party has reason to believe that another country is not fulfilling the obligations of the Treaty may request consultations with that party and try to find a friendly settlement of the different positions. However, in case of difficulties, "any State Party may seek the assistance of the Secretary-General… in order to resolve the controversy".

Though the overall landscape of the space sector has undergone a deep-rooted change in the last decades, the inability of the international community to agree on an organic revision of the bunch of the Treaties briefly recalled make the governance model just outlined the current structure that governs the field at the international level. Therefore, beside a concise international organization, the system turns out to be characterized by a substantial lack of prescriptive provisions: the concrete governance of space activities definitely relies on sovereign and free choices of the national governments, which have adopted different regulations of the system of authorization, as well as the other organizational issues.

[2] Art. XV provides that each party select a member, while the Chairman of the Commission is chosen jointly. In case of no agreement on the position of the Chairman, the Treaty confers the last say to the UN Secretary General.

3. *The advent of the European Union's interest in the space sector and the spawn of a hazy and approximate governance*

The substantial lack of specific binding international regulations left European countries free to decide if and how to intervene in the space sector. As referred to by the European reports, currently 13 states of the EU have adopted space legislation. Based on their specific internal framework, other countries have simply established national space agencies and/or established the national registry of the launched objects. Other countries are still on the path toward a first normative intervention.

Within this context, it must be noted that since the beginning, the most active governments pushed and strove for the establishment of structural cooperation at supranational level. After two initial attempts, represented by ELDO (European Launcher Development Organization) and ESRO (European Space Research Organization)[3], in 1975 a restricted group of countries agreed on the creation of the European Space Agency (ESA)[4].

Established with the ultimate purpose "to provide for and to promote, for exclusively peaceful purposes, cooperation among European States in space research and technology and their space applications, with a view to their being used for scientific purposes and for operational space applications systems", since the origins ESA has tightly cooperated with Member States to set a unified space and related industrial policy, identifying space objectives, and supporting and integrating national programmes.

In an overall framework still characterized by state driven activities, ESA played a not irrelevant role in the system's governance. Through the establishment of mandatory programmes, to which each member has to contribute, and on the other hand, the provision of a "georeturn" policy, upon which funds that Member States provide to ESA are

[3] An historical analysis of the evolution of the European space programmes can be found in B. Doboš, *Geopolitics of the Outer Space* 90 (2019)

[4] The founding Member States that signed and ratified the Convention were Belgium, Denmark, France, West Germany, Italy, the Netherlands, Spain, Sweden, Switzerland, and the United Kingdom. On the history and role of ESA in the space sector, see e.g. C. Panico, *La cooperazione europea in campo spaziale* (1983) J. Tapio – A. Soucek, *The European Space Agency's Contribution to National Space Law, International Actors and the Formation of Laws*, 119 (K. Karjalainen et al. (eds.), 2022); C. Cellerino, *EU Space Policy and Strategic Autonomy: Tackling Legal Complexities in the Enhancement of the 'Security and Defence Dimension of the Union in Space, 8 European Papers*, 491 (2023); ESPI, *"A Field Guide on European Space Governance: European Institutional Actors, Their Roles, Responsibilities and Processes"*, 17 (2026).

returned as contracts to companies in those countries, the international organization concretely influenced the development of the space sector in the different Member States. From an organizational perspective, it must be observed that national governments take part in the Agency's Council via ministerial representatives. The tight interaction between the national and supranational levels is also granted by the participation of national delegates in several operational organisms.

That given, the birth of an advanced cooperation in the space sector operating in the traditional field of multilateral relations, coped with the attempt of the EU institutions to take part in the overall system. A few years later, in 1979 and 1981, the European Parliament passed two resolutions on the participation of the EU in space research and, above all, on the definition of a European space policy[5]. Another step towards direct intervention by the European institutions in space activities can be found in the subsequent 1986 Sigle European Act, which paved the way for the establishment of programmes and financial support in research and technological development.

At the beginning of the new century, the interest of the EU in the sector became even more blatant. Having started working on European space programmes like Copernicus, based on a cooperation with the ESA, in 2003 the EU Commission published a white paper on space as the new frontier for the EU, accentuating the importance of a European Space policy[6]. With specific regard to the governance issue, the document

[5] European Parliament Resolution on Community participation in space research, OJ C 127, pp. 42–43, 21 May 1979 and European Parliament Resolution on European space policy, OJ C 102, pp. 102-104, 12 October 1981. On the evolution of the EU approach to the space sector, see, Ö.E. Bayır – K.M. Akmaz – Ö. Aktaş, *New Space: The European Union's Evolving Space Policy and Changing European Space Ecosystem, Studia Europejskie – Studies in European Affairs* 115 (2021); E. Sigalas, *The European Parliament on Space: From Promoting Scientific Research to Supporting the Common Security and Defence Policy, The Emergence of EU Defense Research Policy*, 247 (N. Karampekios et al. (eds.), 2018); Id., *The European Union Space Policy, Oxford Research Encyclopedia of Politics* (2017); P. Breccia, *La regolamentazione dell'Unione europea in materia di attività spaziali: aspetti giuridici relativi alla proposta di Space Act a margine del lancio dei satelliti Galileo dal territorio degli Stati Uniti d'America, Eurojus.it* 261 (2025); M. Messina, *Quali futuri rapporti tra Unione Europea e Agenzia Spaziale Europea (ESA)?*, *Ordine internazionale e diritti umani,* 90, (2018).

[6] *The White Paper on Space: A New European Frontier for an Expanding Union*, COM(2003) 673. The document was the result of a green paper presented in January 2003 (COM(2003) 17) elaborated on the ground of the work carried out by a joint task force EU/ESA created in 2001 (COM(2001) 718). On this topic, see, S. Marchisio, *L'Europa e la politica spaziale*, *Affari esteri*, 641 (2003); M. Messina, *Quali futuri rapporti tra Unione Europea e Agenzia Spaziale Europea (ESA)?*, cit., 92

recognized the need to identify and encourage the most effective sharing of tasks and responsibilities between the EU, ESA, Member States, national agencies, and industry. In particular, considering the constituent process of that period, the Commission affirmed that ESA should eventually acquire legitimacy within the EU framework. In general, a new system of governance should have been enacted, in which "the roles of the players are clearly understood, the tools for careful coordination properly provided, and processes of accountability put in place. Duplication of work and structures must obviously be avoided. This system should command the consent and support of all stakeholders, respect their mission objectives and complementarity and conform to the principle of subsidiarity" (par 5.1). In this framework, besides the role played by ESA and European institutions, Member States, and their relevant space/research agencies should have been associated with the revision and ultimate adoption and implementation of the European Space Programme.

Even though integrating ESA in the EU framework was substantially halted because of the repeal of the 2004 EU constitutional Treaty, the cooperation agreement entered into force in 2004[7] provided for a joint organism, the so called "Space Council" made up of the member of the EU Council and the ESA Council, that would ease coordination between the two institutions (art. 8). The idea of structured cooperation was subsequently confirmed by the Treaty of Lisbon. Leaving aside any remark on the margins of the competences on space activities conferred to the European Union, it must be highlighted that art. 189, par. 3, TFEU expressly affirms that "the Union shall establish any appropriate relations with the European Space Agency"[8].

A similar framework has undergone a thorough update in the last decade. Starting from the Space Strategy approved in 2016, EU institutions strove to enhance their involvement in the sector's governance. As a result, in 2018 the Commission submitted a proposal for the EU Space Programme 2021-2027. Based on the art. 189 TFEU, the Regulation (EU) 2021/696 passed by the Council and Parliament on 18th April

[7] Council Decision of 29 April 2004 on the Conclusion of the Framework Agreement between the European Community and the European Space Agency Framework Agreement between the European Community and the European Space Agency (2004/578/EC).

[8] For an exam of the provision, among others, see e.g. F. Siniscalchi, *Commento all'articolo 189 TFUE*, *Codice dell'Unione europea operativo*, 1451 (C. Curti Gialdino, 2012); M.E. De Maestri, *Commento all'articolo 189 TFUE*, *Commentario breve ai trattati dell'Unione europea,* 1101 (F. Pocar, M.C. Baruffi, 2014); A. Conzutti, *La New Space Economy: profili costituzionali dell'integrazione europea in materia spaziale, DPCE Online*, 3363 (2021).

2021, aims at granting the capability of the Union to act in the space sector autonomously and competitively[9]. Indeed, among the purposes of the Regulations there was the simplification of the European regulatory framework, even by establishing a more definite system of governance[10]. In this context, it comes up the replacement of the GNSS Agency with the new EU Agency for the Space Programme[11].

In its Title IV, the Regulation tries to identify all the subjects involved in the Programme. First, the document defines the role of EU Member States by clarifying that the participation of national governments is not compulsory. Indeed, according to Art. 27, "The Member States may participate in the Programme". Those who participate are asked to "contribute with their technical competence, know-how and assistance, in particular in the field of safety and security". Through specific agreements, the Commission, and the newborn Agency may even entrust definite tasks to Member State organisations that have been designated by the related governments.

That given, the Regulation confers on the Commission the overarching responsibility for implementing the Programme. Within this framework, it is up to the Commission to determine the priorities and long-term evolution of the Programme and supervise its implementation, as well as ensure a clear division of tasks and responsibilities between the various entities involved in the Programme and coordinate the activities of those entities (Art. 28). As a residual clause, the Regulation confers on the Commission the direct management of any components or sub-

[9] Regulation (EU) 2021/696 of the European Parliament and of the Council of 28 April 2021 Establishing the Union Space Programme and the European Union Agency for the Space Programme and Repealing Regulations (EU) No 912/2010, (EU) No 1285/2013 and (EU) No 377/2014 and Decision No 541/2014/EU.
On this topic, see e.g. L. Orešković, S. Grgić, *The New EU Space Regulation: One Small Step or One Giant Leap for the EU?*, *Croatian Yearbook of European Law and Policy*, 92 (2021); P. Breccia, *La regolamentazione dell'Unione europea in materia di attività spaziali: aspetti giuridici relativi alla proposta di Space Act a margine del lancio dei satelliti Galileo dal territorio degli Stati Uniti d'America*, cit., 267; M. E. De Maestri, *L'UE nello spazio: competenze e opportunità*, *Liber amicorum Sergio Marchisio*, 129 (2022).

[10] Art. 26, par 1, lett. a) affirms that among the principles governing the Programme it comes up a "clear distribution of tasks and responsibilities between the entities involved in the implementation of each of the Programme's components and measures, in particular between the Member States, the Commission, the Agency, ESA and EUMETSAT, building on their respective competences and avoiding any overlap in tasks and responsibilities".

[11] On this specific topic, see M.K. Davis Cross, *'United Space in Europe'? The European Space Agency and the EU Space Programme*, *European Foreign Affairs Review*, 31 (2021).

components of the Programme "not entrusted to another entity".

Besides the Commission, the new Agency is the main actor in the Programme. Generally speaking, EUSPA has to undertake communication, market development, and promotion activities for all three main programmes: Galileo, EGNOS, and Copernicus, as well as the development of downstream applications based on the Programme's components (Art. 29). Further, the Agency is called upon to provide an overarching coordination of user-related aspects of GOVSATCOM and, above all, through its Security Accreditation Board, ensure the security accreditation of all the Programme's components.

Last but not least, the document expressly regulates the role of the ESA[12]. The solution, which represents an innovation compared to the past, confirms the tight relationship between the EU activities and the broader European cooperation on space. Nevertheless, the role of ESA results substantially downsized. From a first aspect, Art. 30 opens to the contribution of ESA but emphasizes the need that, the Agency being an organization participated by non-EU countries, the interests of the Union are protected. Second, the advent of EUSPA has triggered a restriction of the tasks previously performed by the Agency. In particular, ESA should deal with technical tasks such as the design, development, and construction of infrastructures and upstream research and development activities. Further competences could be envisaged, but the Regulation expressly rules out any duplication of activities already entrusted or performed by other subjects of the governance system.

At first sight, the intervention of the Regulation has brought to a more structured governance of the space sector in Europe. The provisions clearly identify the different roles and responsibilities of the various actors concerned. Relying on the traditional approach of international cooperation in space activities, the Union strives to take advantage of the expertise of ESA. Within the ultimate goal of pursuing efficiency and enhancing competitiveness of the European actors, such a solution should prompt a simplification of the previous system.

However, from an in-depth analysis, it comes to light a more complex framework. Actually, Regulation (EU) 2021/696 specifically focuses on the EU Space Programme, which is the result of the systematization of various specific programmes. After all, the ultimate source of legitimacy of the Regulation is represented by Art. 189 TFEU, which insists on the

[12] On the new relationship between ESA and EU, see A. Conzutti, *La New Space Economy: profili costituzionali dell'integrazione europea in materia spaziale*, cit. 3370.

management of specific space activities. The same provision does not provide for a general competence in space for the European institutions; rather, it expressly excludes "any harmonisation of the laws and regulations of the Member States".

As a result, the current landscape is characterized by a multifaceted system. On the one hand, it can be observed that the referred complex multilevel structure operates in the field of EU programmes. On the other hand, national governments retain full jurisdiction over the overall development of the space sector in their systems[13]. In this regard, they continue cooperating with each other and, in particular, with ESA[14].

4. *The proposal for the Regulation of space activities and the new governance of the space sector*

The current landscape is characterized by two different levels of activity. Regarding the European Space Programme, the system confers the main role to the Commission and EUSPA, letting Member States and ESA in a secondary position. On the contrary, concerning the overall development of the commercial sector, national governments and ESA still continue to play a fundamental role. Such a framework stems from the specific legal basis which EU institutions identified when it was elaborated the Regulation (EU) 2021/696. As already mentioned, Art. 189 TFEU limits the competence of the EU to promote joint initiatives, support research, and technological development by adopting policies and measures that can take the form of specific programmes.

The Regulation proposal submitted by the Commission assumes to prompt a comprehensive change[15]. Moving from the recognition of the

[13] For instance, it can be mentioned the participation in the Artemis Accords. On the topic, among others, see V. S. Mosteshar, *Artemis: the Discordant Accords, 44 Journal of Space Law*, 591 (2020); R. Deplano, *The Artemis Accords: Evolution or Revolution in International Space Law?, 70 International and Comparative Law Quarterly*, 799 (2021).

[14] On the steady cooperation between ESA and national governments, among others, see e.g. J. Tapio – A. Soucek, *The European Space Agency's Contribution to National Space Law*, cit. 124 (2022)

[15] For a first analysis of the proposal, even with some remarks on the governance issue, see K. Muti, A. Grillo, S. Marchisio M. Nones, *The Proposal for an EU Space Act: An Italian Perspective, 25/13 IAI Documents* (2025); L.J. Smith, *Constitutionalisation of Space by the European Union: A Perspective, 50/ Special Issue Air & Space Law*, 584 (2025); A. Miglio, *The EU Space Act proposal: A GDPR for outer space?, European Law Blog* (2025);

increasing demand for space-related data and services and, above all, the most recent boost to the space economy, acknowledged the risks of a fragmented market characterized by the different approaches of the single Member States, in order to support and consolidate the development of the European space sector, the Commission decided to move towards a single European legal framework related to the supply of data and services. Recalling the recommendations expressed in Mario Draghi's Report on the future of the European competitiveness, the Commission proposes to establish the single market for space-related activities[16]. Within this overarching purpose, the legal basis of the intervention no longer relies on Art. 189 TFEU; rather it calls directly for the application of Art. 114 TFEU on the establishment of the single market.

As a result, except for specific subjects like national security and defence, which keep being reserved jurisdiction of national governments, the space sector should fall into the bunch of competences attracted to the EU level. Similar reforms should prompt direct consequences to the current system of governance, with a new definition of roles and competences of the different actors involved.

From a general point of view, the draft Regulation contains a specific title on the governance structure. However, upon deeper analysis, it must be observed that several provisions on organizational aspects can be found even in other parts of the document.

In the attempt to sketch out the new overall structure, some preliminary remarks must be addressed to the subjects concerned. The main actor is represented by the Commission. Unlike the model still in force, the EU institution improves its relevance by combining the traditional regulatory and supervision competences with new administrative/management tasks. An enhanced role can be detected even regarding EUSPA, on which it is expressly recognized the need to update the governance and operational structures (recital n. 115).

Besides the EU actors, the overarching position is played by the Member States. Differently from the layout of the Regulation on the European Space Programme, here the role of national governments is

C. Cellerino, *La proposta di "EU Space Act" e il suo possibile impatto, Diritto Costituzionale,* 87 (2026); M Matassa, *I profili giuridici della cibersicurezza nello Spazio, ivi*, 117; M.E. De Maestri, *La governance multilivello delle attività spaziali, ivi*, 79.

[16] Proposal for a Regulation of the European Parliament and of the Council on the Safety, Resilience and Sustainability of Space Activities in the Union, Explanatory Memorandum, par. 1. On the Mario Draghi's Report, see A. Miglio – C. Nota, *Implementing the Draghi Report's Recommendations on Defence and Space* (2024).

not merely prospective: it is upon the states the management of the activities internal to their jurisdiction. To this purpose, the document expressly focuses on the national governance by identifying two different bodies: the "competent authorities", in charge of the overall control of the application of this Regulation (recital n. 106), and the "qualified technical bodies for space activities", responsible for the technical assessment of sensitive issues (recitals 110 ff.).

In this framework, the position of ESA seems to be even more downsized. Even though both ESA and EUMETSAT are identified as "key partners to the Commission, the Agency, and the Member States", they are attracted by the regulatory intervention of the Commission inasmuch as, being responsible for implementing some elements of the European programmes, they are expected to comply with the Regulation (recital 31). Indeed, despite being autonomous international entities, they no longer stay at the same level as the EU institutions: in order to proceed with the 'mutual' cooperation, they have to abide by the rules enforced by the EU.

That given, the draft Regulation splits the different areas of intervention by classifying space activities depending on the different nature of the subjects and the ownership of the assets concerned.

So, delving into the new governance sketched out, regarding space operator established in the Union, so called "Union space operators", which do not operate using nor taking advantage of Union-owned assets, the Regulation provides for a general responsibility of Member States on the delivery of authorizations for carrying out space activities. In particular, the document states that authorizations are issued by the national "competent authorities", which have to take into due account the advice of the qualified technical bodies (Art. 6-7). Regarding the technical assessment, states may even choose to confer the task to international organizations (like ESA) or the EU Agency (Art. 8).

Within the Title dedicated to the governance, the Regulation specifies that each Member State has to designate or establish a public authority to act as the competent authority. Such organism handles the authorisation and supervision of Union space operators and any appropriate market surveillance activity directed at granting the overall compliance of this Regulation (art. 28, par 1). To this end, the authorities are given specific tasks and related powers.

In particular, they shall control the application of the requirements laid down in the Regulation; carry out audits, conduct investigations and ask qualified bodies for a technical assessment (Art. 29). Correlated to the

supervisory task, they also have corrective and sanctioning powers (Art. 30). Among the other competences, they may order to cease conducts which they deem to be contrary to the Regulation and even to remedy the shortcomings identified. In case of non-compliance with the Regulation, the authorities may impose or request a relevant administrative or judicial body to impose an administrative fine, as well as temporarily suspend or withdraw the authorization already issued.

In order to grant the efficiency of these organisms, the Regulation expressly states that Member States shall provide them with the independence, expertise, financial and human resources, operational capacity, and powers necessary to perform the functions conferred (art. 28, par. 2).

Besides these organisms, it is up to the public authorities to designate, assess and monitor the qualified technical bodies for space activities established to deal with technical evaluations (art. 32)[17]. Once more, to grant the well-functioning of the system, these organisms must be public entities (Art. 35) and the public authorities that designate them must avoid any form of conflict of interest and operate fairly.

These two organisms should ensure the implementation of the Regulation at the national level. Because of the expanding level of interconnection between national jurisdictions, gendered by the cross-boundary activity of the space operators, the proposal of the Commission stresses the need for steady cooperation between the different subjects concerned. In this context, it can be referred the idea of the establishment of sectorial groups of technical bodies, which could enhance the dissemination of best practices, and the structured cooperation between the national competent authorities aimed at pursuing coherence in the Regulation's enactment (Art. 29, par 1, lett. d)).

That given, regarding all the other cases, that is EU space operators that operate using or taking advantage of Union-owned assets, non-EU space operators and international organizations, the main roles are played by the Commission and the EUSPA. Regarding the first case, EU operators have to be formally authorized by the Commission upon a preliminary

[17] This occurs in case the Member State prefer carrying out the technical assessment internally, without referring to the Agency or other international organizations. In case of technical evaluation on issues related to the resiliency of space infrastructures, the competent authorities must refer to the technical bodies established upon the Directive (EU) 2022/2555. With regard to the assessment and monitoring of these bodies, the Regulations admits the possibility to take advantage of the activity performed by the national accreditation body established on the basis of the Regulation (EC) 765/2008.

technical evaluation carried out by the Agency (Artt. 11-12). For non-EU operators and international organizations, a formal authorization is not required. Nevertheless, they can be admitted operating only if they comply with the same requirement settled regarding EU operators. Even in this case, the Regulation asks the Commission and the Agency to assess the fulfilment of the conditions (Artt. 14 ff.).

In particular, the Agency has to set up a Union Register of Space Objects (URSO) for the registration of all kinds of operators (Art. 24)[18]. Upon registration, the Agency issues and delivers an electronic certificate ('e-certificate') that identifies the space missions and space objects by attesting their conformity with the requirements laid down in the Regulation (Art. 25). Besides such tasks, the Regulation gives the Agency a general supervisory duty, as well as technical assistance and advisory functions towards the Commission (Art. 40). In order to accomplish its tasks, the Regulation provides for the establishment of two internal organisms: the Compliance Board and the Board of Appeal.

The first body is in charge of technical evaluation activities preparatory to the delivery of authorizations and ongoing supervision. It is composed of one representative of each Member State and one representative of the Commission (Art. 45). Representatives of the Union agencies or bodies and third countries or international organisations may, exceptionally, be invited to attend the meetings of the Compliance Board as observers. However, the real engine of the Board is represented by its internal organisms. Indeed, as provided by Art. 44, the body works in three technical board configurations: (a) the Safety Compliance Technical Board; (b) the Resilience Compliance Technical Board; (c) the Environmental Sustainability Compliance Technical Board. These bodies are composed of joint examination teams comprising staff members from the Agency, competent authorities and qualified technical bodies for space activities, and are supported by a technical secretariat. After completing their technical assessments, these committees submit reports to the Compliance Board.

A different structure characterizes the second body. The Board of Appeal is composed of six members and six alternates which shall be appointed, from a list of qualified candidates established by the Commission, by the Administrative Board, on the basis on their relevant expertise in the fields of space law or space activities (Art. 46). Its members are independent and

[18] On this specific issue, see K. Muti, A. Grillo, S. Marchisio, M. Nones, *The Proposal for an EU Space Act,* cit. 4; C. Cellerino, *La proposta di "EU Space Act" e il suo possibile impatto*, cit., 98.

are not bound by any instructions. Similar requirements are fundamental to grant the effectiveness of their activity. According to Art. 46, the Board is responsible for deciding on appeals against decisions of the Agency.

Besides the identification of the different tasks to be conducted by the Agency and the Commission, the Regulation focuses on the functions conferred. Indeed, regarding the overarching supervision of the system, the document gives significant powers to both organisms. In addition to the possibility to request information, in the exercise of their duties, the Commission and the Agency can independently carry out investigations at the space operators under their surveillance (Art. 50), even by conducting on-site inspections (art. 51). In case the investigations reveal an infringement of the requirements laid down by the Regulation, the Agency may make a proposal to the Commission to formally establish the existence of an infringement of the rules and consequently adopt one or more of provisional and sanctioning measures (Artt. 54-55). In particular, the Commission may order the infringement to cease, even by adopting interim measures to avoid any irreparable damage. Within this framework, upon the proposal of the Agency, the Commission may suspend or even withdraw the authorization, as well as inflict an administrative fine or a periodic penalty payment[19] and, in case, issue a public notice indicating the Union space operator responsible for the infringement and the nature of the infringement.

Complementary to the system briefly sketched out is the establishment, provided for in the Title on Technical Rules, of a Union Space Resilience Network ('EUSRN'). Such a body aims at supporting coordination and exchanges between the Agency and the competent authorities in fulfilling their respective mandates as regards Union-owned assets with specific attention to sensitive issues related to security (Art. 94). To this end, the organism is composed of representatives of competent authorities, the Commission, the Agency and the European External Action Service (EEAS)[20].

From a different perspective, even if the proposal does not establish any specific body in charge of the related tasks, it must be mentioned the establishment of a Union Space Label (Art. 111). Aimed at promoting

[19] In case of pecuniary fine, the Regulation confers the Court of Justice of the European Union unlimited jurisdiction to review decisions of the Commission (Art. 56).

[20] According to the Regulation, where appropriate, the Commission or the Agency may invite representatives of other Union institutions, agencies or bodies, in particular the European Union Agency for Cybersecurity (ENISA), the European Defence Agency (EDA), or the European Union Military Staff (EUMS) to attend specific sessions of the Network.

enhanced voluntary adherence to high standards of protection of space activities and, in the end, supporting the achievement of goals and purposes pursued by the Commission, the European Space Label shall be managed by the Agency and the Commission. Likewise, the authorization process, space operator that intends to get the Label, applies to the Agency. Upon the technical evaluation carried out, the Agency provides a detailed opinion to the Commission, which has the final decision on the merit.

Within the overall framework, a closing remark must be addressed to the role of the ESA. Identified as one of the fundamental partners of the other subjects involved in the governance system, the Regulation demands the signature of an agreement between the EU and the Agency to advance the objectives pursued by the Regulation. In particular, due to its technical expertise, ESA should play an important role in implementing the requirements laid down in Title IV on technical standards, and the practical and operational arrangements for ensuring the control of the application of such requirements (Art. 108). Besides these functions, the Agency may be requested by Member States to act as a qualified technical body for space activities. Lastly, it may be required by the Commission to participate as an observer in technical bodies and organisms that may be established under the umbrella of the Regulation.

5. *The consequences of the Regulation on national governance systems: the cases of France and Italy*

The new model of governance of the space sector, briefly depicted, is going to prompt a significant change in the organizations enacted at the national level. As previously mentioned, based on the general clauses provided in the international Treaties still in force, national governments have progressively established and consolidated autonomous systems. Because of the lack of any specific provision, they have been substantially free to choose the most appropriate solution. As the current EU Regulation on the European Union Space Programme is circumscribed to the management of the European programmes, states maintain their original full jurisdiction.

As analysed before, in case of approval of the current proposal, the landscape could deeply change. In a nutshell, regarding the national systems, the Regulation demands the establishment of a 'competent

authority', which should be responsible for the delivery of authorizations and the overarching supervision of the internal system, and an array of qualified technical bodies for space activities, which should cooperate with the first subjects by performing technical evaluation. Unlike the previous legal interventions, the current proposal fixes some requirements for those bodies. First of all, they must be public entities. Second, they must be and act as independent organisms and must be given the competences and financial and human resources necessary to carry out their tasks. Beyond the administrative and technical evaluation, they have to be able to conduct investigations, inspections, and also impose, or request, sanctions in case of infringement of the rules.

In this framework, those countries that have already established an organic system of governance may be compelled to reconsider it. It is the case, among others, of France and Italy.

Throughout Europe, the French system represents a paramount case. Since 2008, the country has been characterized by a comprehensive regulatory framework. The Act n. 2008-518 on space operations, the statutory instrument Decree n° 2009-643 on authorizations, and its subsequent amendments have established a complex system where the national government plays an overarching role[21]. In particular, space activities must be authorized by the Minister who is in charge of space sector, currently the Minister of Higher Education and Research. The delivery of the authorization results from a complex process that involves different subjects. Indeed, the application must be submitted to the Ministry of Armed Forces for assessment of possible conflicts with national defence interests. In parallel, the dossier is submitted to the Centre National d'Etudes Spatiales (National Center for Space Studies) (CNES) for technical evaluation.

The CNES was established in 1961 under the authority of the Prime Minister and, first of all, is a research institution with financial autonomy[22]. In general, the system has conferred pivotal tasks to the Centre. Among the different functions, the CNES collects information on national and international space activities, prepares and proposes to the Interministerial

[21] On the French regulatory framework, see e.g. L. Rapp, *Une loi spatiale française, AJDA* 1755 (2008); B. Schmidt-Tedd – I. Arnold, *The French Act relating to space activities from international law idealism to national industrial pragmatism, ESPI Perspectives No 11* (2008); G. Carminati, *French National Space Legislation: A Brief "Parcours" Of A Long History, 36-1 Houston Journal of International Law*, 6 (2014).

[22] About the role of the CNES, see G. Carminati, *French National Space Legislation*, cit. 2 (2014).

Committee for Scientific and Technical research programs of national interest in space, supervises on implementing the research projects, gives support to the Minister of Foreign Affairs on problems of international cooperation.

That given, such a system risks to be not consistent with the model outlined by the EU proposal. Indeed, the CNES may well be considered a "qualified technical body for space activities", but is not independent from the national government and can undergo severe conflicts of interest. On the other hand, eventually the competent authority in charge of the delivery of authorizations is the national government itself. This appears to be not in line with the requirements established by the EU regulation, which demands a public entity independent from any kind of external influence.

Last, it must be observed that, according to the French law, the national system applies to any person or company that: (1) launches from the French territory or from a facility under the jurisdiction of France, or who plans to return an object into national territory or onto a facility under French jurisdiction; (2) any French operator, regardless of where they launch from; (3) any French person or corporation headquartered in France, operator or not, that will launch or even just command a space object; and (4) anyone previously authorized under French law who wants to transfer control or command of a space object. Such a solution could prompt severe problems regarding the express provision of Art. 6 of the EU Regulation proposal, that affirms that the authorization "shall be issued by the competent authority of the Member State in which the applicant is established and, as applicable, by the competent authority of the Member State where that applicant intends to operate or respectively launch, if different from the Member State of establishment" and every Member State shall recognise the authorisations issued by another Member State.

Similar remarks can be made regarding the Italian model. Unlike France, only last year Italy passed a statute aimed at providing comprehensive regulation of the space sector. However, the current legal framework represents the last step of a progressive evolution. Since the end of the 50s, the national government established a centre for space research as part of the Research National Centre (CNR). In 1988, the Centre for Space Research was replaced by the Italian Space Agency (ASI). The Law n. 186/1988 identified the Agency as a public entity characterized by technical expertise, placed under the supervision and direction of the Minister of Research with competence on the implementation and supervision of the national space plans. During the decades ASI underwent several reforms and eventually was put under the authority

of the Prime Minister. Gradually, the Agency was identified as the point of contact for international cooperation in the space sector, establishing significant relationships with ESA and other institutions as well as national stakeholders. In addition, it was empowered with the management of the registry of space objects provided by the 1975 Treaty.

Besides the reform of ASI, the Italian system has gradually structured a steering body. Originally, a steering board composed of the Ministers more directly concerned by space activities was created, that is research, foreign affairs, defence, industry, transport, and communications. Such a body was empowered to identify the policy guidelines of the government. In 2018, the organism was renamed Interministerial Committee on Space Policies and Research and rose to include the head of the government and representatives of 11 further ministries. Besides the definition of the general strategies on space, the body should coordinate and direct the activity of ASI as well as promote space programmes.

Similar organization has not been revised by the recent law regulating the sector[23]. Indeed, as it occurs in France, even in Italy space operators must be authorized. The license is delivered at the end of a complex process. The application must be addressed to the ASI, which is requested to carry out a technical assessment of the fulfilment of the requirements laid down in the statute. The Agency submits its advice to the Interministerial Committee on Space, which has to issue its evaluation. Such advice is preliminary to the final decision of the authority in charge of the delivery of the license, that is the Prime Minister or the Minister by him/her expressly delegated. In case of contrasts inside the Interministerial Committee, the evaluation of the application may be deferred to the Council of Ministers.

The framework briefly depicted shows that, likewise the French model, the Italian case is characterized by an organism that could be classified as a "qualified technical body for space activities". However, the authority responsible for the delivery of authorizations and the overall supervision of the system is not a public entity, autonomous and independent; rather being the government itself.

[23] Law n. 89/2025. On this topic, among others, see S. Zolea, *La legge n. 89/2005 e la proposta di Space Act dell'Unione Europea nell'assetto di governance spaziale del mondo multipolare, 47 Rivista italiana di Diritto del turismo*, 76 (2026); S. Marchisio, *Le novità del 25 giugno 2025: la Legge n. 89 sull'economia dello spazio e la proposta di Space Act dell' Unione europea", 20 federalismi.it,* IV (2025); T. E. Frosini, *Il nomos dello spazio, 27 Federalismi.it,* 52 (2025); and all the contributions hosted in S. Zolea - T. Graziani (eds.), *Spazio: verso una legge italiana,* (2025).

Last but not least, similar to France, even the Italian statute provides for the application of the current regime to all the space activities carried out by operators of all the nationalities that operate in Italy as well as Italian operators that want to act outside the national territory (Art. 3).

6. *Conclusive remarks*

The object of the Regulation proposal issued by the EU Commission is the adoption of a set of provisions on "the safety, resilience and sustainability of space activities in the Union". Since its first words, the explanatory memorandum focuses on the idea of the need to regulate space activities and operators with the aim of building up an internal market that could foster the European space economy. In other words, the initial declarations insist on a regulation of the market.

Nevertheless, in the preliminary description of the original framework, the memorandum refers to the national regulations already adopted, as well as the international rules that demand from the states a comprehensive surveillance of space activities. Later on, the introductory analysis recalls two communications from the EU Commission that focus on the need to govern space traffic. Hence, besides the substantial aspects, the Commission proves to be aware of the significance of the issue related to the governance of the system.

Within this framework, the proposal expressly deals with the structure of bodies and organisms that should assume the responsibility for the well-functioning of the system. The present contribution has tried to identify the most important aspects of the model proposed. As it has been shown, the reform pursued by the Commission prompts a radical change in the current system.

At the end of the analysis, beyond any inquiry into the substantial competence of the Union on the space sector, the underlying question is whether such a solution is coherent with the overall European regulatory framework and, above all, shall be efficient.

Moving from the consideration of the pervasive limitation of the margin of autonomy of national governments, regarding the first aspect, it is interesting to provide a comparison of the proposal with other interventions in correlated fields or on subject matters that are currently topical. For instance, it can be mentioned the Regulation (EU) 2018/1139

that provides common rules in civil aviation and establishes the European Union Aviation Safety Agency. Even though both regulations insist on issues related to the activities of aerial vehicles, in the intervention on civil aviation, the Regulation establishes a European organism with monitoring and management powers but does not intervene on the system that Member States have to adopt. In other words, the Regulation confines itself to identifying the division of competences between states and EU institutions, establishing a new European Agency with supporting functions, but does not interfere with national structures.

Likewise, the most recent regulations on AI and digital services give a wider margin of appreciation to national governments regarding management issues[24]. Both Regulation (EU) 2022/2065 on the single market of digital services and Regulation (EU) 2024/1689 on AI systems establish new bodies operating at European level and demand the establishment of competent authorities by the different Member States. From a different point of view, the Regulations identify the need for structured coordination and cooperation between the two levels of government. However, these documents seem to confer national governments more space to detect the most appropriate solution at the national level.

In this framework, the efficiency of the solution adopted by the Commission in the Regulation proposal can be assessed. Indeed, the division of competences based on the nature of the operator or the 'ownership' of the assets risks multiplying the authorization procedures. In a context characterized by the intervention of a plurality of subjects in the same space mission, such a division of competences risks being burdensome. From a different perspective, the detailed provisions on the national governance system could compel Member States to revise solutions that have already been adopted and may have proven to be efficient for the development of the internal market.

In this context, the idea of giving national governments a wider manoeuvre margin to find their route, maybe even by adopting a Directive instead of a Regulation, could result not only more consistent with the principle of subsidiarity but also more appropriate to foster the development of the internal market on space activities.

[24] An express reference to the AI Act can be found in C. Cellerino, *La proposta di "EU Space Act" e il suo possibile impatto,* cit., 95

Stefania Paladini

The Geopolitics of Space Sustainability. Reviewing the EU Space Act

Summary: 1. Introduction - 2. The EU regulatory framework in the space sector pre-2025 Act - 3. The Three Pillars of the EU Space Act - 4. The Geopolitical Consequences: The Brussels Effect in Orbit - 5. Geopolitical implications of the EU Space Act - 6. Conclusion: Towards a GDPR for Outer Space?

1. *Introduction*

The global space economy has been in full expansion over a decade, with technical breakthroughs and achievements growing even faster than record investments and sector's revenues[1]. The landscape has radically transformed. And if space was traditionally the exclusive domain of a few nation-states, it is now driven by private companies and over 80 countries joining the once-exclusive 'space club'[2]. The regulatory framework, however, did not follow.

To fully appreciate the extent and the impact of the EU Space Law of June 2025, it is important to understand where the sector stands at present.

At the international level, space law still relies on five core United Nations treaties developed in the 1960s and 1970s, overseen by the United Nations Office for Outer Space Affairs (UNOOSA). The foundational instrument is the 1967 Outer Space Treaty (OST), which established space as a "*province of mankind,*" prohibited national sovereignty over

[1] While estimates vary considerably due to the well-known challenges of measuring space economy, forecast for 2026 shows + 48% to $12.4 billion in private investments compared to 2025 and about *$626 billion* in 2025 in market size (Space Foundation, 2026; Novaspace, 2026)

[2] Paladini, S. (2024) 'The Case for Space Sustainability: Crowded Orbits, Debris Crisis, and Global Space Governance', in S. Zolea (ed.) Comparative visions in space law. Rome: Roma TrE-Press, pp. 243-252. Available at: https://doi.org/10.13134/979-12-5977-376

celestial bodies, and forbade weapons of mass destruction, a provision that legally challenges the deployment of anti-satellite weapons[3] but did not stop countries to experimenting them[4]. This was followed by the 1968 Rescue Agreement (RA), the 1972 Liability Convention (LC), the 1976 Registration Convention (RC), and finally the largely failed 1979 Moon Agreement, which creates difficulties today due to its definition of the Moon as the "Common Heritage of Mankind"[5].

This existing legal framework is increasingly insufficient and outdated.

A primary flaw is its inability to regulate non-state actors and commercial activities. Today, many active satellites belong to private companies that operate virtually outside the direct remit of international treaties. This severe legal vacuum was exposed on February 10, 2009, when the privately owned American Iridium 33 satellite collided with the derelict Russian Cosmos 2251 satellite[6], an issue the 1972 Liability Convention was unable to resolve. In addition to liability, the lack of enforceable international law exacerbates environmental crises in the Earth's orbits. The surge in space objects has accelerated the accumulation of non-functional debris, heightening the risk of the Kessler Syndrome, i.e., a cascading series of collisions with severe environmental and economic repercussions[7]. While bodies like the United Nations Committee on the Peaceful Uses of Outer Space (UN COPUOS) routinely suggest best practices, these measures are neither legally binding nor universally adopted

[3] Freedland, S., & Blake, D. (2017, July 9). As the World Embraces Space, the 50 Year Old Outer Space Treaty Needs Adaptation. The Conversation. Available at https://theconversation.com/as-the-world-embraces-space-the-50-year-old-outer-space-treaty-needs-adaptation-79833; Chatterjee, P. (2014). Legality of anti-Satellites under the space law regime. *Astropolitics*, *12*(1), 27-45.

[4] Paladini, S. (2023) 'Unsustainable Wars? The Use of Weapons in Lower Earth Orbit', in B. Cayli Messina (ed.) Environmental injustice and catastrophe: how global insecurities threaten the future of humanity. De Gruyter. Available at: https://doi.org/10.1515/9783111081687.

[5] Listner, M. J. (2011). International space law: an overview of law and issues. *New Hampshire Bar Journal*, *52*(1), 62-71.

[6] Paladini, S. (2019) The New Frontiers of Space. London: Macmillan; Baker, H. A. (1988). Liability for damage caused in outer space by space refuse. *Annals Air & Space L.*, *13*, 183.

[7] Kessler, Donald J.; Cour-Palais, Burton G. Collision frequency of artificial satellites: The creation of a debris belt. *Journal of Geophysical Research: Space Physics*, 1978, 83.A6: 2637-2646; Virgili, B. B., Dolado, J. C., Lewis, H. G., Radtke, J., Krag, H., Revelin, B., ... & Metz, M. (2016). Risk to space sustainability from large constellations of satellites. *Acta Astronautica*, *126*, 154-162

by space players. The United States' Orbital Debris Mitigation Standard Practices (ODMSP), established in 2001, provides a framework, but it was not designed for operations beyond geosynchronous orbit (GEO) and lacks universal international enforcement. Thus, without unified legal mechanisms, managing crowded orbits and the environmental impacts of rocket launches[8] remains largely dependent on voluntary compliance.

In this context, private actors operate in a partial legal vacuum, which is the reason sovereign states have been developing their own national space laws to attract business and establish favourable legal ecosystems. This has led to a highly diversified landscape with competing national visions, starting from the U.S. and its Commercial Space Launch Competitiveness Act of 2015 (Title 51), which explicitly entitles U.S. citizens to possess, transport, use, and sell commercial space resources. Crucially, this legislation filled a legal vacuum and allowed private participation in resource extraction without technically infringing upon the 1967 OST provisions[9]. Recognising the economic potential, Luxembourg enacted its own 2017 Space Law (Loi du 20 Juillet 2017), which mirrored the U.S. text regarding resource ownership.

Other countries followed soon after, each one with their own legal approaches, target activities and often conflicting framework, contributing to create additional risks to global space governance.

If states possess vastly different legal regimes, it hinders the existence of a level playing field for operators and creates legal uncertainties. Moreover, in a situation of diverse national legal frameworks, the sector will likely witness phenomena familiar to the law of the sea, such as "flags of convenience" and "forum shopping," where commercial networks establish themselves in terrestrial jurisdictions that offer the most advantageous or lenient regulations[10].

It is in this context that the European Commission published, on 25 June 2025, the proposal for the first-ever EU Space Act, COM(2025)335), a groundbreaking regulation under any perspective. It was by no means the first deliberation or programmatic act of the EU Commission on space, quite the opposite. But nothing else had the breadth, the coherence, let alone the weight and far-reaching impact the Space Act is going to have.

[8] Ryan, R. G., Marais, E. A., Balhatchet, C. J., & Eastham, S. D. (2022). Impact of rocket launch and space debris air pollutant emissions on stratospheric ozone and global climate. *Earth's Future*, *10*(6), e2021EF002612.

[9] Hobe, S. (2007). Legal aspects of space tourism. *Nebraska Law Review*, *86*(2), 6.

[10] Marboe, I. (2015). National space law. In *Handbook of space law* (pp. 127-204). Edward Elgar Publishing.

In formal terms, the Act is structured around Article 114 of the Treaty on the Functioning of the European Union (TFEU), regulating the internal market.

The Space Act objectives are to establish a Union legal framework for the provision of space-based data and services by Union space operators; to ensure the trackability of space objects and reduce the generation of space debris; to create a risk assessment framework tailored to space infrastructure; and to establish a common method for calculating the environmental impact of space activities in the Union. These objectives translate into three operational pillars, safety, resilience, and sustainability, resulting, when taken together, in a comprehensive and overarching space governance architecture.

Importantly, the Space Act is modelled on the regulatory philosophy underlying the General Data Protection Regulation (GDPR). Like the GDPR, it applies not only to operators established in the EU but to any operator providing space-based services to end-users located within the Union, regardless of where the operator is domiciled. This extra-territorial logic, combined with a GDPR-style enforcement framework featuring significant financial penalties and the involvement of a designated supervisory authority (the European Union Agency for the Space Programme, EUSPA, in this specific case), has drawn attention from non-EU stakeholders, particularly in Washington, Beijing, and New Delhi, for the numerous and not immediately obvious geopolitical repercussions of some of its norms.

Discussing them is the objective of this article.

Section 2 goes through a quick summary of what has been the EU regulatory framework before the Act, to be able to put it into the correct perspective. Section 3 provides a structured analysis of all three pillars of the Act. Section 4 examines in specific the sustainability pillar in depth, tracing its connections to the broader EU Circular Economy Regulatory Stack, including the ESPR, the CRM Act, the CEAP II framework, and the Digital Product Passport (DPP) regime, highlighting its potential wide-ranging, disruptive effects. Section 5 analyses the geopolitical consequences of the regulation, with a focus on the Brussels Effect and strategic autonomy implications. Section 6 provides a dedicated discussion of the reactions from the United States, China, and India, while Section 7 concludes with reflections about the future of global space governance.

2. *The EU regulatory framework in the space sector pre-2025 Act*[11]

Nothing better than the following statement in 2016 synthesises the heart of the matter in the European space sector before 2025: the presence of different and at time contrasting interests, especially on the security side, and the need of a more effective integration strategy to exploit synergies. *Space services can strengthen the EU's and Member States' capacity to tackle growing security challenges and improve the monitoring and control of flows which have security implications. Most space technologies, infrastructure and services can serve both civilian and defence objectives. Although some space capabilities have to remain under exclusive national and/or military control, in a number of areas synergies between civilian and defence can reduce costs, increase resilience and improve efficiency. The EU needs to better exploit these synergies.* [12]

Historically, the European Union's approach to regulating space activities (among others) has been characterised by fragmentation, soft law instruments, and a heavy reliance on member state legislation rather than a unified super-national framework. This governance model has evolved incrementally over decades, shaped by early geopolitical dynamics, industrial policy ambitions, security concerns, and, more recently the growing commercial importance of space infrastructure.

The roots of this fragmented landscape date back to the early days of the European Community, when space was viewed strictly as the business of individual countries and domestic policies [13].

Nonetheless, the successful creation of CERN in 1959 paved the way for a more comprehensive regional cooperation strategy on high-tech activities, including space. This led soon after to the establishment of two separate organizations in 1964: the European Space Research Organisation (ESRO) and the European Launcher Development Organisation (ELDO), which led the preeminent segments of the early space industry. The formation of these institutions relied on two distinct factors: a decisive political drive for ELDO, in both a military and commercial function, and the scientific community's recognition of the high research stakes

[11] The content of this section is in large part an abridged and updated synthesis of Chapter 3, in Paladini, 2019, The New Frontier of Space, London: Macmillan.

[12] European Commission. (2016). Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions 'European Defence Action Plan'. COM 2016/ 950.

[13] Paladini, S. (2019) cited.

for ESRO[14]. These foundational organisations ultimately evolved into the European Union Space Programme and the European Space Agency (ESA), respectively. Established in 1975, ESA represents even today a unique, highly successful multinational space agency and a working model for regional scientific and political cooperation [15].

However, a crucial distinction defined the pre-2025 landscape: while ESA worked closely with the EU on a series of flagship projects and received a substantial portion of its operational budget from EU allocations, it remained an independent multinational institution with its own leadership, budget, and mission, rather than a formal EU agency[16]. To bridge this institutional divide, the EU and ESA entered into a Framework Agreement in 2004, resulting in the publication of the joint European Space Policy (ESP) in 2007. This policy was the first major attempt to establish clear leadership by reconciling ESA's purely technical and commercial procurement approach with the broader political and military implications prioritised by the EU, though this synthesis achieved mixed success[17].

This historical dualism heavily influenced the formal legal basis for EU space involvement established by the Lisbon Treaty (2009), which introduced Article 189 of the Treaty on the Functioning of the European Union (TFEU). While Article 189 granted the EU a shared competence to elaborate a European space policy and support research and technological development, it had some important limitations. Member states retained the right to exercise their own national space policies, and the EU was barred from harmonising national laws on the basis of Article 189 alone. This structural limitation represented a fundamental institutional constraint for the Union when attempting to govern global spaces such as outer space[18].

[14] Paladini, S. (2019), cited; Gaubert, A., & Lebeau, A. (2009). Reforming European space governance. *Space Policy*, *25*(1), 37-44.

[15] Hoerber, T. C. (2009). The European Space Agency and the European Union: The Next Step on the Road to the Stars. Journal of Contemporary European Research, 5(3), 405–414. https://doi.org/10.30950/jcer.v5i3.133

[16] Buesa, A. (2024). Titanium metal in the EU: Strategic relevance and circularity potential. Publications Office of the European Union. Joint Research Centre (JRC) Technical Report. https://publications.jrc.ec.europa.eu/repository/handle/JRC137082

[17] Paladini, S. (2019) cited; Venet, C., & Baranes, B. (Eds.). (2012). *European identity through space: Space activities and programmes as a tool to reinvigorate the European identity* (Vol. 9). Springer Science & Business Media; Venet, C. (2012). European space governance: The outlook. *Space Policy*, *28*(1), 59-60.

[18] Larik, J. (2023). EU law and the governance of Global Spaces: ambitions, constraints and legal creativity. Journal of European Integration, 45(7), 1125–1142. https://doi.org

The EU's operational presence in space was embodied more in infrastructure management than centralised regulations, although some of them were important to consolidate the sector, like the Regulation (EU) 2021/696. This act established the Union Space Programme and the European Union Agency for the Space Programme (EUSPA) to govern the EU's space flagship initiatives to date: Galileo (satellite navigation), Copernicus (Earth observation), and EGNOS. Reflecting an expanding understanding of critical space infrastructure, the program also incorporated Space Situational Awareness (SSA) and Government Satellite Communications (GOVSATCOM).

Despite these operational achievements, the most significant gap in the pre-2025 framework remained the absence of a harmonised EU-level licensing and authorisation regime.

As for the Article VI of the Outer Space Treaty (1967), sovereign states bear international responsibility for national activities in space, which obligated individual EU member states to establish their own domestic authorisation regimes instead of acting as one sovereign body.

The result was a patchwork of divergent national laws, including such as France's Space Operations Act (2008), the Netherlands' Space Activities Act (2007), the already mentioned Luxembourg's Law on the Exploration and Use of Space Resources (2017), and last but not the least, the Italian Space Law (2025). All of which pioneered a highly specialised regulatory environment permitting commercial entities to legally own resources harvested from celestial bodies and other forms of private initiatives in outer space[19]. This inconsistency created regulatory arbitrage risks and complicated the governance of multi-national commercial missions.

In the absence of a dedicated, unified space statute, cross-cutting horizontal EU instruments applied to space-derived services by extension. The General Data Protection Regulation (GDPR) governed the processing of personal data collected via satellite systems; the NIS Directive addressed cybersecurity obligations relevant to space-based infrastructure operators; and export control frameworks under Regulation (EU) 2021/821 imposed strict constraints on the transfer of space technologies with potential military applications.

/10.1080/07036337.2023.2270670

[19] Petroni, G., Petroni, A., Bigliardi, B., Filippelli, S., & Pini, B. (2023). From Science to Market: The Business Orientation as Dominant Trait of Some Newly Created Space Agencies' Profile. New Space, 11(3), 190–200. https://doi.org/10.1089/space.2022.0021; Castellucci, I. (2024). The Dark Side of (Space) Law. In *Comparative Visions in Space Law* (pp. 117-131). RomaTre Press.

From early 2022, however, calls for a more structured approach at a supernational level began to appear.

In 2022, a Joint Communication on Space Traffic Management (STM) was issued by European Commission and the High Representative of the Union for Foreign Affairs and Security Policy, aimed at establishing a single-market perspective in the space sector and securing EU technological sovereignty. This initiative was expanded via the 2023 EU Space Strategy for Security and Defence [20] into a broader legislative goal to ensure safety, resilience, and sustainability of space operations alongside the NIS 2 and CER Directives (both of which enacted in 2024). A subsequent Commission Call for Evidence revealed, in case it was even needed, that a multiplying patchwork of divergent national laws was fracturing the single market and harming EU competitiveness [21].

The Commission decided eventually to create a binding EU framework focused on three main pillars, the same that are now at the centre of the Space Act, and namely, Safety, Resilience, Sustainability. After a few false starts, the project got into the drawing board with the newly appointed 2024–2029 Commission and its new Commissioner for Defence and Space, Kubilius.

The outcome is now under everybody's eyes.

3. *The Three Pillars of the EU Space Act*

The EU Space Act proposal (COM(2025) 335) in its June 2025 version (this point is worth mentioning, and it will be discussed in the last section of this article) is articulated in 118 articles and ten annexes. To achieve its goal of building a single regulatory framework while mitigating the hazards of an overcrowded orbit and together secure an Eu space sovereignty, the articles are meticulously partitioned between the preliminary single-market architecture and the three core technical pillars.

[20] Council of the European Union. 2023a. "Council Conclusions on the EU Space Strategy for Security and Defence." doc. 14509/23.

[21] Baumann, I. and Prall, K., 2025. The EU Space Act's Global Reach: Legal Considerations on the Envisaged Application to Non-EU Operators. Global Policy.

3.1 *Safety: Tracking, Debris Mitigation, and Secure Access*

The safety pillar of the EU Space Act is grounded in (justified) concerns about orbital congestion. According to ESA's Space Environment Report 2025 [22], there are currently more than 54,000 trackable objects in Earth orbit larger than 10 centimetres, while smaller, and untraceable, ones are far more numerous. The actual number of debris objects larger than 1 centimetre, sufficient to cause catastrophic damage to an operational satellite, is estimated at over 1.2 million. In the critically populated 500–600 km altitude band, the density of active satellites is almost matching the density of big debris, and non-deliberate satellite fragmentations continue at a rate of approximately 10.5 per year, with more than 3,000 newly catalogued debris fragments added in 2024 alone.

This without even accounting for ASAT tests, which have proved to be most devastating in terms of consequences[23]

Against this background, the Act requires all space operators providing services in the Union to obtain authorisations from a competent national authority, or, in the case of non-EU operators, from the European Commission acting through EUSPA [Art 6; Art. 17]. Authorisation applications must include detailed mission safety analyses [Art 58], orbital debris mitigation plans compliant with or stricter than the IADC Space Debris Mitigation Guidelines (i.e., 25 years for GEO), and commitments for rocket bodies and spacecraft at end-of-life. Specific requirements on the compliance rate for post-mission disposal are required. Current data from ESA indicates that only 40–70% of payload mass reaching end-of-life in LEO meets the established 25-year disposal guideline, and only

[22] European Parliament Research Service. (2025). China's rare-earth export restrictions. European Parliament Think Tank. Policy Briefing PE 768.120. https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA(2025)779220; European Space Agency (2025). ESA Space Environment Report 2025. Darmstadt: ESA Space Debris Office. https://www.esa.int/Space_Safety/Space_Debris/ESA_Space_Environment_Report_2025; New Space Economy (2025). The European Union Space Act: A New Regulatory Framework. December 2025. https://newspaceeconomy.ca/2025/12/10/the-european-union-space-act-a-new-regulatory-framework/; Office of Space Commerce (2025). U.S. Government Provides Feedback on Draft E.U. Space Act. 20 November 2025. https://space.commerce.gov/u-s-government-provides-feedback-on-draft-e-u-space-act/

[23] Paladini, S. (2023) 'Unsustainable Wars? The Use of Weapons in Lower Earth Orbit', in B. Cayli Messina (ed.) Environmental injustice and catastrophe: how global insecurities threaten the future of humanity. De Gruyter. Available at: https://doi.org/10.1515/9783111081687.

20–55% meets the stricter 5-year standard introduced by ESA in 2023[24].

The Act establishes mandatory reporting requirements on disposal compliance and creates an escalating enforcement regime for repeated failures, consistent with the GDPR's graduated penalty structure. For the operators of large constellations, collision avoidance protocols are requested [Art 73]. The Act also introduces a Union Register of Space Objects supported by e-certificates issued through EUSPA [Art 24-25].

The Act also contains provisions for light and radio pollution.

Operators of constellations are required to submit plans to minimise light reflection from satellites (which disrupts astronomical observations) and radio frequency interference with radio astronomy [Art 71-72; Art 73]. This represents the first time the European Union has formally legislated against light pollution from orbital assets, responding to scientific community concerns about the impact of mega-constellation, particularly Starlink, on ground-based astronomy [25][26].

3.2 *Resilience: Cybersecurity, Infrastructure Protection, and Business Continuity*

The resilience pillar responds to the growing strategic vulnerability of space infrastructure in the context of geopolitical competition. The Act's Explanatory Memorandum notes the growing involvement of commercial space actors in geopolitical conflicts and the corresponding threats of kinetic and cyber-attacks on satellite systems[27]. There have recently been

[24] While not specifically detailed in this article, there are important notifications and information requirements for the space operators, in addition to compliance and authorisation procedures, which of course add to the administrative burden. Examples are the databases, such as the one for high-interest events [art 67] and Environmental Footprint [art 99].

[25] Hainaut, O. R. & Williams, A. P. (2020). "Impact of satellite constellations on astronomical observations with ESO telescopes in the visible and infrared domains." A&A 636 https://www.aanda.org/articles/aa/abs/2020/04/aa37501-20/aa37501-20.html; Boley, A. C. & Byers, M. (2021). "Satellite mega-constellations create risks in Low Earth Orbit, the atmosphere and on Earth." Scientific Reports. 11-10642 https://www.nature.com/articles/s41598-021-89909-7

[26] See the dark skies debate now ongoing; IAU, 2024.

[27] Abels, J. (2024). "Private infrastructure in geopolitical conflicts: the case of Starlink and the war in Ukraine." European Journal of International Relations, 30(4), 842–866. https://doi.org/10.1177/13540661241260; Farrell H, Newman AL (2019) Weaponized interdependence: how global economic networks shape state coercion. International Security 44(1): 42–79.

a few, well documented incidents (jamming, spoofing, and cyberattacks targeting satellite navigation and communication systems) in the ongoing armed conflicts (Ukraine first).

The resilience pillar requires operators of Critical Space Infrastructure (CSI), here defined as satellite navigation, Earth observation, and secure communications systems, to implement measures aligned with the EU's NIS 2 Directive and the Cyber Resilience Act. These include mandatory risk assessments, incident reporting within 24 hours of a significant disruption, the implementation of a zero-trust security architecture for satellite and ground segment systems, and capabilities for rapid asset replacement. The Act mandates the development of in-orbit servicing (IOD/IOV) options as default design considerations and business continuity planning[28].

The resilience framework also addresses Radio Frequency Interference (RFI), both natural (solar flares) and man-made (jamming, spoofing), which poses risks to aviation and maritime safety. Operators must demonstrate that their systems incorporate anti-jamming and anti-spoofing protections and participate in coordinated spectrum management procedures administered through EUSPA. These provisions create obligations that align broadly with the objectives of the EU's IRIS2 secure connectivity constellation[29], while simultaneously creating compliance requirements that affect competitors to that programme.

3.3 *The third Pillar. Sustainability: A Deep Dive*

It is however the Act's third pillar, Sustainability (Title IV, Chapter III), even more than the other two, to have provoked some of the outcries from outside the EU in terms of geopolitical implications it can, and probably, will produce.

An attentive reading of the letter of the articles however shows that there is nothing surprising about this. The application of those measures

[28] This is one the points where the Space Act attracted internal criticism. While SMEs and academic operators are granted specific exemptions from some of the more demanding security certification requirements until 2031, to mitigate disproportionate burden these provisions would otherwise impose on smaller actors, there are calls for a more lenient implementation (Bratu & von Der Dunk, 2025).

[29] IRIS2 coverage is outside the scope of this article. However, it represents one of the most critical responses in the EU to address the digital sovereignty conundrum. See The EU Commission, 2023; Paladini, 2025

is going to produce systemic changes in the way the sector functions, with far-reaching consequences even outside the EU.

The Pillar requires all space operators providing services in the Union to calculate and publicly declare the environmental footprint of their activities across the full mission lifecycle, from satellite and launcher manufacturing through operational phase to end-of-life disposal. This requirement is operationalised through a mandatory space-specific Life Cycle Assessment (LCA) methodology, drawing on the EU's Product Environmental Footprint (PEF) approach. It also, and crucially, establishes the connection between the LCA obligations and existing EU product sustainability legislation, creating interlocking requirements that effectively extend the EU's circular economy framework into the space supply chain.

3.4 *The LCA Mandate and Its Regulatory Logic*

This section of the Space Act is arguably the most novel for a space regulation.

The Commission's Explanatory Memorandum [30] to the Act acknowledges that methodologies for evaluating the environmental impacts of space activities are 'underdeveloped.'

No existing international framework, not the IADC guidelines, not the ISO standards, not even the ESG framework, addresses the specific and complex environmental impacts of space activities, which is what the Act instead attempts, i.e., a mandatory Life Cycle Assessment covering propellant production, satellite manufacturing, launch operations, on-orbit operations, and end-of-life disposal, both atmospheric re-entry and controlled deorbiting included.

The LCA framework covers the full value chain.

Under Articles 96–100 of the Space Act, space operators must submit an Environmental Footprint Declaration (EFD) as part of their authorisation application, accompanied by supporting environmental footprint studies verified by a qualified technical body [Art 96]. Operators are required to obtain relevant data from their suppliers and transmit aggregated and disaggregated datasets to the Commission for inclusion in the Environmental Footprint Database. This provision is particularly

[30] European Commission (2025d) Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on the safety, resilience and sustainability of space activities in the Union -COM/2025/335 final

important, and it is something already found in NIS2[31]: the extension of the requirements beyond the EU actors to included non-EU subjects as well. And, as in the case of NIS2, the obligation to collect data from suppliers and transmit it to a centralised EU database has significant implications for supply chain transparency and for the ability of non-EU operators to maintain control over proprietary technical information (NIS 2, Art 21-23).

The EU Space Act Articles require mandatory LCA for satellites [Art 96], from propellant production to launch, through orbital operations and finally to deorbit and disposal, complementing the broader ESPR (*Eco-design for Sustainable Products Regulations*) and Circular Economy Action Plan framework. This is critical: to date, no global standard currently exists for this full-chain satellite LCA. The Commission intends the LCA requirements to complement product sustainability measures laid down in the Eco-design for Sustainable Products Regulation and the Circular Economy Action Plan framework, creating a coherent but legally complex multi-regulation overlay for the space manufacturing sector.

3.5 *The ESPR-DPP Extension into Aerospace*

An underappreciated but not for this less crucial dimension of the sustainability pillar is its relationship with the Ecodesign for Sustainable Products Regulation and the Digital Product Passport (DPP) regime Regulation (EU) 2024/1781[32].

The ESPR, which entered into force in July 2024, applies to all physical products of the EU market, requiring them to carry structured digital records of material composition, environmental performance, and end-of-life guidance. The ESPR's initial product priority sectors include textiles, furniture, iron and steel, and aluminium, but the working plan for 2025–2030 previews the extension of the provisions to aerospace and

[31] For more, see the consolidated version of NIS2 -Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on measures for a high common level of cybersecurity across the Union, amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and repealing Directive (EU) 2016/1148. At http://data.europa.eu/eli/dir/2022/2555/oj

[32] Suzuki-Klasen, A. K. (2025). The Digital Product Passport under the EU Ecodesign Regulation 2024: Empowerment of the Customer-Consumer on the Road to Sustainability in a Circular Economy or Information Overload?. *Journal of Law, Market & Innovation*, *4*(3), 444-469.

space intermediates after 2027[33].

This extension means that aerospace-grade components, structural aluminium alloys, titanium fasteners, carbon fibre composite panels, rare earth permanent magnets used in electric propulsion systems, will be subject to DPP obligations requiring traceability from mine to manufacturer to satellite to end-of-life recycler. For a typical small satellite with hundreds of sourced components, implementing full DPP compliance could require redesigning procurement systems, renegotiating supply contracts, and building new data infrastructure.

The EU Critical Raw Materials Act (CRMA) specifies that from 24 May 2027, or two years from the entry into force of the relevant delegated act, products containing permanent magnets above 0.2 kg must disclose the share of critical raw materials recovered from post-consumer waste [CRMA, art 29]. Given the use of rare earth permanent magnets in satellite electric propulsion and reaction wheel systems, this provision directly affects the space sector's supply chain.

The combined impact of the ESPR-DPP and EU Space Act LCA obligations will create a full-stack industrial transparency requirement for space manufacturing: every component batch traceable from mine to manufacturer to satellite to end-of-life recycler. This represents a qualitative shift in regulatory ambition, in the same way the GDPR transformed data governance from a compliance exercise to a foundational requirement.

3.6 *The CRMA Interface: Domestic Content, Circular Economy, and Trade Tensions*

The formidable impact of the Space Act is due to the fact these provisions do not stand alone.

The Sustainability pillar operates in conjunction with the Critical Raw Materials Act (CRMA) which entered into force in May 2024 and establishes binding targets for domestic extraction, processing, and

[33] Blot, E. (2025). External impacts of new EU sustainable product standards: Exploring spillovers of the Ecodesign for Sustainable Products Regulation. Institute for European Environmental Policy (IEEP). https://ieep.eu/wp-content/uploads/2025/04/External-impacts-of-new-EU-sustainable-product-standards-IEEP-2025.pdf; Chawla, K., Chirvasuta, T., Wolf, M.-A., Wolf, K., & Rongen, S. (2026). Methodology for defining data requirements for the Digital Product Passport under the ESPR framework. Publications Office of the European Union. Joint Research Centre (JRC) Technical Report.

recycling of strategic raw materials. The CRMA sets the following 2030 benchmarks: at least 10% of the EU's annual consumption of each strategic raw material from EU extraction; at least 40% from EU processing; and at least 25% from domestic recycling [CRMA, Art 1].

No single third country is to supply more than 65% of EU consumption of any strategic raw material at any relevant stage of processing, which, on the other hand, makes CRMA a formidable geopolitical instrument even before of a commercial one[34].

For the space sector[35], the materials most directly affected are space-grade titanium (used extensively in launch vehicles and satellite structures), carbon fibre (structural components), rare earth elements used in permanent magnets (electric propulsion, reaction wheels, deployable mechanisms), and gallium and germanium (solar cell and semiconductor applications). The EU currently sources 100% of its heavy rare earth elements from China, which also provides the vast majority of the EU's supply of gallium, tungsten, and bismuth [36]. Since 2024, China has implemented a series of escalating export control measures covering these materials, dramatically increasing supply risk for European space manufacturers [37].

There is also another, often overlooked point: more than 65%+ of EU aerospace-grade titanium scrap is currently exported to the US under buy-back agreement, a fact that illuminates a structural absurdity[38].

[34] Correia, V., Rosendo, L., & Falck, E., Hamor T., Cescon, M. (2026). From Principles to Practice: Operationalising Europe's Critical Raw Materials Strategy. Intereconomics: Review of European Economic Policy, 61(2), 74–82.

[35] The CRMA explicitly lists "aerospace" as a strategic technology sector. Large space and defence companies operating within the EU are legally required to conduct comprehensive supply chain risk assessments every three years [Art. 24 & 25]

[36] European Parliament Research Service. (2025). China's rare-earth export restrictions. European Parliament Think Tank. Policy Briefing PE 768.120. https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA(2025)779220; European Space Agency (2025). ESA Space Environment Report 2025. Darmstadt: ESA Space Debris Office. https://www.esa.int/Space_Safety/Space_Debris/ESA_Space_Environment_Report_2025

[37] Wachmeinster, H. (2025). China's mineral export restrictions: Market impacts and implications. Swedish Institute of International Affairs (UI). UI Briefing Report No. 4. https://www.ui.se/globalassets/ui.se-eng/publications/other-publications/chinas-mineral-export-restrictions_market-impacts-and-implications_nkk_2025.pdf

[38] Buesa, A. (2024). Titanium metal in the EU: Strategic relevance and circularity potential. Publications Office of the European Union. Joint Research Centre (JRC) Technical Report. https://publications.jrc.ec.europa.eu/repository/handle/JRC137082; Jakimów, M., Samokhalov, V., & Baldassarre, B. (2024). Achieving European Union strategic autonomy: circularity in critical raw materials value chains. *International affairs*, *100*(4), 1735-1748.

European space manufacturers produce titanium-rich waste streams that are exported for processing in the United States and then reimported as finished aerospace components, one of the things the Act addresses, since the combined effect of the CRMA and Space Act's sustainability obligations implicitly require this loop to be closed within Europe. And this is where things risk creating international issues, as these new provisions might violate existing trade agreements, including WTO rules on national treatment and most-favoured-nation obligations, as well as bilateral agreements that have structured US-European aerospace supply chains for decades.

CRMA sets a 25% recycled content target from domestic sources by 2030.

For space hardware (e.g. satellites and launchers) compliance with this target creates a domestic content requirement. A satellite that incorporates titanium, carbon fibre, or rare earth magnets sourced from non-EU recycling streams would fail to meet the CRMA's recycled content benchmarks, while the Space Act's LCA obligations would make this non-compliance visible and enforceable. The interaction of the two regulations functions as a full-stack industrial sovereignty tool: the CRMA controls who can supply recycled space-grade materials, and the Space Act controls who can operate in the EU market, creating a Brussels-governed gateway to both the supply chain and the market.

This dynamic is further amplified by the Space Act's sustainability reporting obligations, which require operators to reduce virgin material use, as specified in the Recitals 94 and 95. Operators using virgin titanium or non-recycled rare earth materials will likely face escalating reporting burdens and, once performance classifications are introduced through following acts, a potential competitive disadvantage in EU procurement processes. The combination effectively incentivises the entire space supply chain to source from CRM Act-compliant domestic or partner-country suppliers, regardless of whether existing international supply chains are more cost-efficient or violate existing commercial treaty.

3.7 *SME Exemptions and the Risk of Market Concentration*

In terms of mitigation, the Space Act does indeed provide exemptions from SMEs and other institutions, and this also include the environmental footprint obligations. It also explicitly excludes In-Orbit Demonstration (IOD/IOV) missions from Environmental Footprint calculations [Art 96;

Art 10(4)]. Supporting measures are foreseen to help start-ups, small mid-caps and research institutions offset compliance costs. There measures reflect a recognition of how compliance burden of full LCA and DPP implementation could negatively impact smaller operators, who lack the data infrastructure and supply chain leverage of large prime contractors.

There is also the possibility of intended consequences, and more than one [39] remarked how these exemptions may accelerate market concentration. If LCA and DPP compliance become *de facto* prerequisites for EU market access, as seems likely once the regulatory infrastructure becomes operative, companies that cannot afford compliance may be forced to exit the market or consolidate with larger players capable of absorbing overheads. The net effect could be a market structure dominated by large EU primes and compliant non-EU majors, in what seems a recurring feature of the EU regulatory initiatives in high-technology sectors [40].

4. *The Geopolitical Consequences: The Brussels Effect in Orbit*

4.1 *The Brussels Effect and Its Space Dimension*

The *Brussels Effect*, otherwise defined the tendency for EU regulations to become de facto global standards through market power rather than formal treaty negotiation[41] has been extensively documented in areas including data protection (GDPR), competition law, chemicals regulation (REACH), and financial services. The EU Space Act has been designed,

[39] Jørgensen, B. N. (n.d.). Impact of EU laws and regulations on the adoption of artificial intelligence in cyber–physical systems: A review of regulatory barriers, technological challenges, and cross-sector implications. Electronics 2026, 15(10), 2184; https://doi.org/10.3390/electronics15102184

[40] Quaritsch, L. (n.d.). The EU's Digital Omnibus is heading in the wrong direction. Jacques Delors Centre Policy Brief. https://www.delorscentre.eu/en/publications/detail/publication/the-eus-digital-and-ai-omnibus; Vipra, J., & Korinek, A. (2023). Market concentration implications of foundation models. arXiv. https://doi.org/10.48550/arxiv.2311.01550

[41] Bradford, A. (2020). *The Brussels effect: How the European Union rules the world*. Oxford University Press; Gunst, S., & De Ville, F. (2021). The Brussels effect: how the GDPR conquered Silicon Valley. *European Foreign Affairs Review*, *26*(3); Bendiek, A., & Stuerzer, I. (2023). The Brussels effect, European regulatory power and political capital: Evidence for mutually reinforcing internal and external dimensions of the Brussels effect from the European digital policy debate. *Digital Society*, *2*(1), 5.

whether explicitly or structurally, to replicate the same approach. By conditioning market access to EU-based customers on compliance with its safety, resilience, and sustainability requirements, the Act creates powerful incentives for non-EU operators to align their global practices with EU standards.

The conditions for the Brussels Effect to work also for the space sector also exist for the space sector.

The EU does represent a significant market for downstream space services, accounting for about 10% of the world's space sector[42]. European institutional customers, as much as government agencies, defence procurement bodies, and publicly regulated industries such as aviation and the maritime sector, are subject to EU procurement rules that increasingly require compliant services. Third-country operators seeking access will have to obtain authorisation from the EU and comply with the Act's technical requirements, creating a direct pathway for EU standards to shape global space operations, the way it already happened with the GDPR.

What creates issues is the different treatment the space operators have depending on their residency status. While the EU-established companies apply for licences through their home member state's national competent authorities, the non-EU (a wide and diverse category that includes corporations such as SpaceX, Amazon Kuiper, OneWeb's non-EU shareholders, and virtually all Chinese and Indian operators) must obtain authorisation directly from the EU authorities, upon review by a Compliance Board within EUSPA.

The conflict of interests is structural, as the EUSPA is simultaneously the regulatory gatekeeper for non-EU operators and the planned operator of the EU's own IRIS2 secure connectivity constellation[43], which will compete directly with the operators it is tasked with regulating. The Act also authorises the EU to request inspections of non-EU operators' facilities located outside the Union as a condition of market access [Art 52]. This specific provision is in direct conflict with US export control

[42] Novaspace, 2025, cited.

[43] The IRIS2 constellation represents the most recent EU project, and it is especially important in terms of the EU's digital sovereignty. IRIS2 will have approximately 264 satellites (therefore below the 1,000-satellite threshold that triggers the Act's 'giga-constellation' category and its associated enhanced obligations, applicable instead to SpaceX's Starlink with its 8,100 satellites, and Amazon's Project Kuiper with more 3,236 satellites. According to the critics, this represents yet another evidence the Act functions as a non-tariff barrier.

laws, the International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR), which restrict foreign access to certain categories of technical information and physical assets[44]. A (hypothetical for the moment) demand of inspection for access to a US satellite facility could create an irresolvable compliance conflict for US operators, who would face the choice of breaching US law or forfeiting EU market access.

The impact of these specific norms on the space sector is impossible to overstate, and it has been noted in no ambiguous terms from the non-European actors.

As US-based ITIF observed in its August 2025 analysis, if enacted in its present state, the Space Act would establish a precedent for other jurisdictions to create their own protectionist space regulations, fragmenting the global regulatory environment and multiplying compliance burdens for satellite operators worldwide [45]. The key question is whether the Brussels Effect in space will represent a force for convergence on high standards or for regulatory protectionism. The answer may well differ depending on the perspective of the observer.

Moreover, substantial as it may be, the authorisation regime for non-EU operators is not the only issue in competition terms that the Space Act creates. As already mentioned, the Sustainability pillar has the potential to represent a formidable hurdle, due to the combination of regulations that includes, among others, the Ecodesign for Sustainable Products Regulation (ESPR) with its Digital Product Passport requirements, the Critical Raw Materials Act (CRMA), the Cyber Resilience Act with the NIS 2 Directive and, last but not the least, the planned Circular Economy Action Plan II (CEAP II).

In these conditions, any space operator seeking to operate in the EU space services market faces substantial hurdles, and namely, a cumbersome application process, supply chain transparency requirements, materials sourcing, cybersecurity certification, environmental reporting. This cumulative compliance burden for non-EU operators has been estimated by industry groups [46] at several million euros per year for a medium-sized satellite operator, and even higher for large satellite constellations.

[44] Blount, P. J. (n.d.). The ITAR Treaty and Its Implications for U.S. Space Exploration Policy and the Commercial Space Industry. Journal of Air Law and Commerce. https://scholar.smu.edu/cgi/viewcontent.cgi?article=1180&context=jalc

[45] Information Technology and Innovation Foundation (ITIF) (2025). The EU's Space Act. 26 August 2025. https://itif.org/publications/2025/08/26/the-eu-space-act/

[46] Gleason, M.P. (2024) Anticipating the New European Union Space Law, Center for Space Policy and Strategy. Available at: https://csps.aerospace.org (Accessed: 8 June 2026).

5. *Geopolitical implications of the EU Space Act*

The United States government's response to the EU Space Act has been both rapid and highly critical. The Office of Space Commerce, in coordination with the Department of State, conducted a formal consultation involving over 70 US commercial space companies following the Act's publication in June 2025 [47]. The resulting November 2025 feedback to the European Commission identified several structural concerns: the lack of regulatory clarity in the Act's text itself (with critical operational parameters deferred to implementing acts); the need for exemptions for civil governmental activities conducted by US federal agencies and NATO allies; and the requirement to ensure that European member states could continue to leverage US commercial space capabilities in support of transatlantic defence industrial cooperation.

US industry's concerns focused on three main aspects.

One is the threshold set for the 'giga-constellation' of the kind of Starlink, i.e. the 1,000-satellite criterion that lacks grounding in established international safety standards developed by IADC and ISO[48] and seems a bespoken criteria based on commercial not technical logics. The authorisation procedure mentioned in the previous section is another obvious target as it creates a structural conflict of interest, with the double role of EUSPA. Last but not the least, the Act's bespoke technical requirements deviate from globally accepted IADC and ISO standards, and they are seen as creating parallel compliance systems and defying the economic benefits of global standardisation.

There are other, less obvious impact, which arise from the linkage between the Act and the other regulation. At a trade policy level, the Act's requires compliance with the CRM Act's 25% recycled content target. The buy-back agreements that currently govern the recycling of EU-origin aerospace titanium scrap in US facilities risk being nullified by a regulatory environment that penalises the use of non-domestically recycled materials. The July 2025 US-EU trade agreement excluded (so far) aerospace from tariff escalation, but the Act's non-tariff dimensions were not addressed in that framework. Considering that over 8,500 active satellites are

[47] Dede, G. 2025 "Toward an equitable governance of space resources: A policy-oriented framework for sustainability and resilience", JASS, vol. 5, no. 2, pp. 111–122, Aug. 2025, doi: 10.52995/jass.1736362.

[48] IADC, for instance, set the risk parametres based on operational and orbital characteristics rather than arbitrary satellite counts.

owned or operated by US-based companies (the number represents about three-quarters of the approximately 11,000 active satellites currently in orbit) a regulatory framework that imposes enhanced compliance burdens specifically on the operators of large constellations is perceived in Washington as a regulatory tool for industrial policy 'greenwashed' in environmental and safety language.

China's position is somehow more nuanced and based on a different logic.

Beijing emphasised the divergence between the Act's technical requirements and those of internationally agreed frameworks, specifically the IADC Space Debris Mitigation Guidelines and ISO standards, arguing that the EU's unilateral rulemaking undermines the authority of multilateral bodies, in addition of creating discriminatory market access conditions. China's space industry, represented by the two state-owned conglomerate CASC, CASIC, and the emerging commercial constellation operators (Galaxy Space, GeeSpace), has at the moment limited current exposure to the EU market but anticipates growing downstream service ambitions in Europe.

The sustainability pillar, however, presents particular complications for Chinese operators. China's domestic regulatory framework has no space-specific LCA requirement and no equivalent to the EU's Product Environmental Footprint methodology.

Full compliance with the Act's Environmental Footprint Declaration requirements would require Chinese satellite manufacturers to restructure their supply chain data management systems, obtain third-party verification from qualified technical bodies (presumably EU-accredited), and transmit detailed manufacturing and operational data to EUSPA's Environmental Footprint Database, raising obvious concerns about industrial espionage and technology transfer[49].

The rare earth dimension is sensitive.

China dominates the global supply of heavy rare earth elements and since October 2025 has dramatically expanded its export control regime to cover not only raw materials but also midstream and downstream manufacturing processes. The EU Space Act's LCA requirements, combined with the CRM Act's domestic content targets, push European space manufacturers away from Chinese rare earth supply chains and toward domestic or partner-country alternatives. From Beijing's perspective, this

[49] White & Case (2026). Regulating Space: A Closer Look at the Proposed EU Space Act. March 2026. https://www.whitecase.com/insight-our-thinking/regulating-space-closer-look-proposed-eu-space-act

constitutes an attempt to weaponise environmental regulation against Chinese industrial supply chains [50]. The EU's debris and sustainability standards deviate from IADC/ISO global norms in ways that create China-specific compliance costs, and the bespoke compliance overhead relative to the IADC/ISO baseline represents a structural market access barrier.

China's response has been to accelerate the development of domestic standards through participation in the International Telecommunication Union and to promote IADC guidelines as the appropriate international baseline, while simultaneously developing its own national space sustainability framework. The risk is one of bifurcation: a global space regulatory environment divided between an EU-aligned framework and a China-aligned alternative, with other jurisdictions forced to choose sides, or to navigate both simultaneously [51].

Finally, India.

The country occupies a distinctive position in the global space economy. ISRO has established India as a credible launch services provider, with the LVM3 and SSLV launch vehicles increasingly competitive in the international small satellite launch market [52]. The Indian commercial space sector, liberalised through the Indian Space Policy 2023 [53], is expanding rapidly, with companies such as Skyroot Aerospace, Agnikul Cosmos, and Pixxel active in launch, manufacturing [54], and Earth observation. India's

[50] Jones Day. (2025). China imposes extraterritorial export control measures over rare earth items. Jones Day Legal Insights. https://www.jonesday.com/en/insights/2025/10/china-imposes-extraterritorial-export-control-measures-over-rare-earth-items#; Taylor Wessing. (2025). China's expanded export controls on rare earths – What European companies need to know. Taylor Wessing Regulatory Briefing. https://www.taylorwessing.com/en/insights-and-events/insights/2025/10/chinas-expanded-export-controls-on-rare-earths

[51] Stroikos, D. (2026). China, the 'rise of the rest' and the remaking of the international space order in a multiplex world. International Affairs, 102(2), 389–408. https://doi.org/10.1093/ia/iiaf275

[52] Kumar, M.J. (2026) 'From Missions to Markets: India's Leadership in Space Technology', IETE Technical Review, 43(3), pp. 293–295. doi: 10.1080/02564602.2026.2670851.; Stroikos, D. (2024) India's Space Policy: Between Strategic Autonomy and Alignment With the United States, New York: Council on Foreign Relations. Available at: https://cdn.cfr.org/sites/default/files/pdf/Stroikos%20-%20India%27s%20Space%20Policy.pdf

[53] Indian Space Research Organisation [ISRO] (2023) Indian Space Policy 2023, Bengaluru: Department of Space. Available at: https://www.isro.gov.in/media_isro/pdf/IndianSpacePolicy2023.pdf (Accessed: 8 June 2026).; Kawai, M. (2025) 'Strategic National Objectives in Space: Strategic Alignment Framework', Space Policy, 41(2), pp. 73–85.

[54] Jain, H. (2025) 'Unlocking India's Space Economy: Frugal Innovation and Private

ambitions in the European downstream market are real and growing, creating direct exposure to the EU Space Act's requirements.

India's fundamental challenge is structural.

The country has no space-specific LCA framework. Full compliance with the Act's Environmental Footprint Declaration requirements for an Indian launcher or satellite manufacturer would require building a bespoke engineering compliance capability from scratch [55], i.e., an end-to-end investment that industry estimates suggest could cost several million euros for a medium-sized Indian space company. The absence of any existing Indian regulatory framework for space sustainability means there is no equivalence pathway through which Indian operators could demonstrate compliance with EU standards without building parallel systems.

Otherwise said, this is a 'framework gap'.

The Act's equivalence mechanism, through which the Commission may recognise third-country regulatory frameworks as meeting EU standards, could theoretically create a faster compliance pathway for India, but only if India develops a domestic space LCA framework aligned to the EU's PEF approach. This creates a structural incentive for the Indian Space Research Organisation and the newly established IN-SPACE regulatory body to develop LCA standards partly or wholly shaped by EU methodological choices, a form of regulatory export that serves European interests regardless of whether it was intentionally designed to do so.

So far, India's response has been cautious. Indian space industry representatives have participated in EU consultation processes and flagged the compliance cost implications for developing-country space actors. More in general, there is a widely shared, legitimate concern that a global space regulatory architecture built on EU sustainability standards, however well-intentioned, could create substantial barriers to entry for space sectors in the Global South that lack the established industrial infrastructure to absorb compliance costs.

Sector Integration', International Journal on Science and Technology (IJSAT), 15(1), pp. 115–124.

[55] Gleason, M.P. (2024) Anticipating the New European Union Space Law, Center for Space Policy and Strategy. Available at: https://csps.aerospace.org (Accessed: 8 June 2026).

6. *Conclusion: Towards a GDPR for Outer Space?*

The EU Space Act represents the most ambitious attempt since the Outer Space Treaty of 1967 to establish a comprehensive governance framework for space activities. Its three-pillar architecture of Safety, Resilience, Sustainability is a coherent as a response to the genuine challenges facing the orbital environment. The debris statistics are alarming, the cybersecurity vulnerabilities are real, and the absence of any mandatory LCA framework for space activities is a genuine regulatory gap. The Act's ambition to fill this gap should not be dismissed as mere protectionism.

However, the Act's design choices, in particular the extra-territorial scope, the GDPR-style enforcement architecture, the asymmetric authorisation procedures, and the interaction with the CRM Act's domestic content implications, do create challenges, as they translate into an industrial policy tool that, if left as they are at present, advantages EU operators. The 'giga-constellation' threshold is the most visible signal, but it is not isolated: the cumulative effect of the full EU C.E. regulatory stack on space manufacturers creates barriers to entry that are compliance-shaped rather than tariff-shaped.

The geopolitical consequences are already visible. The US government has formally raised concerns about ITAR conflicts and discriminatory market access. China has responded by accelerating domestic standards development and expanding export controls on rare earth supply chains. India faces structural framework gaps that translate into bespoke compliance costs. None of these reactions suggests a trajectory toward convergence on common global space sustainability standards. Instead, the more plausible near-term outcome is a fragmented global regulatory environment in which operators must navigate multiple, partially incompatible frameworks simultaneously, which risks create even more issues in terms of sustainability.

The analogy to the GDPR is instructive in both its positive and cautionary dimensions.

The GDPR successfully raised global data protection standards and triggered regulatory convergence in many jurisdictions. But it also created compliance costs that systematically disadvantaged smaller operators and non-EU companies, and inspired imitators whose frameworks are not always genuinely equivalent. A GDPR for outer space could similarly raise the global floor for orbital sustainability but only if it is designed

addressing to the interests and capabilities of all countries, not just those of the European space industrial complex. And, as pointed out, the equity dimension of the Brussels Effect in space is a significant consideration that has received insufficient attention in European policy discussions.

The current text does not meet those standards (yet). The legislative procedure is still ongoing. The amendments adopted by the European Parliament and Council will determine whether the final Act emerges as a valid and inspiring instrument of global space sustainability governance or as a sophisticated and geopolitically oriented non-tariff barrier.

Vincenzo Zeno-Zencovich

Postface

Summary: 1. What a pity! – 2. Too much, too late.

1. *What a pity!*

What a pity the European Union did not exist already in the 15th and 16th centuries. The world would now be much smaller, close-knit and probably more peaceful apart from a few skirmishes with the Turks.

One has only to think that Vasco da Gama, Christopher Columbus and Ferñao de Magalhães instead of rambling over the seas could and would have been enjoined by an over-compassing "Outer-Seas Discovery Act" – 140 recitals and 119 articles – from leaving the Pillars of Hercules and the Baltic Sea, with a very limited range for cabotage along the European Atlantic and the North Sea. Their ESG caravels would have kept them from venturing more than a few miles from the coast, and needless to say the strict protection of tropical flora would have prevented the invasion of our agricultural markets by potatoes, tomatoes, pineapples and bananas. And how healthier we would all be if these wretched discoveries hadn't brought us the scourge of tobacco and coffee. And clearly Europeans would have not been led onto the criminal path of slavery from Africa to the New World.

However the EU now exists and is, coherently, setting the rules to curb a reckless discovery of Space, to ensure a quiet and safe sky, environmentally compliant, lit at night only by the moon and the stars. The further step, that no doubt will follow, is the ban on discriminatory and stereotyped hate-speech narratives such as the "Man on the Moon" (much worse than "Little Black Sambo"), and the countless shockingly racist (towards alien species) novels and short stories by Isaac Asimov. And here enters the proposal for a EU Regulation "on the safety, resilience and sustainability of space activities in the Union".

2. *Too much, too late*

The proposed EU "SRSSA" Act is a typical product of the EU institutions, which live – and thrive – in a bubble convinced that hundred of pages filled with normative texts can change the world and the behaviour of humans. This superstitious view is not dissimilar from that of the counsellors of King Canut who assured him that his command could withhold the tide. I will not insist on the most obvious flaws – to put it mildly – of the proposed Regulation, which have in detail been highlighted by several of the papers that are published in this book, and will focus on a few policy points.

a) *Space as an already colonized context.*
While in the 15th and 16th century European navigators competitors' where only in the East (dhows and sampans), today's orbital space is entirely colonized by US companies (Starlink and Amazon Leo) with over ten thousand satellites, to which one must add the thousand launched by PRC state owned companies. The comparison is embarrassing: over the last decades Eutelsat, the European company that provides telecommunication services, has launched a mere 600. In other, and rather brutal, words the EU is trying to regulate a market in which it holds around 5%. As most of the recent EU legislation (whether on goods, on services or on agri-food) also the SRSSA has a clear protectionist function: foreign companies may not provide satellite-driven services in the EU if they do not comply with its regulations.
However the sheer economic fundamentals tell us that there can be no reasonable competitive dynamics if your competitors hold (today) 95% of the market (and even more tomorrow) and are able to offer higher quality services and much lower prices. What the proposed Regulation is proposing and promoting is that European end users (businesses and consumers) should foot the bill of a subsidized and lame domestic industry.

b) *Kill telecoms.*
Tout se tient. The proposed Regulation follows the footsteps of a decades old enmity of the EU institutions against telecommunication infrastructures.
On the one side burdening infrastructure managers with the unsustainable weight of the EU Electronic Communications

Code (Directive 2018/1972) made of 326 recitals, 125 articles and 12 annexes. But most importantly weighed down by the purely ideological mantra of "common carrier" through which foreign OTT's have been free-riding European infrastructures. Although the ECD does not apply to satellites, the complete dominance of the terrestrial orbit market by foreign companies simply will lead to the progressive extinction of European telecommunications service providers. In a near future one can expect that data, voice, images will be literally in the clouds of orbital space and will mostly be received, stored, accessed by and on satellites.

c) *Of masters and servants.*
Stephane Grumbach quite aptly entitles his book "*L'empire des algorithmes*". And develops with a broad geopolitical analysis the forecast that under the Empires (USA and PRC) there are servants and slaves. Europe is one of them. All its mouth-washing on EU sovereignty hides the fact – obvious to anybody not living under the EU spell – that Europe is entirely dependent from a technological, financial and military point of view from foreign countries and industries. If one only thinks of the miserable failure of the Gaia-X project and of how all European public and para-public big data are stored on foreign facilities or thanks to foreign computer hardware and software the claim of EU sovereignty appears to be a sham.

d) *The only good Regulation is…*
One of the most baffling aspects of the EU law-making process is that of setting up a complex procedure called "Better Regulation" which inevitably ends up by saying that the proposed legislation is compliant with all the fundamental rights and principles set in the Treaties and in EU Charter of Fundamental Rights, that it will bear negligeable costs for individuals and businesses, and that it will benefit society. When a few years later one discovers that the legislation did not achieve practically any of its objectives, the finding is silently swept under the carpet of the C Series of the Official Journal.
What is totally lacking – but this is no surprise, considering the bubble in which EU institutions live – is the notion that what should and could be done is remove existing obstacles to European space industries, favouring their cooperation, get rid of the BBB (Brussels Big Brother) effect from their shoulder, promoting

investments in education and training in the many facets of space technology and economics. A simple moratorium for the next ten years might, maybe, prevent Europe from falling under a complete space servitude, which inevitably will be replicated on the terrestrial ground.

Roma Tre

The publication of the European Commission's proposal for a Regulation on the safety, resilience and sustainability of space activities in the Union — COM(2025) 335 final, of June 25, 2025, immediately designated by commentators as the EU Space Act — marks a turning point in the governance of outer space that extends well beyond the internal market rationale invoked in its explanatory memorandum. The EU Space Act is a regulatory instrument conceived in the tradition of the GDPR and the AI Act, designed to impose European standards on a global scale through the leverage of market access. While the institutional debate on the proposal at European level is ongoing, the contributions collected in this instant book examine the text and its potential impact and implications considering several complementary disciplinary vantage points and, taken together, offer a critical anatomy of an instrument that is simultaneously ambitious and structurally constrained.

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